CHMD11 · 课程讲义复习 · COURSE LECTURE REVIEW · 幻灯片 1–80 · SLIDES 1–80

CHMD11 高级NMR方法 · 讲义复习版

CHMD11 Advanced NMR Approaches · Lecture Review

溶剂峰抑制 · 多维 NMR · NUS 与超快 NMR · MAS / HR-MAS / CMP · 探头与微线圈 · 混合物选择性实验 · 活体 NMR · SSFP 与 CRAFT

Solvent suppression · higher-dimensional NMR · NUS and ultrafast NMR · MAS / HR-MAS / CMP · probes and microcoils · selective mixture experiments · in vivo NMR · SSFP and CRAFT

KaTeX 公式 · KaTeX formulas | 中英对照 · Bilingual | 每页考点速记 · Key points on every page | 逐张对应课件 · Slide by slide

本讲义按课件顺序讲解全部 80 张幻灯片：每张给出双语标题、机理讲解、图的读法和考点速记。课件之外的标准知识标为「背景」，课件本身的疑点放在「编者注」中。

> This review walks through all 80 slides in the deck's order: each slide gets a bilingual title, an explanation of the mechanism, a reading of its figure and key points. Standard knowledge not on the slides is marked "Background"; questionable points in the deck itself go into "Editor's note" boxes.

为 Jiasheng Fu 准备 · 2026 年 11 月

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考点速记 KEY POINTS

- 大溶剂峰占满接收机动态范围 → 样品峰 SNR 下降（S2）。
  A huge solvent peak fills the receiver's range → sample SNR drops (S2).
- NUS 25 %：64 个增量 $3.75\ \mathrm{min}$；常规 256 个需 $15\ \mathrm{min}$（S20）。
  NUS 25 %: 64 increments in $3.75\ \mathrm{min}$ vs 256 in $15\ \mathrm{min}$ (S20).
- 魔角 $\theta_{\mathrm{m}} \approx 54.74^{\circ}$，满足 $P_2(\cos\theta) = 0$（S25）。
  Magic angle $\theta_{\mathrm{m}} \approx 54.74^{\circ}$, where $P_2(\cos\theta) = 0$ (S25).
- 小线圈质量灵敏度好；大线圈浓度灵敏度好（S39、S47）。
  Small coils: mass sensitivity; large coils: concentration sensitivity (S39, S47).

全书地图 · 怎样读这本讲义

Book map · how to read this review

§	主题 Topic	幻灯片 Slides	学习目标 Learning goal
1	溶剂峰抑制 Solvent suppression	2–13	大溶剂峰为何损失 SNR；presat、WET、PURGE、WATERGATE、W5 的原理；按溶剂、可交换质子、近溶剂信号和弛豫选方法。 Why a solvent peak costs SNR; how presat, WET, PURGE, WATERGATE and W5 work; choosing by solvent, exchangeables, nearby signals, relaxation.
2	多维 NMR Higher-dimensional NMR	14–18	峰容量；三维实验与投影如何构建（HMQC-COSY、H–C–Si）；三维何时值得花时间。 Peak capacity; how 3D experiments and projections are built (HMQC-COSY, H–C–Si); when 3D is worth the time.
3	NUS 与超快 NMR NUS and ultrafast NMR	19–23	线性 t_1 采样、NUS 与单扫描空间编码：各省多少时间，付出什么 SNR / 伪峰代价。 Linear t_1 sampling vs NUS vs single-scan spatial encoding: time saved vs SNR and artifact costs.
4	MAS、HR-MAS、固体与 CMP MAS, HR-MAS, solids, CMP	24–36	魔角；转速与转子尺寸的影响（边带、分辨率、样品存活）；选 HR-MAS、固体或 CMP（谱编辑）。 The magic angle; what spin rate and rotor size do (sidebands, resolution, survival); picking HR-MAS, solids or CMP (spectral editing).
5	探头与线圈尺寸 Probe and coil size	37–49	质量与浓度灵敏度；小探头、微线圈（阵列）、大线圈何时有利，及其 B_1、操作和加热限制。 Mass vs concentration sensitivity; when small probes, microcoils (arrays) or large coils help; B_1, handling and heating limits.
6	混合物与选择性实验 Mixtures, selective experiments	50–66	选择性 TOCSY、单重态、仅氨基酸 NMR、GEMSTONE、DREAMTIME 如何挑出目标物，各损失多少信号。 How selective TOCSY, singlet, amino-acid-only NMR, GEMSTONE and DREAMTIME pick targets out, and their signal cost.
7	活体 NMR In vivo NMR	67–74	让生物在磁体中存活并测量（流动系统、活体 CMP）；监测胁迫、恢复、PCA、同位素过滤代谢。 Keeping organisms alive and measured in the magnet (flow systems, in vivo CMP); monitoring stress, recovery, PCA, isotope-filtered metabolism.
8	SSFP 与 CRAFT SSFP and CRAFT	75–80	SSFP 的稳态与分辨率–SNR 权衡；CRAFT 用振幅–频率表代替 FT 来校相、找弱峰。 SSFP's steady state and resolution–SNR trade-off; CRAFT's amplitude–frequency table replaces the FT to phase and find weak peaks.

阅读说明 / How to read

- S2、S3……= 课件幻灯片编号（S1 为封面）；内容多的另有（续）页。中文在前，英文随后。
  S2, S3… = slide numbers in the deck (S1 is the cover); dense slides get a (cont.) page. Chinese first, English after.
- **背景** = 幻灯片上没有的标准知识；**蓝框「编者注」** = 课件的疑点或未标注处；「约」= 从图上估读。
  Background = not on the slide; **blue "Editor's note"** = doubtful or unlabelled in the deck; "≈" = read off the plot.

考点速记 KEY POINTS

- 8 节 · 80 张幻灯片：按 S 编号与课件对照。
  8 sections, 80 slides: match pages to the deck by S number.
- 「背景」不是课件原话，考试按课件表述作答。
  "Background" ≠ deck wording; answer in the deck's terms.
- 复习先读每页底部的考点速记。
  Revise from each page's key points first.
- 「约」= 从图上估读，不是课件给出的精确值。
  "≈" = read off the plot, not an exact value from the deck.

S2 · 为什么要抑制溶剂峰

S2 · Why suppress the solvent peak

幻灯片要点 / What the slide says

正如「采集与处理」一讲所说，样品中出现很大的溶剂峰既很常见，又是相当大的难题。谱中若有一个极大的信号，接收机（receiver，接收机增益）必须按这个峰来优化，结果是你真正关心的峰损失大量信噪比（SNR）。

> As discussed in the acquisition and processing lecture, a large solvent peak is both very common and a considerable challenge. If there is one very large signal, the receiver (receiver gain) must be optimized for that peak, and the peaks you actually care about lose considerable signal-to-noise ratio (SNR).

机理：动态范围 / Mechanism: dynamic range

背景 接收机增益要设到最大的信号刚好不让数字化器溢出。溶剂峰占满了这个量程，样品峰只占其中极小一部分，只分到很少的有效位数，被数字化噪声和基线畸变淹没。动态范围（dynamic range）就是可同时测量的最强与最弱信号之比，所以「溶剂比样品浓得多」直接变成「样品的 SNR 变差」。

> **Background** The receiver gain is set so that the largest signal just avoids overflowing the digitizer. The solvent fills that range; the sample peaks use a tiny fraction of it, get only a few effective bits and sink into digitization noise and baseline distortion. Dynamic range is the ratio of the strongest to the weakest signal measurable together, so "the solvent is far more concentrated than the sample" becomes "the sample has poor SNR".

$$\frac{[^1\mathrm{H}]_{\mathrm{solvent}}}{[^1\mathrm{H}]_{\mathrm{sample}}} \approx \frac{6 \times 11.2\ \mathrm{M}}{4 \times 0.020\ \mathrm{M}} \approx 8 \times 10^2$$

背景估算：纯苯 ≈ 11.2 M × 6 H；样品 4 个芳香 H / background estimate: benzene ≈ 11.2 M × 6 H; sample 4 aromatic H

图怎么读 / Reading the figure

- 样品：20 mM 邻苯二甲酸二乙酯（diethyl phthalate，图中画有结构）溶于非氘代（质子化）苯。横轴 f1 (ppm)，约 8.3–5.9 ppm。
  > Sample: 20 mM diethyl phthalate (structure drawn) in protonated benzene. The x-axis is f1 (ppm), ≈ 8.3–5.9 ppm.
- 红线「1D standard」（普通一维谱，纵向放大 ×500）：溶剂峰（约 7.15 ppm）冲出图框，宽大的峰脚铺满大半个窗口；约 8.2–8.1 和 6.3–6.1 ppm 是溶剂的 $^{13}\mathrm{C}$ 卫星峰。
  > Red "1D standard" (vertical zoom ×500): the solvent peak (≈ 7.15 ppm) runs off scale and its broad wings fill most of the window; the solvent's $^{13}\mathrm{C}$ satellites sit at ≈ 8.2–8.1 and 6.3–6.1 ppm.
- 青线「1D solvent suppression」：基线平坦，只剩一条尖锐的残留溶剂峰（Residual Solvent，约 7.15 ppm）；样品的两组多重峰（约 7.6–7.5 和 7.0–6.9 ppm）清楚可见。
  > Teal "1D solvent suppression": a flat baseline with one sharp residual solvent peak (≈ 7.15 ppm); the sample's two multiplets (≈ 7.6–7.5 and 7.0–6.9 ppm) stand out clearly.
- **背景** $^{13}\mathrm{C}$ 卫星峰来自含 $^{13}\mathrm{C}$（天然丰度约 1.1 %）的溶剂分子，被 $^{1}J_{\mathrm{CH}}$ 劈裂、左右对称；这约 1 % 的溶剂已经和样品峰一样大甚至更大。
  > **Background** The $^{13}\mathrm{C}$ satellites come from solvent molecules carrying $^{13}\mathrm{C}$ (≈ 1.1 % natural abundance), split symmetrically by $^{1}J_{\mathrm{CH}}$; that ≈ 1 % of the solvent is already as large as the sample peaks or larger.

考点速记 KEY POINTS

- 大溶剂峰决定接收机增益 → 样品峰 SNR 下降。
  A huge solvent peak sets the receiver gain → sample SNR drops.
- 损失来自动态范围：样品只占数字化量程的一小部分。
  The loss is dynamic range: the sample uses a sliver of the digitizer.
- $^{13}\mathrm{C}$ 卫星峰对称分布在溶剂峰两侧，不是样品峰。
  $^{13}\mathrm{C}$ satellites flank the solvent symmetrically; not sample peaks.
- 抑制后仍有残留溶剂峰：抑制从不彻底。
  A residual solvent peak remains: suppression is never complete.

S3 · 第一条路：选一个不出峰的溶剂

S3 · Route 1: use a solvent with no signal

幻灯片要点 / What the slide says

解决这个问题主要有两条路。第一条最简单：用一种在你关心的谱里不出峰的溶剂。做氢谱时氘代溶剂（deuterated solvent）非常常见，其他核则有别的选择；做氟谱（$^{19}\mathrm{F}$ NMR）时，可能根本不需要氘代溶剂。第二条路是用实验手段抑制溶剂峰，从幻灯片 4 开始。

> There are two main approaches to this problem. The first is simply to use a solvent that shows no signals in the spectrum you care about. Deuterated solvents are very common for proton work, and other options exist for other nuclei; for fluorine ($^{19}\mathrm{F}$) NMR a deuterated solvent may not be necessary. The second approach, suppressing the solvent experimentally, starts on slide 4.

为什么有效 / Why it works

背景 $^{2}\mathrm{H}$（D）的共振频率与 $^{1}\mathrm{H}$ 相差很远，把溶剂中的 H 换成 D，溶剂在氢谱里就「隐身」了。普通溶剂分子里一般没有氟，所以 $^{19}\mathrm{F}$ 谱里本来就没有溶剂峰。氘代溶剂通常还为谱仪提供氘锁场（deuterium lock）。

> **Background** $^{2}\mathrm{H}$ (D) resonates far from $^{1}\mathrm{H}$, so replacing the solvent's H by D makes the solvent invisible in the proton spectrum. Ordinary solvents contain no fluorine, so a $^{19}\mathrm{F}$ spectrum has no solvent peak to begin with. A deuterated solvent usually also provides the spectrometer's deuterium lock.

背景 氘代从来不是 100 %：$\mathrm{CDCl_3}$ 中残留的 $\mathrm{CHCl_3}$ 会给出一个小的残留溶剂峰。另外，有些样品必须留在水里，例如要观察会与 $\mathrm{D_2O}$ 交换的 NH（幻灯片 4、10 用的是 $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$），这时只能走第二条路。

> **Background** Deuteration is never 100 %: residual $\mathrm{CHCl_3}$ in $\mathrm{CDCl_3}$ gives a small residual solvent peak. And some samples must stay in water, for example to observe NH protons that would exchange with $\mathrm{D_2O}$ (slides 4 and 10 use $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$); then only the second approach works.

图怎么读 / Reading the figure

- 上图：$\mathrm{CDCl_3}$（氘代氯仿）的结构，一个 C 连着一个 D 和三个 Cl，即氯仿的 C–H 换成了 C–D。
  > Top: the structure of $\mathrm{CDCl_3}$ (deuterated chloroform), one C bonded to one D and three Cl, i.e. chloroform with C–H replaced by C–D.
- 下图：玻璃安瓿瓶装的氘代溶剂照片，只作示意，没有数据。
  > Bottom: a photo of deuterated solvent in glass ampoules; illustrative only, no data.

考试提醒 / Exam reminder

关键不是「氘」本身，而是溶剂在被观测核的谱里不出峰。对 $^{19}\mathrm{F}$ 谱，普通溶剂往往已经满足。

> The point is not deuterium as such but a solvent that is silent for the observed nucleus. For $^{19}\mathrm{F}$ spectra, ordinary solvents often already are.

考点速记 KEY POINTS

- 两条路：换无信号溶剂，或用实验抑制。
  Two routes: a signal-free solvent, or experimental suppression.
- 氢谱常用氘代溶剂，如 $\mathrm{CDCl_3}$。
  Proton work: deuterated solvents, e.g. $\mathrm{CDCl_3}$.
- $^{19}\mathrm{F}$ NMR 可能不需要氘代溶剂。
  $^{19}\mathrm{F}$ NMR may not need a deuterated solvent.
- 样品必须留在 $\mathrm{H_2O}$ 中时，只能实验抑制。
  Samples that must stay in $\mathrm{H_2O}$ need experimental suppression.

S4 · 预饱和

S4 · Presaturation

幻灯片要点 / What the slide says

溶剂信号也可以用实验方法抑制，最简单的是预饱和（presaturation, presat）。做法是施加一个相对弱但很长的射频脉冲，把溶剂自旋「饱和」；溶剂自旋于是指向随机方向，对谱图不再有贡献。

> The solvent signal can also be suppressed experimentally, and the simplest way is presaturation (presat). A relatively weak but long RF pulse "saturates" the solvent spins; they then point in random directions and do not contribute to the spectrum.

机理 / Mechanism

背景 弱的射频场 B_1 只能激发很窄的频带，放在溶剂频率上就只打到溶剂。持续照射使溶剂上下能级的布居趋于相等，净磁化趋于零，即「饱和」。随后一个短而强的非选择性硬脉冲激发其余所有信号并采集 FID，被饱和的溶剂几乎不给信号。

> **Background** A weak RF field B_1 excites only a narrow band, so on the solvent frequency it hits only the solvent. The long irradiation equalizes the solvent's level populations and drives its net magnetization towards zero: saturation. A short, strong non-selective hard pulse then excites everything else and the FID is recorded; the saturated solvent gives almost no signal.

翻转角 / flip angle

$$\theta = \gamma B_1 t_\mathrm{p}$$

弱 B_1 → 需要长 t_p / weak B_1 → long t_p

激发带宽（背景） / excitation bandwidth (background)

$$\Delta\nu_\mathrm{exc} \sim \frac{1}{t_\mathrm{p}}$$

脉冲越长越有选择性 / longer pulse → more selective

背景 代价：溶剂附近的信号也会被部分饱和；与水交换的 OH、NH 质子会接收被饱和的 H，强度下降（幻灯片 10）。

> **Background** The cost: signals close to the solvent are partly saturated too, and OH/NH protons exchanging with water take up saturated H and lose intensity (slide 10).

图怎么读 / Reading the figure

- 标题 Solvent Suppression by Presaturation。脉冲序列：先是宽而矮的黑块「Long Low Power Selective Pulse」（长、低功率、选择性脉冲），接着一条细高的「Short High Power Non-selective Pulse」（短、高功率、非选择性脉冲），然后是 FID。

  > Title: Solvent Suppression by Presaturation. Sequence: a wide, low block "Long Low Power Selective Pulse", then a thin, tall "Short High Power Non-selective Pulse", then the FID.

- 样品：苯丙氨酸（phenylalanine，图中结构）溶于 $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$，横轴约 7.5–2.5 ppm。

  > Sample: phenylalanine (structure shown) in $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$; x-axis ≈ 7.5–2.5 ppm.

- 红线「Standard Spectrum」：约 4.7 ppm 的水峰冲出图框，宽峰脚铺到两侧，水峰右侧的三组峰骑在峰脚上；约 7.3 ppm 是芳香峰。蓝线「Presaturation」：水峰只剩一条细小的残留线，所有峰都立在平坦基线上。

  > Red "Standard Spectrum": the water peak at ≈ 4.7 ppm runs off scale with broad wings, and the three multiplets to its right ride on the wing; the aromatic peak is at ≈ 7.3 ppm. Blue "Presaturation": water is reduced to a thin residual line and every peak stands on a flat baseline.

考点速记 KEY POINTS

- presat：长、弱、选择性脉冲照射溶剂，再用硬脉冲激发。
  Presat: long, weak selective pulse on solvent, then a hard pulse.
- 弱 B_1 → 窄带宽 → 只打溶剂；$\theta = \gamma B_1 t_\mathrm{p}$。
  Weak B_1 → narrow band → solvent only; $\theta = \gamma B_1 t_\mathrm{p}$.
- 被饱和的自旋方向随机，不贡献信号。
  Saturated spins point randomly and give no signal.
- 陷阱：可交换质子（OH、NH）也被衰减。
  Trap: exchangeable protons (OH, NH) are attenuated too.

S5 · WET：比预饱和更强

S5 · WET: stronger than presat

幻灯片要点 / What the slide says

对大多数样品，预饱和（presat）往往已经足够；但如果溶剂峰确实很糟，就可能需要别的方法。WET（基于 T_1 效应增强的水峰抑制，Water suppression Enhanced through T_1 effects）是另一种方法，它使用梯度（脉冲场梯度，PFG），通常比 presat 更有效。

> Presaturation is often sufficient for most samples, but if the solvent peak is truly bad, additional approaches may be required. WET (Water suppression Enhanced through T_1 effects) is another approach; it uses (pulsed field) gradients and is typically more effective than presat.

机理 / Mechanism

背景 WET 在激发前对溶剂频率施加几个选择性脉冲，每个之后跟一个梯度，把刚翻到横向的溶剂磁化散相。几个翻转角是配好的，使溶剂的 T_1 即使不同，到激发时其纵向磁化 M_z 也接近零，名字里的「T_1 效应」即指这一点。最后用硬脉冲激发样品。

> **Background** Before excitation WET applies a few selective pulses at the solvent frequency, each followed by a gradient that dephases the solvent magnetization just tipped into the transverse plane. The flip angles are chosen so that the solvent's longitudinal magnetization M_z is close to zero at excitation even if its T_1 varies; that is the "T_1 effects" of the name. A hard pulse then excites the sample.

$$\left[\theta_{\text{sel}} \rightarrow G\right]_{\times n} \rightarrow 90^\circ \rightarrow \text{FID}$$

背景：WET 序列示意，选择性脉冲 + 散相梯度重复几次

/ background: WET scheme, selective pulse + dephasing gradient, repeated

图怎么读 / Reading the figure

- 标题「Non-alcoholic beverage · 1H 1D versus WET」：无酒精饮料的普通一维 ^{1}H 谱（红）对比 WET（青），横轴 14 至 –5 ppm。

  > Title "Non-alcoholic beverage · 1H 1D versus WET": the ordinary 1D ^{1}H spectrum (red) vs WET (teal) of a non-alcoholic beverage; x-axis 14 to –5 ppm.

- 全谱：红线只看得到一个巨大的水峰（略低于 5 ppm），其余都是平线，样品峰小到看不见。

  > Full view: the red trace shows only a huge water peak (just below 5 ppm); everything else is flat and the sample peaks are too small to see.

- 圆形放大框（4.0–1.0 ppm）：淡红色的普通谱向 4 ppm 一侧陡升（水峰的峰脚）；青色 WET 谱基线平坦，可见约 3.7–3.2 ppm 的一簇峰、约 3.0 ppm 的一个高峰和 1.5–1.0 ppm 之间的一对峰。

  > Circular inset (4.0–1.0 ppm): the pale red ordinary spectrum climbs steeply towards 4 ppm (the water peak's wing); the teal WET spectrum has a flat baseline with a cluster at ≈ 3.7–3.2 ppm, a tall peak at ≈ 3.0 ppm and a pair between 1.5 and 1.0 ppm.

考点速记 KEY POINTS

- presat 往往够用；溶剂峰「真的很糟」才上更强的方法。
  Presat is often enough; stronger methods for a truly bad peak.
- WET 通常比 presat 更有效。
  WET is typically more effective than presat.
- WET（基于 T_1 效应增强的水峰抑制）使用梯度。
  WET = Water suppression Enhanced through T_1 effects; uses gradients.
- 背景：选择性脉冲 + 梯度散相，在激发前完成。
  Background: selective pulses + dephasing gradients, before excitation.

S6 · WET 的优点

S6 · Advantages of WET

幻灯片要点 / What the slide says

- 不干扰任何可交换峰（exchangeable peaks）。
  > It does not disturb any exchangeable peaks.
- 所有脉冲序列操作都在激发样品之前完成，所以弛豫不是问题。
  > All the pulse-sequence work is done before pulsing the sample, so relaxation is not a concern.
- 可以同时抑制多个溶剂峰；这一点使它非常适合分析酒精饮料。
  > It can suppress several solvent peaks at once, which makes it great for analysing alcoholic beverages.

为什么 / Why

背景 presat 要长时间照射溶剂，饱和会通过化学交换传给 OH、NH；WET 只用几个短的选择性脉冲和梯度，交换来不及把饱和带过去（可交换质子的衰减见幻灯片 10）。抑制在激发前完成、激发后马上采集，样品信号不会因为抑制而在横向多停留。酒精饮料里除了水，乙醇也有很强的峰；能同时压住几个峰，才看得清微量成分。

> **Background** Presat irradiates the solvent for a long time, and saturation passes to OH and NH by chemical exchange; WET uses only a few short selective pulses and gradients, so exchange has little time to carry saturation over (exchangeable-proton loss is on slide 10). Suppression ends before excitation and acquisition follows at once, so the sample signal spends no extra time in the transverse plane. In alcoholic beverages ethanol gives strong peaks besides water; suppressing several at once is what makes minor components visible.

图怎么读 / Reading the figure

- 上下两组叠放谱，每组 5 种饮料：HN、YM、TS、MF、MJ（样品代号，幻灯片未解释），两组都标「×1」，横轴 $^{1}\mathrm{H}$ (ppm)，约 5.5–0.6 ppm。
  > Two stacked sets of 5 beverages each, HN, YM, TS, MF, MJ (sample codes, not explained on the slide), both marked "×1"; x-axis $^{1}\mathrm{H}$ (ppm), ≈ 5.5–0.6 ppm.
- 上组（幻灯片未注明方法）：约 4.1–3.2 ppm 一大簇密集的强峰，另有约 5.25、4.9、4.65 ppm 的峰；3 ppm 以下的峰很小。
  > Upper set (method not stated): a dense cluster of strong peaks at ≈ 4.1–3.2 ppm plus peaks at ≈ 5.25, 4.9 and 4.65 ppm; peaks below 3 ppm are tiny.
- 下组：箭头标出「Region suppressed by the broadband WET method」（宽带 WET 抑制的区域），约 5.6–3.0 ppm；区内只剩约 5.25 ppm 的一条尖峰和小的残余起伏，约 2.7–2.4 ppm 的一组峰和约 1.15 ppm 的高峰（HN）变得醒目。
  > Lower set: an arrow marks the "Region suppressed by the broadband WET method", ≈ 5.6–3.0 ppm; inside it only a sharp line at ≈ 5.25 ppm and small residual wiggles remain, and the peaks at ≈ 2.7–2.4 ppm and a tall one at ≈ 1.15 ppm (HN) now stand out.

注意 / Caution

宽带 WET 压掉的是一整段区域，区内的样品峰也一起丢失（对比上下两组 4.1–3.2 ppm）。

> Broadband WET removes a whole region, and the sample peaks inside it go too (compare 4.1–3.2 ppm in the two sets).

考点速记 KEY POINTS

- WET 不干扰可交换峰（OH、NH）。
  WET leaves exchangeable peaks (OH, NH) undisturbed.
- 抑制在激发前完成 → 弛豫不是问题。
  Done before excitation → relaxation is no concern.
- 可同时抑制多个溶剂峰 → 适合酒精饮料。
  Several solvent peaks at once → alcoholic beverages.
- 宽带 WET 抹掉整段区域，区内样品峰也丢失。
  Broadband WET clears a whole region, sample peaks included.

S7 · PURGE：简单的「中等」方法

S7 · PURGE: a simple "medium" method

幻灯片要点 / What the slide says

PURGE 是一种相对简单、同样使用梯度的方法。它的主要优点是很容易做：序列中几乎不需要调整什么，所以对非专家很有用。它是「中等」强度的方法：比 presat 好，但大概不如最强的序列。它在序列中使用预饱和，因此会衰减可交换质子。

> PURGE is a relatively simple approach that also uses gradients. Its key advantage is that it is quite easy to run: very little needs adjusting in the sequence, so it is powerful for non-experts. It is a "medium" approach, better than presat but likely not as good as the most intense sequences. It uses presaturation during the sequence, so it attenuates exchangeable protons.

名称 / The name

背景 全称不在幻灯片上：Presaturation Utilizing Relaxation Gradients and Echoes（利用弛豫、梯度与回波的预饱和）。名字说明它把预饱和与梯度回波结合在一起；幻灯片只强调三点：用梯度、序列内含 presat、几乎不用调参。

> **Background** The full name is not on the slide: Presaturation Utilizing Relaxation Gradients and Echoes. The name says it combines presaturation with gradient echoes; the slide stresses only three points: it uses gradients, it contains presat, and it needs almost no adjustment.

考试提醒 / Exam reminder

用了梯度不等于保住可交换质子：PURGE 内部有 presat，OH、NH 仍被衰减。

> Using gradients does not save the exchangeables: PURGE contains presat, so OH and NH are still attenuated.

图怎么读 / Reading the figure

- 四条叠放的 $^{1}\mathrm{H}$ 谱 A–D，横轴约 5.5–3.2 ppm（刻度 5.0、4.5、4.0 ppm）。
  > Four stacked $^{1}\mathrm{H}$ spectra A–D; x-axis ≈ 5.5–3.2 ppm (ticks at 5.0, 4.5, 4.0 ppm).
- A：约 4.7 ppm 的水峰两侧都冲出图框，基线带噪声。B：水峰仍冲出图框，右侧拖着宽峰脚。C：水峰收窄成一条高而细的线，脚部畸变。D：只剩约 4.7 ppm 处一个很小的色散状残留，基线平坦。
  > A: the water peak at ≈ 4.7 ppm runs off scale on both sides; noisy baseline. B: still off scale, with a broad wing on the right. C: water narrowed to a tall thin line with a distorted foot. D: only a tiny dispersive residual at ≈ 4.7 ppm; flat baseline.
- 约 5.3、4.1、3.9 ppm 的峰和 3.8–3.3 ppm 的一组峰在四条谱里位置相同；从 A 到 D 水峰残留越来越小。
  > Peaks at ≈ 5.3, 4.1 and 3.9 ppm and a group at 3.8–3.3 ppm sit at the same positions in all four; the water residual shrinks from A to D.

编者注 / Editor's note

幻灯片没有说明 A–D 各用了哪种方法，也没有说哪条是 PURGE。这里只按图描述，考试中不要假定某条谱就是 PURGE。

> The slide does not say which method produced A–D, nor which trace is PURGE. They are described as shown; do not assume in an exam that a given trace is PURGE.

考点速记 KEY POINTS

- PURGE：用梯度、简单、几乎不用调参 → 适合非专家。
  PURGE: gradients, simple, little to adjust → for non-experts.
- 强度「中等」：好于 presat，不及最强的序列。
  "Medium": better than presat, below the most intense sequences.
- 序列内含 presat → 可交换质子被衰减。
  Contains presat → exchangeable protons attenuated.
- 图中 A–D 未标方法，只比较水峰残留。
  A–D are unlabelled; compare only the water residual.

S8 · WATERGATE：翻转除水以外的一切

S8 · WATERGATE: flip everything but the water

幻灯片要点 / What the slide says

在 WATERGATE 中，水峰靠梯度来抑制。本质上，有一个翻转除水以外一切的脉冲；随后用梯度把水散相，而真正的信号不受影响。这通常能得到极好的水峰抑制。**背景** 全称不在幻灯片上：WATER suppression by GrAdient-Tailored Excitation（梯度定制激发水峰抑制）。

> In WATERGATE the water is suppressed by gradients. Essentially, a pulse flips everything but the water; the water is then dephased by a gradient while the real signals are unaffected. This typically gives excellent water suppression.
> **Background** The full name is not on the slide: WATER suppression by GrAdient-Tailored Excitation.

图怎么读：脉冲序列 / Reading the figure: the pulse sequence

- 标题「Water Suppression with WATERGATE」。RF 行：先是激发脉冲 $(\pi/2)_x$，然后是六个脉冲 p_1–p_6，相邻两个之间隔一个延迟 τ，这一段标注「π for all but water」（对除水以外的一切是 π）；最后采集 FID（标 x）。

  > Title "Water Suppression with WATERGATE". RF row: the excitation pulse $(\pi/2)_x$, then six pulses p_1–p_6 separated by delays τ, labelled "π for all but water"; then the FID (marked x).

- 梯度行：G_1 在 $(\pi/2)_x$ 与 p_1 之间，G_2 在 p_6 与采集之间，脉冲串夹在两个梯度之间。图下给出角度与相位：

  > Gradient row: G_1 between $(\pi/2)_x$ and p_1, G_2 between p_6 and acquisition, so the train sits between the two gradients. Angles and phases below the diagram:

脉冲串翻转角 / train flip angles

$$p_1 = p_6 = 20.79^\circ$$
$$p_2 = p_5 = 62.28^\circ$$
$$p_3 = p_4 = 131.58^\circ$$

相位 / phases

$$p_1, p_2, p_3 = x$$
$$p_4, p_5, p_6 = -x$$

$$(\pi/2)_x \; - \; G_1 \; - \; p_1 \; \tau \; p_2 \; \tau \; p_3 \; \tau \; p_4 \; \tau \; p_5 \; \tau \; p_6 \; - \; G_2 \; - \; \mathrm{FID}$$

机理：梯度回波 / Mechanism: the gradient echo

背景 $(\pi/2)_x$ 把所有磁化（包括水）翻到横向。G_1 让样品各层带上与位置有关的相位 ϕ。脉冲串对样品信号相当于 π 脉冲，把 ϕ 变成 $-\phi$；G_2（图中与 G_1 画得一样）再加 ϕ，正好抵消，信号重聚成回波。对水，脉冲串的净翻转为零、相位不反转，G_2 再加一份 ϕ，水被散相，平均为零。

> **Background** $(\pi/2)_x$ tips all magnetization, water included, into the transverse plane. G_1 gives each slice of the sample a position-dependent phase ϕ. For the sample signals the train acts as a π pulse and turns ϕ into $-\phi$; G_2 (drawn the same as G_1) adds ϕ and cancels it, so the signal refocuses as an echo. For water the train's net rotation is 0°, the phase is not reversed, G_2 adds a second ϕ, and the water is dephased and averages to zero.

$$\text{signals: } \phi \xrightarrow{\pi} -\phi \xrightarrow{G_2} 0, \qquad \text{water: } \phi \xrightarrow{0^\circ} \phi \xrightarrow{G_2} 2\phi$$

背景：信号重聚，水被散相 / background: signals refocus, water is dephased

考点速记 KEY POINTS

- 脉冲串对除水以外是 π，对水净 0°。
  The train is π for all but water; net 0° for water.
- G_1、G_2 夹住脉冲串：信号重聚，水被散相。
  G_1, G_2 bracket the train: signals refocus, water dephases.
- $p_1 = p_6 = 20.79^\circ$，$p_2 = p_5 = 62.28^\circ$，$p_3 = p_4 = 131.58^\circ$；前三个 x，后三个 $-x$。
  $p_1 = p_6 = 20.79^\circ$, $p_2 = p_5 = 62.28^\circ$, $p_3 = p_4 = 131.58^\circ$; first three x, last three $-x$.
- 不长时间照射水 → 可交换质子保留（S10）。
  No long irradiation of water → exchangeables survive (S10).

S8（续）· 背景：3-9-19 脉冲串

S8 (cont.) · Background: the 3-9-19 train

背景 本页整页是背景：解释幻灯片 8 的角度从何而来，幻灯片本身没有这一推导。三个角度之比约为 3 : 9 : 19，所以这种脉冲串叫 3-9-19 脉冲串，是一种类二项式脉冲串。

> **Background** This whole page is background: it explains where the slide-8 angles come from; the slide itself gives no derivation. The three angles are in the ratio ≈ 3 : 9 : 19, so this is called the 3-9-19 pulse train, a binomial-like pulse train.

$$3\alpha,\ 9\alpha,\ 19\alpha,\quad 26\alpha = 180^\circ\ (\alpha \approx 6.92^\circ)$$

$3\alpha \approx 20.8^\circ$，$9\alpha \approx 62.3^\circ$，$19\alpha \approx 131.5^\circ$，与幻灯片的 20.79°、62.28°、131.58° 相符

/ $3\alpha \approx 20.8^\circ$, $9\alpha \approx 62.3^\circ$, $19\alpha \approx 131.5^\circ$, matching the slide's 20.79°, 62.28°, 131.58°

共振处：水 / On resonance: water

水的频率就是载波频率，脉冲之间不进动。前三个脉冲（相位 x）共转 $(3+9+19)\alpha$，后三个（相位 $-x$）反向转同样的角度，净翻转为零，水留在 G_1 给它的散相状态。

> Water sits at the carrier frequency and does not precess between pulses. The first three pulses (phase x) rotate it by $(3+9+19)\alpha$ and the last three (phase $-x$) rotate it back by the same amount: the net rotation is zero, and water stays dephased by G_1.

$$(3+9+19)\,\alpha \ - \ (19+9+3)\,\alpha \ = \ 0^\circ$$

偏共振 $\Delta\nu = 1/(2\tau)$ 处 / Off resonance at $\Delta\nu = 1/(2\tau)$

在每个 τ 内恰好进动 180° 的频率处，进动让相邻脉冲的有效方向交替变号；再乘上后三个脉冲的 $-x$，六个脉冲的效果相加而不是相消：

> At the offset where spins precess exactly 180° during each τ, precession flips the effective sign of successive pulses; combined with the $-x$ phase of the last three, the six pulses add instead of cancelling:

$$3 - 9 + 19 + 19 - 9 + 3 = 26 \ \Rightarrow\ 26\alpha = 180^\circ = \pi$$

每个 τ 进动 $180^\circ \Leftrightarrow \Delta\nu = 1/(2\tau)$ / 180° per $\tau \Leftrightarrow \Delta\nu = 1/(2\tau)$

所以这些频率处的信号被完全反转，能在 G_1/G_2 回波中重聚。在 $\Delta\nu = n/\tau$（$n = 0, 1, 2, \ldots$）处每个 τ 进动 360° 的整数倍，等同于共振，出现零点。零点附近的信号只被部分翻转，会和水一起部分散相，这就是幻灯片 11 的溶剂附近衰减；τ 决定这些频率的位置，这又引出幻灯片 12 低场需要更长延迟。

> Signals at these offsets are fully inverted and refocus in the G_1/G_2 echo. At $\Delta\nu = n/\tau$ ($n = 0, 1, 2, \ldots$) spins precess whole multiples of 360° per τ, which acts like resonance and gives a null. Signals near a null are only partly flipped and are partly dephased along with the water: the near-solvent loss of slide 11. τ sets where these offsets lie, which leads to the longer delays at low field on slide 12.

考点速记 KEY POINTS

- 3-9-19：$3\alpha, 9\alpha, 19\alpha$，$26\alpha = 180^\circ$。
  3-9-19: $3\alpha, 9\alpha, 19\alpha$ with $26\alpha = 180^\circ$.
- $\Delta\nu = 1/(2\tau)$：$3 - 9 + 19 + 19 - 9 + 3 = 26 \rightarrow \pi$。
  At $\Delta\nu = 1/(2\tau)$: $3 - 9 + 19 + 19 - 9 + 3 = 26 \rightarrow \pi$.
- 共振（水）：前后两半抵消，净 0°。
  On resonance (water): halves cancel, net 0°.
- 零点在 $\Delta\nu = n/\tau$；零点附近信号被衰减。
  Nulls at $\Delta\nu = n/\tau$; signals near a null are attenuated.

S9 · W5-WATERGATE：湖水样品

S9 · W5-WATERGATE on lake water

幻灯片要点 / What the slide says

这是 WATERGATE 的一个变体 SPR-W5-WATERGATE 的例子。样品就是湖水，谱中的峰来自水中的溶解有机物。这一方法的抑制效果非常好，而 presat 会不够用。

> This is an example of a WATERGATE variant called SPR-W5-WATERGATE. The sample is simply lake water, and the peaks are the dissolved organic material in it. This approach has very good suppression, whereas presat would be insufficient.

W5（W5-WATERGATE）是 WATERGATE 的一种变体。SPR（成形脉冲预饱和）在课件中没有展开，幻灯片 12 把它画成 W5 模块之前的一串成形饱和脉冲；整个序列见幻灯片 12。

> W5 (W5-WATERGATE) is a variant of WATERGATE. SPR is not expanded in the deck; slide 12 draws it as a train of shaped saturation pulses before the W5 blocks. The whole sequence is on slide 12.

为什么 presat 不够 / Why presat is not enough

背景 水中的 $^{1}\mathrm{H}$ 浓度约 111 M，而湖水里的溶解有机物极稀，两者相差许多个数量级。presat 留下的那点残余水峰仍比目标峰大得多，还带着宽峰脚，所以需要能把水再压低几个数量级的方法。

> **Background** Water holds about 111 M of $^{1}\mathrm{H}$, while dissolved organic matter in lake water is extremely dilute, many orders of magnitude lower. The residual water left by presat is still far larger than the target peaks and has broad wings, so a method that removes several more orders of magnitude of water is needed.

$$[^{1}\mathrm{H}]_{\mathrm{H_2O}} \approx 2 \times 55.5\ \mathrm{M} \approx 111\ \mathrm{M}$$

背景：每个水分子 2 个 H / background: 2 H per water molecule

图怎么读 / Reading the figure

- 三条叠放谱 B、C、D，横轴约 5.4–3.4 ppm（刻度 5.0、4.5、4.0 ppm）。
  > Three stacked spectra B, C, D; x-axis ≈ 5.4–3.4 ppm (ticks 5.0, 4.5, 4.0 ppm).
- B：约 4.6 ppm 的水峰冲出图框，两侧宽峰脚，基线带噪声。C：水峰仍冲出图框但变窄，基线更平。D：水峰区域（约 5.0–4.2 ppm）完全平坦。
  > B: the water peak at ≈ 4.6 ppm runs off scale with broad wings; noisy baseline. C: still off scale but narrower; flatter baseline. D: the water region (≈ 5.0–4.2 ppm) is completely flat.
- D 中箭头指向离水最近的峰：约 5.3、4.1、3.9 ppm。与 C 相比，它们相对 3.8–3.4 ppm 的高峰明显变小（见幻灯片 11）。
  > In D, arrows point to the peaks nearest the water, at ≈ 5.3, 4.1 and 3.9 ppm. Compared with C they are clearly smaller relative to the tall peaks at 3.8–3.4 ppm (see slide 11).

编者注 / Editor's note

B–D 在幻灯片上没有标出各自的方法；幻灯片只说 SPR-W5-WATERGATE 抑制很好、presat 不够。D 的水区最平，但幻灯片没有明说 D 就是 SPR-W5-WATERGATE。幻灯片 7、11 的谱峰位置与此相同（约 5.3、4.1、3.9 ppm）。

> B–D are not labelled by method on the slide; it only says SPR-W5-WATERGATE suppresses very well and presat is insufficient. D has the flattest water region, but the slide does not state that D is SPR-W5-WATERGATE. Slides 7 and 11 show peaks at the same positions (≈ 5.3, 4.1, 3.9 ppm).

考点速记 KEY POINTS

- SPR-W5-WATERGATE：WATERGATE 的变体，SPR 在 W5 之前。
  SPR-W5-WATERGATE: a WATERGATE variant, SPR before W5.
- 湖水：峰来自溶解有机物；presat 不够用。
  Lake water: peaks are dissolved organics; presat is insufficient.
- 背景：水中 $^{1}\mathrm{H} \approx 111\ \mathrm{M}$，目标物极稀。
  Background: water $^{1}\mathrm{H} \approx 111\ \mathrm{M}$; targets very dilute.
- 箭头标出离水最近、被衰减的峰。
  Arrows mark the attenuated peaks nearest the water.

S10 · 注意事项 1：可交换质子被衰减

S10 · Things to consider 1: exchangeable protons

幻灯片要点 / What the slide says

溶剂峰抑制很强大，但也带来（有时很严重的）问题。第一是可交换质子（OH、NH）被衰减（通常由 presat 引起），即 OH 或 NH 基团的强度会降低。

> Solvent suppression is very powerful, but it comes with (sometimes severe) challenges. The first is the attenuation of exchangeable protons (often by presat): OH or NH groups show lower intensity.

机理：饱和转移 / Mechanism: saturation transfer

背景 OH、NH 上的 H 不断与水中的 H 交换。presat 长时间饱和水，被饱和的 H 交换到 OH、NH 位置上，这些位置也被「饱和」，强度下降；交换越快，损失越大。WATERGATE 不长时间照射水，水只在采集前被散相，所以 NH 保留下来。

> **Background** The H on OH and NH keeps exchanging with water H. Presat saturates water for a long time; saturated H atoms exchange onto the OH and NH sites, which become "saturated" too and lose intensity; the faster the exchange, the bigger the loss. WATERGATE does not irradiate water for long, water is only dephased before acquisition, so the NH survives.

背景 要看到 NH，样品必须留在 $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$ 中：在纯 $\mathrm{D_2O}$ 里 NH 会交换成 ND，在 $^1\mathrm{H}$ 谱中消失；10 % 的 $\mathrm{D_2O}$ 用来锁场。

> **Background** To see NH, the sample must stay in $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$: in pure $\mathrm{D_2O}$ the NH would exchange to ND and vanish from the $^1\mathrm{H}$ spectrum; the 10 % $\mathrm{D_2O}$ is for the lock.

图怎么读 / Reading the figure

- 标题「Watergate vs Presaturation」，副标题「500 MHz $^1\mathrm{H}$ NMR of the NH Region (solvent = $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$)」。结构：两个带 OH 的糖环连着一条含多个酰胺 NH 的短肽链。
  > Title "Watergate vs Presaturation", subtitle "500 MHz $^1\mathrm{H}$ NMR of the NH Region (solvent = $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$)". Structure: two sugar rings bearing OH linked to a short peptide chain with several amide NH.
- 两张谱横轴都约 9.0–7.5 ppm。黄色高亮区（约 8.7–8.1 ppm）是 NH 峰：Watergate 中又高又清楚；Presaturation 中明显变矮，只剩约一半或更少。
  > Both spectra span ≈ 9.0–7.5 ppm. The yellow-highlighted region (≈ 8.7–8.1 ppm) holds the NH peaks: tall and clear with Watergate, clearly shorter with Presaturation, about half or less.
- 高亮区外约 7.85 和 7.7 ppm 的峰在两张谱中高度相近，可作对照：差别集中在高亮的 NH 峰上。
  > Outside the highlight, the peaks at ≈ 7.85 and 7.7 ppm have similar heights in both, a reference: the difference is concentrated in the highlighted NH peaks.

考点速记 KEY POINTS

- 问题 1：可交换质子（OH、NH）被衰减，多由 presat 引起。
  Issue 1: exchangeable OH/NH attenuated, often by presat.
- 500 MHz 对比：WATERGATE 保留 NH，presat 让 NH 峰变矮。
  At 500 MHz: WATERGATE keeps NH; presat shrinks the NH peaks.
- 背景：饱和转移，被饱和的水 H 交换到 OH、NH 上。
  Background: saturation transfer, saturated water H exchanges onto OH/NH.
- 看 NH 用 $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$，不能用纯 $\mathrm{D_2O}$。
  Observe NH in $90\%\ \mathrm{H_2O}/10\%\ \mathrm{D_2O}$, not pure $\mathrm{D_2O}$.

S11 · 注意事项 2：溶剂附近的信号

S11 · Things to consider 2: signals near the solvent

幻灯片要点 / What the slide says

第二，溶剂附近的区域被衰减得有多严重？在最激进的方法中，最多约 0.4 ppm 被完全抹除（obliterated），最多 1.1 ppm 被衰减（attenuated）。

> Second, how badly is the area near the solvent attenuated? In the most aggressive approaches, up to ≈ 0.4 ppm is obliterated and up to 1.1 ppm is attenuated.

被完全抹除 / obliterated	被衰减 / attenuated
≈ 0.4 ppm	up to 1.1 ppm

机理 / Mechanism

背景 「翻转除水以外一切」的脉冲串在水频率处有一个有限宽度的零点（S8 续页）。离水越近，信号被翻转得越不足 π，经过 G_2 后就越像水一样被散相：最近处完全消失，稍远处只被衰减。ppm 换成 Hz 要乘谱仪频率，同样的 ppm 宽度在高场对应更多 Hz。

> **Background** The "flip all but water" train has a null of finite width at the water frequency (S8 cont.). The closer a signal is to water, the further short of π it is flipped and the more it is dephased by G_2 like the water: the nearest signals vanish, slightly farther ones are only attenuated. Converting ppm to Hz multiplies by the spectrometer frequency, so the same ppm width means more Hz at high field.

$$\Delta\nu\ (\mathrm{Hz}) = \Delta\delta\ (\mathrm{ppm}) \times \nu_0\ (\mathrm{MHz})$$

背景换算例：0.4 ppm 在 500 MHz 下为 200 Hz / background example: 0.4 ppm at 500 MHz is 200 Hz

图怎么读 / Reading the figure

- 两条谱 C 和 D，横轴约 5.4–3.4 ppm（刻度 5.0、4.5、4.0 ppm）。C：约 4.6 ppm 的水峰冲出图框，右侧有宽峰脚，其余峰都在。
  > Two spectra, C and D; x-axis ≈ 5.4–3.4 ppm (ticks 5.0, 4.5, 4.0 ppm). C: the water peak at ≈ 4.6 ppm runs off scale with a broad wing on the right; the other peaks are present.
- D：水两侧约 5.0–4.2 ppm 没有任何信号。箭头指向约 5.3、4.1、3.9 ppm 的峰，离水约 0.5–0.7 ppm，落在「至多 1.1 ppm 被衰减」的范围内；它们相对 3.8–3.4 ppm 的高峰明显变小。
  > D: nothing at all at ≈ 5.0–4.2 ppm around the water. Arrows point to peaks at ≈ 5.3, 4.1 and 3.9 ppm, ≈ 0.5–0.7 ppm from water, inside the "up to 1.1 ppm attenuated" zone; they are clearly smaller relative to the tall peaks at 3.8–3.4 ppm.

编者注 / Editor's note

C、D 在幻灯片上没有标出所用方法（看起来与幻灯片 9 的 C、D 相同）。幻灯片也没有说明 0.4 ppm 和 1.1 ppm 是总宽度还是到溶剂峰的单侧距离，按原话记忆即可。

> C and D are not labelled by method on the slide (they appear to be the same traces as C and D on slide 9). The slide also does not say whether 0.4 ppm and 1.1 ppm are total widths or one-sided distances from the solvent; memorise the wording as given.

考点速记 KEY POINTS

- 问题 2：溶剂附近的信号被抹除或衰减。
  Issue 2: signals near the solvent are wiped out or attenuated.
- 背景：零点附近翻转不足 π，信号随水一起散相。
  Background: near the null, flips fall short of π; signal dephases with water.
- 最激进方法：约 0.4 ppm 抹除，至多 1.1 ppm 衰减。
  Most aggressive: ≈ 0.4 ppm obliterated, up to 1.1 ppm attenuated.
- 关心溶剂附近的峰 → 别选最激进的方法。
  Need peaks near the solvent → avoid the most aggressive methods.

S12 · 注意事项 3：长序列中的弛豫

S12 · Things to consider 3: relaxation in long sequences

幻灯片要点 / What the slide says

弛豫很重要，尤其在低场。脉冲序列越长，弛豫的时间越多，可能损害信号。各个黑块（脉冲模块）之间的延迟通常是几百微秒，但在低场需要更长。

> Relaxation is important, especially at low field. The longer the pulse sequence, the more time there is for relaxation, which can hurt signals. The delays between the black boxes (pulse blocks) are usually a few hundred microseconds, but at low field they need to be longer.

图怎么读：SPR + W5 WATERGATE / Reading the figure: SPR + W5 WATERGATE

- 左段「SPR」：一串圆顶形的成形脉冲（成形饱和脉冲串）。
  > Left, "SPR": a train of dome-shaped shaped pulses (shaped saturation pulses).
- 右段「W5 WATERGATE」：先是单独的激发块 ψ_1，然后是两组密集的脉冲串（ψ_2、ψ_3 一组，ψ_4、ψ_5 一组），最后采集 FID；ψ_1–ψ_5 是相位标记。
  > Right, "W5 WATERGATE": a single excitation block ψ_1, then two dense pulse trains (ψ_2, ψ_3 and ψ_4, ψ_5), then the FID; ψ_1–ψ_5 are phase labels.
- 梯度行：G_1、G_1 夹住第一组脉冲串，G_2、G_2 夹住第二组，每组都在一对相同的梯度之间。
  > Gradient row: G_1, G_1 bracket the first train and G_2, G_2 the second, so each train sits between a pair of equal gradients.

为什么低场延迟要更长 / Why low field needs longer delays

背景 由 S8 续页，脉冲串的零点和完全反转点由 τ 决定（$\Delta\nu = 1/(2\tau)$ 处完全反转）。同样的 ppm 范围在低场对应更少的 Hz，要让反转区覆盖同样的 ppm 范围，τ 就要变长，整个序列随之变长。

> **Background** From S8 (cont.), the train's nulls and full-inversion points are set by τ (full inversion at $\Delta\nu = 1/(2\tau)$). The same ppm range spans fewer Hz at low field, so to cover the same ppm range τ must grow, and the whole sequence grows with it.

延迟与场强（背景） / delay vs field (background)

$$\tau = \frac{1}{2\,\Delta\nu}, \qquad \Delta\nu = \Delta\delta \times \nu_0$$

ν_0 越小 → τ 越长 / smaller ν_0 → longer τ

横向弛豫损失（背景） / transverse decay (background)

$$M_{xy}(t) = M_{xy}(0)\, e^{-t/T_2}$$

序列越长，t 越大 / longer sequence → larger t

考点速记 KEY POINTS

- 问题 3：序列越长，弛豫越多，信号受损；低场尤甚。
  Issue 3: longer sequence → more relaxation loss, worst at low field.
- 图：SPR 成形脉冲 → W5 两组脉冲串，各夹在 G_1/G_1、G_2/G_2 之间。
  Figure: SPR shaped pulses → two W5 trains inside G_1/G_1 and G_2/G_2.
- 黑块之间的延迟通常几百 μs，低场更长。
  Delays between blocks: a few hundred μs, longer at low field.
- 背景：低场同样 ppm 对应更少 Hz → τ 更长。
  Background: fewer Hz per ppm at low field → longer τ.

S13 · 该用哪种方法?

S13 · Which technique should I use?

幻灯片要点 / What the slide says

一如既往：看情况！要问三个问题：溶剂峰有多糟？如果很糟，可能需要强的方法。我有多在乎可交换基团？我有多在乎溶剂峰附近的信号?

> As always, it depends! Ask three questions. How nasty is the solvent peak? If it is bad, a strong approach may be needed. How much do I care about exchangeable groups? How much do I care about the signals near the solvent peak?

图怎么读 / Reading the figure

- 方法按抑制强度从左到右排成三栏，底部楔形条标「Increasing water suppression strength」（水峰抑制强度递增）。横轴 $^{1}\mathrm{H}$ Chemical Shift (ppm)，5.8–3.2 ppm。
  > The methods are ranked left to right in three columns, with a wedge labelled "Increasing water suppression strength" underneath. x-axis: $^{1}\mathrm{H}$ Chemical Shift (ppm), 5.8–3.2 ppm.
- 弱（Weak）：Pre-SAT（即 presat）。红色虚线圈出约 5.0–4.6 ppm 的水区，那里留着一个起伏的残余峰。
  > Weak: Pre-SAT (presat). A red dashed circle marks the water region at ≈ 5.0–4.6 ppm, where a wavy residual remains.
- 中（Intermediate）：NOESY-PR（一维 NOESY 预饱和）、g-NOESY-PR（梯度 NOESY 预饱和）、PURGE。约 4.8 ppm 处只剩一条细小的残留线或小凹陷。
  > Intermediate: NOESY-PR (1D NOESY presaturation), g-NOESY-PR (gradient NOESY-PR), PURGE. Only a thin residual line or a small dip remains at ≈ 4.8 ppm.
- 强（Strong）：JRS10、JRS8、JRS6（JRS，跳跃–回复夹心序列，课件未展开）和 W5。水区几乎完全平坦。
  > Strong: JRS10, JRS8, JRS6 (JRS, jump-and-return sandwich, not expanded in the deck) and W5. The water region is almost perfectly flat.
- 各谱的样品峰相同：约 5.4 ppm 的双峰、约 4.2 和 4.05 ppm 的小峰、约 3.85 和 3.75 ppm 的两个高峰。
  > The sample peaks are the same in every trace: a doublet at ≈ 5.4 ppm, small peaks at ≈ 4.2 and 4.05 ppm, and two tall peaks at ≈ 3.85 and 3.75 ppm.

三个问题怎么用 / Using the three questions

溶剂峰越糟，越往右（强）选；越在乎可交换基团，越要避开含 presat 的方法（Pre-SAT、PURGE，S7、S10）；越在乎溶剂附近的信号，越要提防最激进的方法（S11）。三个问题常常互相拉扯，所以答案是「看情况」。

> The nastier the solvent peak, the further right (stronger) you go; the more you care about exchangeable groups, the more you avoid methods containing presat (Pre-SAT, PURGE; S7, S10); the more you care about signals near the solvent, the warier you are of the most aggressive methods (S11). The three questions often pull against each other, hence "it depends".

考点速记 KEY POINTS

- 三问：溶剂多糟？要不要可交换基团？要不要溶剂附近的峰？
  Ask: how bad is the solvent? Need exchangeables? Need nearby peaks?
- 弱：Pre-SAT；中：NOESY-PR、g-NOESY-PR、PURGE；强：JRS6/8/10、W5。
  Weak: Pre-SAT; intermediate: NOESY-PR, g-NOESY-PR, PURGE; strong: JRS6/8/10, W5.
- 强 ≠ 总是更好：越激进，溶剂附近损失越大。
  Stronger ≠ always better: more aggressive, more near-solvent loss.
- 含 presat 的方法会衰减 OH、NH。
  Methods containing presat attenuate OH and NH.

S13（续）· 方法对比表

S13 (cont.) · Comparison of the techniques

下表只用幻灯片 2–13 明确说过或图上看得到的内容，括号里是出处；「—」表示课件没有说。

The table uses only what slides 2–13 state or show, with the source slide in brackets; "—" means the deck says nothing.

强度 Strength	方法 Method	代价 / 限制 Cost	何时选 When to pick
弱 weak	Pre-SAT (presat)	衰减可交换质子（S10）；留有残余水峰（S13 图）；对湖水不够（S9） attenuates exchangeables (S10); residual water hump (S13 fig.); insufficient for lake water (S9)	多数样品往往够用（S5）；最简单（S4） often sufficient for most samples (S5); simplest (S4)
中 medium	PURGE	内含 presat → 衰减可交换质子；大概不如最强的序列（S7） contains presat → attenuates exchangeables; likely below the most intense (S7)	要简单、少调参、非专家操作；好于 presat（S7） easy, little to adjust, non-experts; better than presat (S7)
中 medium	NOESY-PR, g-NOESY-PR	—（只给出强度排名，S13） — (only ranked, S13)	—
强 strong	JRS6, JRS8, JRS10, W5; WATERGATE	最激进时约 0.4 ppm 抹除、至多 1.1 ppm 衰减（S11）；长序列有弛豫损失，低场延迟更长（S12） most aggressive: ≈ 0.4 ppm obliterated, up to 1.1 ppm attenuated (S11); relaxation in long sequences, longer delays at low field (S12)	溶剂峰很糟（S5、S13）；极稀水样，如湖水（S9）；WATERGATE 抑制极好且保留 NH（S8、S10） bad solvent peak (S5, S13); very dilute water samples, e.g. lake water (S9); WATERGATE: excellent and keeps NH (S8, S10)
未排名 not ranked	WET	宽带 WET 抹掉整段区域，区内样品峰也丢失（S6 图） broadband WET clears a whole region, sample peaks included (S6 fig.)	比 presat 更有效（S5）；不扰可交换峰、弛豫不是问题、多个溶剂峰同时抑制，如酒精饮料（S6） more effective than presat (S5); exchangeables untouched, relaxation no concern, several solvent peaks at once, e.g. alcoholic beverages (S6)

考试提醒：怎么选 / Exam reminder: how to choose

- 溶剂峰一般 → presat；溶剂峰很糟或样品极稀 → 强的方法或 WET。
  Ordinary solvent peak → presat; bad peak or very dilute sample → a strong method or WET.
- 需要 OH、NH → 避开 presat 和 PURGE，用 WATERGATE 或 WET。
  Need OH, NH → avoid presat and PURGE; use WATERGATE or WET.
- 需要溶剂附近的峰 → 别用最激进的方法；低场 → 注意长序列的弛豫。
  Need peaks near the solvent → not the most aggressive method; low field → watch relaxation in long sequences.

考点速记 KEY POINTS

- 弱 → 强：抑制越强，溶剂附近损失越大。
  Weak → strong: stronger suppression, more near-solvent loss.
- WET：多峰同时抑制，不扰可交换峰。
  WET: several peaks at once, exchangeables untouched.
- 含 presat（Pre-SAT、PURGE）→ OH、NH 被衰减。
  Presat inside (Pre-SAT, PURGE) → OH, NH attenuated.
- 低场 + 长序列 → 弛豫损失。
  Low field + long sequence → relaxation loss.

S14 · 高维 NMR：为什么要第三维（峰容量）

S14 · Higher-dimensional NMR: why a third dimension (peak capacity)

幻灯片要点 / What the slide says

三维 NMR 在蛋白质研究中相当常用，但在蛋白质之外也有应用实例。三维 NMR 的峰容量（peak capacity）更高。三维以上的 NMR 通常很少见，往往只是为了"秀一下"。

> 3D NMR is quite commonly applied in protein work, but there have been examples outside it. 3D NMR has a higher peak capacity. Going above 3D is usually rare and often just to show it off.

机理：峰容量随维数相乘，时间也一样 / Mechanism: peak capacity multiplies with each dimension, and so does time

背景 峰容量是一张谱最多能分开的峰数，约为谱宽/线宽。新维度的分离与原有各维独立，所以总峰容量近似为各维之积。代价是每个间接维都要逐步增加演化时间，每一步都是一次完整实验，FID 数等于各间接维增量数之积。

> **Background** Peak capacity is the most peaks a spectrum can resolve, roughly spectral width over linewidth. A new dimension separates independently, so the total is roughly the product over dimensions. The cost: each indirect dimension steps its evolution time, every step is a full experiment, and the FID count is the product of the increment counts.

峰容量（背景） / peak capacity (background)

$$n_c \approx \frac{\Delta\nu}{w}, \qquad n_c^{(n\mathrm{D})} \approx \prod_{i=1}^{n} n_{c,i}$$

$\Delta\nu$ 谱宽，w 线宽（Hz） / $\Delta\nu$ spectral width, w linewidth (Hz)

三维实验时间（背景） / 3D experiment time (background)

$$t_{\mathrm{exp}} \approx N_1 N_2 \, N_{\mathrm{scans}} \, (D_1 + t_{\mathrm{aq}})$$

N_1, N_2：两个间接维的增量数

/ N_1, N_2: increments in the two indirect dimensions

背景 示例（非幻灯片数字）：$N_1 = 128$、$N_2 = 64$、每增量 4 次扫描、每次 1.5 s，共 $49\,152$ s ≈ 13.7 h；128 个增量的二维只需约 13 min。再加第四维又乘一次，这就是三维以上少见的原因。

> **Background** Example (not from the slide): $N_1 = 128$, $N_2 = 64$, 4 scans of 1.5 s per increment give $49\,152$ s ≈ 13.7 h; a 2D with 128 increments needs about 13 min. A fourth dimension multiplies again, which is why going above 3D is rare.

图怎么读：两种峰容量的对数坐标图 / Reading the figure: a log–log plot of two kinds of peak capacity

- 横轴 "spectroscopy / spectrometry"（波谱/质谱方法的峰容量），纵轴 "separation"（分离方法的峰容量），都是 1 到 10^8 的对数刻度。
  > x-axis "spectroscopy / spectrometry" (peak capacity of spectroscopic and MS methods), y-axis "separation" (peak capacity of separation methods), both logarithmic from 1 to 10^8.
- 竖实线（约值）：UV/VIS ≈ 20，IR $\approx 5 \times 10^2$，^{1}H NMR $\approx 4 \times 10^3$，^{13}C NMR $\approx 3 \times 10^4$，^{1}H,^{13}C HSQC $\approx 2 \times 10^6$，FTICR MS $\approx 10^7$，NMR³ 接近 10^8（图中最右）。
  > Vertical lines (approx.): UV/VIS ≈ 20, IR $\approx 5 \times 10^2$, ^{1}H NMR $\approx 4 \times 10^3$, ^{13}C NMR $\approx 3 \times 10^4$, ^{1}H,^{13}C HSQC $\approx 2 \times 10^6$, FTICR MS $\approx 10^7$, NMR³ near 10^8 (rightmost).
- 横虚线（约值）：TLC ≈ 70、HPLC $\approx 2 \times 10^2$、cGC、UPLC $\approx 10^3$，CE、TLC² 与二维凝胶电泳 $\approx 5 \times 10^3$，LC/GC、LC/UPLC $\approx 10^5$，GC²、UPLC²、CE² $\approx 10^6$，LC³ $\approx 10^7$。上标 ²、³ 指二维、三维联用（推断）。
  > Dashed lines (approx.): TLC ≈ 70, HPLC $\approx 2 \times 10^2$, cGC and UPLC $\approx 10^3$, CE, TLC² and 2D gel electrophoresis $\approx 5 \times 10^3$, LC/GC and LC/UPLC $\approx 10^5$, GC², UPLC², CE² $\approx 10^6$, LC³ $\approx 10^7$. Superscripts ², ³ denote 2D/3D hyphenation (inference).
- ^{1}H、^{13}C 的点位与"谱宽/线宽"相符（背景：600 MHz 下 10 ppm = 6000 Hz，线宽约 1–2 Hz）。HSQC 低于两者之积：间接维增量有限、分辨率低（背景）。
  > The ^{1}H and ^{13}C positions fit width/linewidth (background: 10 ppm = 6000 Hz at 600 MHz, lines $\approx$ 1–2 Hz). HSQC sits below their product: few indirect increments, lower resolution (background).

考点速记 KEY POINTS

- 峰容量 ≈ 谱宽/线宽；多维时各维相乘（背景）。
  Peak capacity ≈ width/linewidth; it multiplies across dimensions.
- 图：^{1}H $\approx 10^3$–10^4，HSQC $\approx 10^6$，NMR³ $\approx 10^8$。
  Plot: ^{1}H $\approx 10^3$–10^4, HSQC $\approx 10^6$, NMR³ $\approx 10^8$.
- 三维常用于蛋白质；三维以上少见，常为展示。
  3D is common for proteins; above 3D is rare, often for show.
- 代价：时间 ∝ 各间接维增量数之积 $N_1 N_2$。
  Cost: time ∝ the product of increments $N_1 N_2$.

S15 · 三维 NMR 怎样工作：立方体、平面与投影

S15 · How 3D NMR works: cube, planes and projections

幻灯片要点 / What the slide says

对小分子的结构解析（structure elucidation）来说，更高的维数通常没有必要；它们对蛋白质或混合物分析更有用。图中大致解释了三维 NMR 的工作方式。由于真正的数据立方体用起来很笨重，实际中常用投影（projection）。

> Higher dimensions are usually unnecessary for small-molecule structure elucidation; they are more useful for proteins or for mixture analysis. The figure gives a sort of explanation of how 3D NMR works. Projections are often used because actual cubes can be cumbersome.

为什么小分子不需要、蛋白质需要 / Why small molecules do not need it and proteins do

背景 小分子只有几十个峰，COSY、HSQC 等二维谱的峰容量（S14）早已足够。蛋白质有成百上千个化学环境相似的残基，混合物有许多组分，即使在二维谱里也大量重叠。第三维用第三种核的化学位移把重叠的峰拉开。图中用的 $^{15}\mathrm{N}$ 和 $^{13}\mathrm{C}$ 天然丰度很低（约 0.37%、1.1%），所以蛋白质三维实验通常要先做同位素标记。

> **Background** A small molecule has a few tens of peaks; the peak capacity of 2D spectra such as COSY and HSQC (S14) is already enough. A protein has hundreds to thousands of residues in similar environments, and a mixture has many components, so peaks overlap heavily even in 2D. The third dimension uses a third nucleus's chemical shift to pull overlapping peaks apart. The $^{15}\mathrm{N}$ and $^{13}\mathrm{C}$ used in the figure have low natural abundance ($\approx 0.37\%, 1.1\%$), so protein 3D work normally needs isotopic labelling first.

图怎么读：(a)–(e) 一步步搭出三维 / Reading the figure: building 3D step by step, (a)–(e)

- 坐标（左下）：$x = {}^{1}\mathrm{H}$，$y = {}^{13}\mathrm{C}$，$z = {}^{15}\mathrm{N}$。
  > Axes (bottom left): $x = {}^{1}\mathrm{H}$, $y = {}^{13}\mathrm{C}$, $z = {}^{15}\mathrm{N}$.
- (a) 二维 $^{1}\mathrm{H}$–$^{15}\mathrm{N}$ 平面，5 个蓝色峰，每个峰是一对相连的 H 与 N。
  > (a) A 2D $^{1}\mathrm{H}$–$^{15}\mathrm{N}$ plane with 5 blue peaks, each a bonded H–N pair.
- (b) 加上 $^{13}\mathrm{C}$ 轴：每个蓝峰沿黑箭头被"抬"到对应的 $^{13}\mathrm{C}$ 化学位移高度，成为立方体中的红点。在 (a) 中 $^{1}\mathrm{H}$ 位置相近的峰，只要 $^{13}\mathrm{C}$ 不同，在三维中就彼此分开。
  > (b) Adding the $^{13}\mathrm{C}$ axis: each blue peak is lifted along a black arrow to its $^{13}\mathrm{C}$ shift and becomes a red point in the cube. Peaks close in $^{1}\mathrm{H}$ in (a) separate in 3D as long as their $^{13}\mathrm{C}$ differs.
- (c) 只剩三维立方体中的 5 个峰：这就是三维谱本身。
  > (c) Only the 5 peaks in the cube remain: this is the 3D spectrum itself.
- (d)、(e) 立方体中各取出一张粉色平面：平行于 $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ 面，在 $^{15}\mathrm{N}$ 轴上的位置不同。每张平面上有一个点画成深红色（落在该平面上的峰），其余画得较浅。
  > (d), (e) Each takes one pink plane out of the cube, parallel to the $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ face at a different $^{15}\mathrm{N}$ position. One point per plane is dark red (the peak lying in that plane); the rest are drawn paler.

$$\underbrace{S(F_1, F_2, F_3)}_{\text{cube}} \;\rightarrow\; \underbrace{S(F_1, F_2, F_3 = c)}_{\text{plane}}, \qquad \underbrace{\textstyle\sum_{F_3} S(F_1, F_2, F_3)}_{\text{projection}}$$

背景：平面固定一维；投影沿一维叠加全部峰 / background: a plane fixes one axis; a projection sums along it

考点速记 KEY POINTS

- 小分子结构解析通常不需要三维；蛋白质、混合物才需要。
  Small-molecule elucidation rarely needs 3D; proteins and mixtures do.
- 平面固定一维位移；投影沿一轴叠加（背景）。
  Plane = 2D at one fixed shift; projection = sum along an axis.
- 第三维按第三种核的位移拉开二维中重叠的峰。
  The third dimension separates 2D overlaps by a third nucleus's shift.
- 立方体不便显示和阅读，实际多看投影或平面。
  Cubes are cumbersome; projections or planes are read instead.

S16 · 三维 NMR 用于活体生物

S16 · 3D NMR applied to living organisms

幻灯片要点 / What the slide says

这是把三维 NMR 用于活体生物的例子。上排是二维投影，但由于弛豫，信号很弱；下排是离体（ex vivo）数据，好得多。

> Here is an example of 3D NMR applied to living organisms. The top row shows 2D projections, but the signals are low due to relaxation. The bottom row shows ex vivo data, which is a lot nicer.

机理：为什么活体中弛豫吃掉信号 / Mechanism: why relaxation eats the signal in vivo

背景 活体内分子处在黏稠、拥挤的环境中，部分与膜或大分子结合，组织本身也使 B_0 不均匀，所以横向弛豫快（T_2 短）、谱线宽。三维序列含多个相干转移步骤，每步都有毫秒级的延迟；在这段总时间 t_{seq} 内磁化按 T_2 衰减，快弛豫的信号在检测前就所剩无几。离体样品没有这些问题，谱线窄、信号多。

> **Background** Inside an organism, molecules sit in a viscous, crowded medium, some bound to membranes or macromolecules, and the tissue makes B_0 inhomogeneous, so transverse relaxation is fast (short T_2) and lines are broad. A 3D sequence has several coherence-transfer steps with millisecond delays each; over the total time t_{seq} magnetization decays with T_2, and fast-relaxing signals are mostly gone before detection. Ex vivo samples avoid this: narrower lines, more signals.

$$S \approx S_0\, e^{-t_{\mathrm{seq}}/T_2}$$

背景：T_2 越短、序列越长，剩下的信号越少 / background: shorter T_2 and longer sequences leave less signal

图怎么读 / Reading the figure

- A–E 是二维图：横轴 $^{1}\mathrm{H}$ 化学位移（ppm），纵轴 $^{13}\mathrm{C}$ 化学位移（ppm），右下角注明所取的 F1 值；彩色圆点标出目标代谢物的峰。
  > A–E are 2D maps: x = $^{1}\mathrm{H}$ chemical shift (ppm), y = $^{13}\mathrm{C}$ chemical shift (ppm), the F1 value in the lower right corner; coloured dots mark the target metabolite's peaks.
- 上排：A 尿苷（uridine），F1 = 5.7 ppm，4 个蓝点，另有一条约 5.3 ppm 的竖直条纹；B TAG，F1 = 1.37 ppm；C 脯氨酸、丙氨酸、谷氨酸、谷氨酰胺，F1 = 3.51 ppm。A 中只有零星几个峰。
  > Top row: A uridine, F1 = 5.7 ppm, 4 blue dots plus a vertical streak near 5.3 ppm; B TAG, F1 = 1.37 ppm; C proline, alanine, glutamic acid, glutamine, F1 = 3.51 ppm. A shows only a few scattered peaks.
- 下排：D 赖氨酸，F1 = 3.49 ppm；E D-葡萄糖，F1 = 3.41 ppm，两图都布满清晰的等高线峰；F 是三维立方体（红色等值面），轴为 F1 [ppm]（$^{1}\mathrm{H}$，约 3.5–2.0）、F2 [ppm]（$^{13}\mathrm{C}$，约 40–80）、F3 [ppm]（$^{1}\mathrm{H}$，约 3.5–2.5）。
  > Bottom row: D lysine, F1 = 3.49 ppm; E D-glucose, F1 = 3.41 ppm, both full of clear contour peaks; F is the 3D cube (red isosurfaces) with axes F1 [ppm] ($^{1}\mathrm{H}$, ≈ 3.5–2.0), F2 [ppm] ($^{13}\mathrm{C}$, ≈ 40–80), F3 [ppm] ($^{1}\mathrm{H}$, ≈ 3.5–2.5).

编者注 / Editor's note

幻灯片称上排为"二维投影"，但 A–E 每张都标有"F1 = … ppm"，即在一个固定 F1（$^{1}\mathrm{H}$）位移处从立方体中取出的平面（S15 的 (d)、(e)），而不是沿 F1 的投影。按 F 的轴，这是 $^{1}\mathrm{H}$–$^{13}\mathrm{C}$–$^{1}\mathrm{H}$ 三维谱。幻灯片没有给出实验名称、生物种类，也没有展开 TAG，这里不作补充。

> The slide calls the top row "2D projections", yet every panel A–E is labelled "F1 = … ppm": a plane cut from the cube at one fixed F1 ($^{1}\mathrm{H}$) shift (as in S15 (d), (e)), not a projection along F1. From F's axes the data set is a $^{1}\mathrm{H}$–$^{13}\mathrm{C}$–$^{1}\mathrm{H}$ 3D. The slide names neither the experiment nor the organism and does not expand TAG; nothing is added here.

考点速记 KEY POINTS

- 活体三维可行，但弛豫使信号弱（上排）。
  3D works in vivo, but relaxation weakens signals (top row).
- 离体（ex vivo）数据好得多（下排）。
  Ex vivo data are much nicer (bottom row).
- 标"F1 = … ppm"的图是固定 F1 的平面（编者注）。
  Panels marked "F1 = … ppm" are fixed-F1 planes (note).
- 轴：F1 $^{1}\mathrm{H}$、F2 $^{13}\mathrm{C}$、F3 $^{1}\mathrm{H}$（面板 F）。
  Axes: F1 $^{1}\mathrm{H}$, F2 $^{13}\mathrm{C}$, F3 $^{1}\mathrm{H}$ (panel F).

S17 · 三维用于结构解析：HMQC-COSY

S17 · 3D for structure elucidation: HMQC-COSY

幻灯片要点 / What the slide says

有人提出三维 NMR 是一种有效的结构解析方法。但由于实验时间长（因而成本高），它并不常用，对小分子尤其如此。尽管如此，它能给出非常好的信息。

> 3D NMR has been suggested as an effective method for structural elucidation. However, because of the length of the experiment (and thus cost) it is not commonly applied, especially for small molecules. That said, it can give excellent information.

机理：用 ^{13}C 位移把 COSY 拆成几层 / Mechanism: splitting a COSY into layers by ^{13}C shift

HMQC-COSY 把两个二维实验串成一个三维实验。HMQC（异核多量子相干谱）部分让每个质子在 t_1 期间带上与它相连碳的化学位移；COSY（^1H–^1H 化学位移相关谱）部分让质子在 t_2 演化，再经一个脉冲把磁化传给耦合伙伴，在 t_3 检测，同时对 ^{13}C 去耦。结果是按 ^{13}C 位移分层的 COSY：^1H 中挤在一起的多重峰，只要所连的碳不同，就落在不同的层里。

> HMQC-COSY chains two 2D experiments into one 3D. The HMQC part labels each proton during t_1 with the shift of its attached carbon; the COSY part lets the protons evolve during t_2, a pulse passes magnetization to coupled partners, and detection runs in t_3 with ^{13}C decoupling. The result is a COSY layered by ^{13}C shift: multiplets crowded together in ^1H land in different layers as long as their carbons differ.

HMQC 演化 / HMQC evolution	COSY 演化 / COSY evolution	检测（去耦） / detection (dec)
$t_1 \rightarrow F_1\ (^{13}C)$	$t_2 \rightarrow F_2\ (^1H)$	$t_3 \rightarrow F_3\ (^1H)$

图怎么读 / Reading the figure

- 片段：三个相连的碳 C_A、C_B、C_C，各带一个质子 H_A、H_B、H_C。
  > Fragment: three bonded carbons C_A, C_B, C_C, each carrying one proton H_A, H_B, H_C.
- 脉冲序列（窄条 = 90°、宽条 = 180°，背景惯例）：HMQC 为 prep、Δ、t_1、Δ、t_2（采样，C 通道 dec）；COSY 为脉冲、t_1、脉冲、t_2；HMQC-COSY 为 prep、Δ、t_1、Δ、t_2、再一个 ^1H 脉冲、t_3（采样，dec）。背景 $\Delta \approx 1/(2\,{}^1J_{CH}) \approx 3.6$ ms。
  > Pulse sequences (thin bar = 90°, wide = 180°, background convention): HMQC is prep, Δ, t_1, Δ, t_2 (acquire, C channel dec); COSY is pulse, t_1, pulse, t_2; HMQC-COSY is prep, Δ, t_1, Δ, t_2, one more ^1H pulse, t_3 (acquire, dec). **Background** $\Delta \approx 1/(2\,{}^1J_{CH}) \approx 3.6$ ms.
- b（绿，HMQC）：上方 ^{13}C 谱有 C_A、C_B、C_C 三条线，平面中 3 个黑点，每点一对 C–H。a（蓝，COSY）：上方 ^1H 谱中 H_A 与 H_B 几乎挨着，H_C 分开；对角线上 3 个黑点，红色交叉峰只在 A–B、B–C 之间（关于对角线对称）。
  > b (green, HMQC): the ^{13}C trace shows C_A, C_B, C_C; the plane has 3 black dots, one per C–H pair. a (blue, COSY): in the ^1H trace H_A and H_B nearly touch, H_C stands apart; 3 black diagonal dots, and red cross peaks only for A–B and B–C (mirrored about the diagonal).
- c（黄，HMQC-COSY）：立方体沿 ^{13}C 方向排着三张平面。前层画 2 个对角峰与 2 个交叉峰（A–B），中层 3 个对角峰与 4 个交叉峰（A–B 与 B–C），后层 2 与 2（B–C）。每层只含与该碳上质子有关的 COSY 峰，所以在 a 中挤在一起的 A–B 峰在三维中分开了。
  > c (yellow, HMQC-COSY): three planes along the ^{13}C direction. The front one has 2 diagonal and 2 cross peaks (A–B), the middle 3 and 4 (A–B and B–C), the back 2 and 2 (B–C). Each layer holds only COSY peaks involving the proton on that carbon, so the crowded A–B peaks of a separate in 3D.

考点速记 KEY POINTS

- HMQC-COSY = HMQC（^{13}C 标记 H）+ COSY（H–H 相关）。
  HMQC-COSY = HMQC (^{13}C labels H) + COSY (H–H).
- $t_1, t_2, t_3 \rightarrow F_1, F_2, F_3$；去耦在 t_3（采样）期间。
  $t_1, t_2, t_3 \rightarrow F_1, F_2, F_3$; decoupling during t_3.
- 每层 ^{13}C 平面把重叠的 H_A/H_B 分开。
  Each ^{13}C plane separates overlapping H_A/H_B.
- 少用：实验长、成本高，尤其对小分子。
  Rarely used: long and costly, especially for small molecules.

S18 · 加入不同的核：H–C–Si 三维

S18 · Adding different nuclei: an H–C–Si 3D

幻灯片要点 / What the slide says

通常的做法是把几个实验拼在一起（例如 HSQC-TOCSY）。但也可以加入不同的核。本例采集的谱显示碳、硅和氢之间的相关，用来研究一种有机硅聚合物。

> Normally several experiments are added together (e.g., HSQC-TOCSY). However, we can also add different nuclei. In this example a spectrum is acquired that shows correlations between carbon, silicon and hydrogen; it was used to examine an organosilicone polymer.

机理：沿 H→C→Si 的耦合传递 / Mechanism: transfer along H→C→Si couplings

背景 磁化先经 $^{1}J_{\mathrm{CH}}$ 从 H 传到 C，再经 C–Si 耦合传到 Si，在那里按 δ_{Si} 演化后原路返回 H 检测。每一步的延迟按所用耦合常数设定（约 $1/(2J)$），所以选择 $^{1}J_{\mathrm{CSi}}$ 还是 $^{2}J_{\mathrm{CSi}}$，决定了看到的是与 Si 直接相连的碳，还是隔一个原子的碳。只有同时含 $^{13}\mathrm{C}$（1.1%）和 $^{29}\mathrm{Si}$（4.7%）的分子才贡献信号，约 0.05%，这是灵敏度低、实验长的原因之一。

> **Background** Magnetization goes from H to C via $^{1}J_{\mathrm{CH}}$, then to Si via the C–Si coupling, evolves at δ_{Si} and returns the same way to H for detection. Each delay is set from its coupling ($\approx 1/(2J)$), so choosing $^{1}J_{\mathrm{CSi}}$ or $^{2}J_{\mathrm{CSi}}$ decides whether you see carbons bonded to Si or one atom further away. Only molecules carrying both $^{13}\mathrm{C}$ (1.1%) and $^{29}\mathrm{Si}$ (4.7%) contribute, about 0.05%, one reason for low sensitivity and long runs.

图中耦合常数 / couplings in the figure

$$^{1}J_{\mathrm{CH}} = 140\ \mathrm{Hz}, \qquad J_{\mathrm{CSi}} = 59\ \mathrm{Hz}$$

图中写作 J_{CH}，140 Hz 即一键 C–H 耦合

/ printed as J_{CH}; 140 Hz is the one-bond C–H coupling

对应延迟（背景） / matching delays (background)

$$\frac{1}{2\times 140\ \mathrm{Hz}} \approx 3.6\ \mathrm{ms}, \qquad \frac{1}{2\times 59\ \mathrm{Hz}} \approx 8.5\ \mathrm{ms}$$

图怎么读 / Reading the figure

- 顶部化合物 1：以一个 Si 为核心、经 $\mathrm{CH_2CH_2}$ 链连接外层 Si 的有机硅聚合物，Si 上带甲基（Me），链端为 Si–H（按结构式读出）。箭头标 $J_{\mathrm{CH}} = 140\ \mathrm{Hz}$ 与 $J_{\mathrm{CSi}} = 59\ \mathrm{Hz}$；红、黄底色的含义幻灯片未说明。

  > Top, compound 1: an organosilicone polymer with a Si core linked by $\mathrm{CH_2CH_2}$ chains to outer Si atoms, methyl (Me) groups on Si and terminal Si–H (read from the formula). Arrows mark $J_{\mathrm{CH}} = 140\ \mathrm{Hz}$ and $J_{\mathrm{CSi}} = 59\ \mathrm{Hz}$; the slide does not explain the red and yellow shading.

- 六张平面：横轴 F1 (ppm) = $\delta(^{13}\mathrm{C})$，6 到 -6 ppm；纵轴 F3 (ppm) = $\delta(^{1}\mathrm{H})$，0.2–1.0 ppm。三行分别取 $\delta_{\mathrm{Si}} = 9.95,\ 7.99,\ 2.25$ ppm（c 标为 "2.25ppm*"，星号未解释）；左列 d、e、f 为 $^{2}J_{\mathrm{CSi}}$，右列 a、b、c 为 $^{1}J_{\mathrm{CSi}}$。

  > Six planes: x = F1 (ppm) = $\delta(^{13}\mathrm{C})$, 6 to -6 ppm; y = F3 (ppm) = $\delta(^{1}\mathrm{H})$, 0.2–1.0 ppm. Rows are taken at $\delta_{\mathrm{Si}} = 9.95,\ 7.99,\ 2.25$ ppm (c reads "2.25ppm*"; the asterisk is not explained); left column d, e, f = $^{2}J_{\mathrm{CSi}}$, right column a, b, c = $^{1}J_{\mathrm{CSi}}$.

- 右列：a 一个红峰（约 4 / 0.7 ppm）；b、c 各两个红峰，其中一个在 $\delta(^{13}\mathrm{C}) \approx -5$、$\delta(^{1}\mathrm{H}) \approx 0.1$ ppm。左列每张都含同一 δ_{Si} 右列的红峰，另多一个黄峰（约 6 / 0.6–0.7 ppm）。

  > Right column: a has one red peak (≈ 4 / 0.7 ppm); b and c two each, one at $\delta(^{13}\mathrm{C}) \approx -5$, $\delta(^{1}\mathrm{H}) \approx 0.1$ ppm. Each left plane repeats the red peaks of the right plane at the same δ_{Si} plus one yellow peak (≈ 6 / 0.6–0.7 ppm).

- 推断：右列是与该 Si 直接相连的 C–H，左列多出的黄峰是隔两根键的碳；三个 δ_{Si} 对应聚合物中三种 Si 环境。

  > Inference: the right column shows C–H bonded directly to that Si, the extra yellow peak on the left is a carbon two bonds away; the three δ_{Si} values are three Si environments of the polymer.

考点速记 KEY POINTS

- 多维可拼实验（HSQC-TOCSY），也可加第三种核。
  Combine experiments (HSQC-TOCSY) or add a third nucleus.
- 平面取 $\delta_{\mathrm{Si}} = 9.95,\ 7.99,\ 2.25$ ppm。
  Planes at $\delta_{\mathrm{Si}} = 9.95,\ 7.99,\ 2.25$ ppm.
- H→C 经 $^{1}J_{\mathrm{CH}} = 140$ Hz，C→Si 经 $J_{\mathrm{CSi}} = 59$ Hz。
  H→C via $^{1}J_{\mathrm{CH}} = 140$ Hz; C→Si via $J_{\mathrm{CSi}} = 59$ Hz.
- $^{1}J_{\mathrm{CSi}}$ 列 = 直接相连；$^{2}J_{\mathrm{CSi}}$ 列多出隔两键的碳。
  $^{1}J_{\mathrm{CSi}}$: bonded carbons; $^{2}J_{\mathrm{CSi}}$ adds two-bond ones.

S19 · 非均匀采样（NUS）：少采增量，更快采集

S19 · Non-uniform sampling: fewer increments, faster acquisition

幻灯片要点 / What the slide says

非均匀采样（non-uniform sampling, NUS）在多维 NMR 实验中非常常用。二维实验通常让演化延迟 t_1 线性增加；但也可以用别的方式采样，从而少用增量、加快采集。它与线性预测（linear prediction）有几分相似。

> Non-uniform sampling (NUS) is very commonly used in nD NMR experiments. A 2D experiment typically uses a linear increase in the evolution delay t_1; however, it is possible to sample in different ways, which allows fewer increments and thus faster acquisition. It is somewhat analogous to linear prediction.

机理：时间花在哪里，NUS 省在哪里 / Mechanism: where the time goes and where NUS saves it

背景 常规采样中每个 t_1 值都是一次完整实验（一条 FID 加上它的全部扫描）。步长 Δt_1 决定间接维谱宽，最长演化时间 $t_{1,\max}$ 决定间接维分辨率；想要宽谱又要高分辨率，增量数 N 就很大。NUS 保留同一个网格（同样的 Δt_1 与 $t_{1,\max}$），只测其中 M 个点，缺失点在处理时重建，时间约缩短为 M/N。

> **Background** In conventional sampling every t_1 value is a complete experiment (one FID with all its scans). The step Δt_1 sets the indirect spectral width and the longest evolution time $t_{1,\max}$ sets the indirect resolution; wide spectra at high resolution need a large increment count N. NUS keeps the same grid (same Δt_1 and $t_{1,\max}$), measures only M of its points and rebuilds the rest in processing, cutting the time to about M/N.

线性 t_1 采样 / linear t_1 sampling

$$t_1 = n\,\Delta t_1,\quad n = 0, 1, \ldots, N-1$$

谱宽与分辨率（背景） / width and resolution (bkgd.)

$$\mathrm{SW}_1 = \frac{1}{\Delta t_1},\quad \delta\nu_1 \approx \frac{1}{t_{1,\max}}$$

NUS 节省（背景） / NUS saving (bkgd.)

$$\frac{t_{\mathrm{NUS}}}{t_{\mathrm{conv}}} \approx \frac{M}{N}$$

背景 为什么像线性预测：线性预测把测得的 FID 当作若干衰减正弦之和，向后外推出未测的点；NUS 重建同样假设谱中只有少数峰，以此填补网格中的空缺。两者都由模型"补"出数据，模型不符时都会产生伪峰。

> **Background** Why it resembles linear prediction: linear prediction treats the measured FID as a sum of decaying sinusoids and extrapolates the missing later points; NUS reconstruction likewise assumes the spectrum holds only a few peaks and uses that to fill gaps in the grid. Both fill in data from a model, and both create artifacts when the model does not fit.

图怎么读：常规与非均匀二维采样 / Reading the figure: conventional vs non-uniform 2D sampling

- 上（Conventional 2D Sampling）：左边一列蓝色脉冲对，自下而上两脉冲间隔（t_1）逐行增大，每行后接一条沿 t_2 的 FID；沿 t_2 做 FT 得到一叠 F2 谱，峰高随 t_1 起伏（红色虚线包络）；再沿 t_1 做 FT 得到 F1，最后是 F1 对 F2 的二维谱，一个等高线峰。
  > Top (Conventional 2D Sampling): a column of blue pulse pairs whose spacing (t_1) grows row by row from bottom to top, each followed by an FID along t_2; FT along t_2 gives a stack of F2 spectra whose peak height oscillates with t_1 (red dotted envelope); FT along t_1 gives F1, and finally an F1 vs F2 map with one contoured peak.
- 下（Non-uniform 2D Sampling）：只采 4 行（2 个短 t_1、2 个长 t_1），中间留空；重建后中间一栏与上图同样是完整的一叠，最终二维谱与常规采样看起来一样。
  > Bottom (Non-uniform 2D Sampling): only 4 rows are acquired (2 short and 2 long t_1), with a gap between; after reconstruction the middle stack is complete again and the final 2D map looks the same as the conventional one.

考点速记 KEY POINTS

- 常规：$t_1 = n\Delta t_1$ 线性递增，每个增量一次实验。
  Conventional: $t_1 = n\Delta t_1$, one experiment per increment.
- 分辨率看 $t_{1,\max}$；NUS 保留 $t_{1,\max}$（背景）。
  Resolution follows $t_{1,\max}$; NUS keeps it (background).
- NUS：只采部分 t_1 点 → 增量少 → 采集快。
  NUS: sample some t_1 points → fewer increments → faster.
- 缺失点靠重建补全，类似线性预测。
  Missing points are reconstructed, akin to linear prediction.

S20 · NUS：省 4 倍时间，付出什么代价

S20 · NUS: a four-fold time saving and what it costs

幻灯片要点 / What the slide says

本例中 NUS 谱给出的结果与一张长 4 倍的标准谱相当。它在 SNR 上有缺点，可以用加权采样（weighted sampling）缓解；还可能产生伪峰（artifact）。尽管如此，尤其对更高维的实验，它能节省大量时间。

> Here the NUS spectrum gives results comparable to a standard spectrum that is four times longer. There are drawbacks in SNR, which can be helped with weighted sampling, and there is potential for artifacts. That said, especially for higher-dimensional experiments, it is a massive time save.

核对数字 / Checking the numbers

$$\frac{64}{256} = 25\%, \qquad \frac{15\ \text{min}}{3.75\ \text{min}} = 4\times, \qquad \frac{3.75\ \text{min}}{64} = \frac{15\ \text{min}}{256} \approx 3.5\ \text{s}$$

每个增量耗时相同，总时间只看采了几个增量 / equal cost per increment: time depends only on how many are taken

为什么 NUS 像 15 分钟谱 / Why NUS matches the 15-min run

背景 常规 64 增量只走到 256 增量网格的前四分之一，$t_{1,\max}$ 只有四分之一，所以 $^{13}\mathrm{C}$（F1）方向线宽约宽 4 倍。NUS 25 % 也只采 64 个点，但它们散布在整个 256 点网格上，$t_{1,\max}$ 与 15 分钟谱相同，分辨率也相同。

> **Background** The 64-increment conventional run covers only the first quarter of the 256-point grid, so its $t_{1,\max}$ is a quarter as long and its $^{13}\mathrm{C}$ (F1) lines are about 4 times broader. NUS 25 % also takes 64 points, but spread over the whole 256-point grid, so its $t_{1,\max}$, and hence its resolution, equal the 15-min run's.

背景 三个代价：(1) SNR：总测量时间少、平均少，粗略按 $\mathrm{SNR} \propto \sqrt{t}$，$1/4$ 时间约剩一半 SNR。(2) 加权采样：信号随 t_1 衰减，短 t_1 处多采、长 t_1 处少采，把时间花在信号最强处。(3) 伪峰：重建误差表现为沿 F1 的噪声条纹或小假峰，强峰旁最明显。多维网格点数是各维之积，同样比例省下的绝对时间更多（S14 的 13.7 h 示例按 25 % 约 3.4 h）。

> **Background** Three costs: (1) SNR: less measuring time, less averaging; roughly $\mathrm{SNR} \propto \sqrt{t}$, so $1/4$ of the time leaves about half the SNR. (2) Weighted sampling: the signal decays with t_1, so sample densely at short and sparsely at long t_1, spending time where the signal is strongest. (3) Artifacts: reconstruction errors show as noise ridges along F1 or small false peaks, worst next to strong peaks. In nD the grid is a product, so the same fraction saves far more absolute time (S14's 13.7 h example becomes about 3.4 h at 25 %).

图怎么读：蔗糖的部分 HSQC / Reading the figure: a partial HSQC of sucrose

- 标题：部分 600 MHz $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ HSQC，蔗糖溶于 $\mathrm{D_2O}$，常规对比非均匀采样；上方为一维 $^{1}\mathrm{H}$ 谱。刻度太小无法读出，不引用。

  > Title: partial 600 MHz $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ HSQC of sucrose in $\mathrm{D_2O}$, conventional vs non-uniform sampling; a 1D $^{1}\mathrm{H}$ trace on top. Axis ticks are unreadable and not quoted.

- 每个交叉峰画三次，上下错开（显示上的平移，推断）：紫 = NUS，25 %，3.75 min（64 增量）；蓝 = 常规，3.75 min（64 增量）；黑 = 常规，15 min（256 增量）。

  > Each cross peak is drawn three times, offset vertically (a display shift, inference): purple = NUS, 25 %, 3.75 min (64 increments); blue = conventional, 3.75 min (64); black = conventional, 15 min (256).

- 紫色与黑色等高线都是扁平的窄椭圆，几乎一样；蓝色在纵向（F1，$^{13}\mathrm{C}$）明显更高更圆，即 F1 分辨率差。

  > Purple and black contours are flat, narrow ellipses and nearly identical; blue ones are clearly taller and rounder in the vertical (F1, $^{13}\mathrm{C}$) direction, i.e. poorer F1 resolution.

考点速记 KEY POINTS

- NUS 25 %（64 增量）3.75 min ≈ 常规 256 增量 15 min。
  NUS 25 % (64 increments, 3.75 min) ≈ conventional 256 (15 min).
- 同为 64 增量的常规谱 F1 峰更宽：$t_{1,\max}$ 短（背景）。
  Conventional 64: broader F1 peaks, short $t_{1,\max}$ (background).
- 代价：SNR 降低（加权采样可补救），可能出现伪峰。
  Costs: lower SNR (weighted sampling helps) and artifacts.
- 维数越高，NUS 省时越多。
  The higher the dimension, the bigger the saving.

S21 · 超快 NMR：把增量放到样品管的不同位置

S21 · Ultrafast NMR: increments in different parts of the tube

幻灯片要点 / What the slide says

通常，二维 NMR 用不同的增量来编码第二维，每个增量都是实验的一次重复。超快 NMR（ultrafast NMR, UF）则在样品管的不同部位同时采集这些增量。

> Typically, 2D NMR encodes the second dimension in different increments, which are repeats of the experiment. Ultrafast NMR (UF) gathers the increments at the same time in different parts of the tube.

机理：用空间换时间 / Mechanism: trading space for time

常规二维中，第 k 个增量的演化时间是 $k\Delta$，n 个增量就要 n 次扫描，每次扫描之间还要等弛豫。超快 NMR 只扫描一次：管中不同高度的薄层同时经历 $\Delta, 2\Delta, \ldots, n\Delta$ 的演化，再一起检测。这叫空间编码（spatial encoding）。

> In a conventional 2D the k-th increment evolves for $k\Delta$, so n increments need n scans, with a relaxation wait between scans. Ultrafast NMR scans once: thin slices at different heights in the tube evolve for $\Delta, 2\Delta, \ldots, n\Delta$ at the same time and are detected together. This is spatial encoding.

常规二维 / conventional 2D

$$t_1 = k\Delta,\ \ k = 1, \ldots, n \ \Rightarrow\ n \text{ scans}$$

超快二维 / ultrafast 2D

$$t_1 = t_1(z)\ \ (\Delta \ldots n\Delta \text{ along the tube}) \ \Rightarrow\ 1 \text{ scan}$$

背景 怎样做到：沿管轴 z 加脉冲场梯度，使共振频率随位置变化；同时用频率扫描的激发脉冲，让各薄层依次被激发，于是演化时间随 z 线性变化（$t_1 \propto z$）。检测时再用来回切换的梯度在一次 FID 中读出这段空间信息，空间位置就是 F1 坐标。由此也能预见代价（S22–S23）：每层只含一小部分样品，读出带宽又宽，SNR 低；样品若沿 z 不均匀，"增量"之间就不再只差演化时间。

> **Background** How: a pulsed field gradient along the tube axis z makes the resonance frequency depend on position, and a frequency-swept excitation pulse excites the slices one after another, so the evolution time varies linearly with position, $t_1 \propto z$. During detection, a gradient switched back and forth reads out this spatial information within one FID, and position becomes the F1 coordinate. The costs follow (S22–S23): each slice holds a small part of the sample and the readout bandwidth is wide, so SNR is low; and if the sample is not uniform along z, the "increments" differ by more than their evolution time.

图怎么读 / Reading the figure

- a) Traditional 2D NMR，"n scans"：彩色背景分出准备期（Preparation）、演化期（Evolution）、混合期（Mixing）、检测期（Detection）。三行画出 p → Δ → m → FID、p → 2Δ → m → FID、p → $n\Delta$ → m → FID；每行右侧整支样品管都参与，分别标 Δ、2Δ、$n\Delta$。

  > a) Traditional 2D NMR, "n scans": coloured bands mark Preparation, Evolution, Mixing and Detection. Three rows show p → Δ → m → FID, p → 2Δ → m → FID, p → $n\Delta$ → m → FID; at the right of each row the whole tube takes part, labelled Δ, 2Δ, $n\Delta$.

- b) Ultrafast 2D NMR，"1 scan"：同样三组 p–Δ–m、p–2Δ–m、p–$n\Delta$–m，但虚线都指向同一支管的不同高度：样品上部为 Δ，往下 2Δ，箭头向下直到底部 $n\Delta$。图中没有画出梯度。

  > b) Ultrafast 2D NMR, "1 scan": the same three blocks p–Δ–m, p–2Δ–m, p–$n\Delta$–m, but dashed lines point to different heights of one tube: Δ at the top of the sample, 2Δ below, an arrow down to $n\Delta$ at the bottom. The gradients are not drawn.

考点速记 KEY POINTS

- 常规：每个增量 $\Delta, 2\Delta, \ldots, n\Delta$ 重复一次 → n 次扫描。
  Conventional: each increment $\Delta \ldots n\Delta$ is a repeat → n scans.
- 空间编码靠梯度：t_1 随位置 z 线性变化（背景）。
  Gradients encode space: t_1 varies linearly with z (background).
- 超快：所有增量同时在管内不同位置完成 → 1 次扫描。
  Ultrafast: all increments at once across the tube → 1 scan.
- 代价伏笔：每层样品少 → SNR 低；怕样品不均匀。
  Costs: little sample per slice → low SNR; needs uniformity.

S22 · 超快 COSY 与常规 COSY

S22 · Ultrafast vs conventional COSY

幻灯片要点 / What the slide says

右图是常规 COSY（^{1}H–^{1}H 化学位移相关谱）与超快（UF）COSY 的对比。本例中 UF 的 SNR 看起来与普通 COSY 相当接近。但情况并非总是如此，而且产生伪峰的可能性相当大。

> To the right is an example of a conventional COSY vs UF. Here the UF looks quite similar in SNR to the normal COSY. However, this is not always the case, and there is considerable potential for artifacts.

核对数字：快了多少 / Checking the numbers: how much faster

$$\frac{21\ \text{min}}{0.2\ \text{s}} = \frac{1260\ \text{s}}{0.2\ \text{s}} = 6300$$

由图中两个时间算出：约快 6300 倍 / computed from the two times in the figure: about 6300 times faster

为什么超快谱容易出伪峰 / Why ultrafast spectra are prone to artifacts

背景 超快谱的 F1 不是由 FT 一个个真实增量得到，而是从空间编码中读出（S21）。梯度切换不完美、激发层不够理想、正负读出梯度之间不完全对齐，都会使峰形畸变，或在别处出现"鬼峰"。F1 的谱宽和分辨率还受梯度强度和样品长度限制。所以一张漂亮的超快谱是好结果，不是普遍规律。

> **Background** In an ultrafast spectrum F1 does not come from a Fourier transform over real increments but is read out of the spatial encoding (S21). Imperfect gradient switching, non-ideal slice excitation and slight mismatch between the positive and negative readout gradients distort peak shapes or create "ghost" peaks elsewhere. F1 width and resolution are also limited by gradient strength and sample length. A clean ultrafast spectrum is therefore a good result, not the rule.

背景 用途：当整张二维谱必须在一秒内完成时（例如跟踪快速变化的过程），超快 NMR 几乎是唯一选择；只是要更快，常规采样加 NUS（S19–S20）通常更稳妥。

> **Background** Use: when a whole 2D spectrum must be done within a second (for example, to follow a fast-changing process), ultrafast NMR is nearly the only option; when you merely want it faster, conventional sampling with NUS (S19–S20) is usually safer.

图怎么读：两张三维曲面图 / Reading the figure: two 3D surface plots

- 上："Conventional 2D COSY, 21 min"；下："Ultrafast 2D COSY, 0.2 s"。二维 COSY 画成曲面：两条水平轴为 F_2 (ppm) 与 F_1 (ppm)，刻度 3.5、2.5、1.5（超快图另有 0.5）；高度 = 峰强度。

  > Top: "Conventional 2D COSY, 21 min"; bottom: "Ultrafast 2D COSY, 0.2 s". Each 2D COSY is drawn as a surface: the horizontal axes are F_2 (ppm) and F_1 (ppm), ticks 3.5, 2.5, 1.5 (plus 0.5 in the ultrafast plot); height = peak intensity.

- 绿色底面是基线与噪声，黄、橙色尖峰是信号；最高的峰在对角线两端（约 3.5 与 1.5 ppm 一侧），较小的交叉峰散布在底面上。两图的峰型分布相似，底面同样平坦，这就是"SNR 相近"所指。

  > The green floor is baseline and noise; yellow–orange spikes are signals. The tallest peaks stand at the two ends of the diagonal (towards 3.5 and 1.5 ppm), with smaller cross peaks scattered over the floor. The two plots show a similar peak pattern and an equally flat floor, which is what "similar SNR" refers to.

- 幻灯片没有说明样品是什么化合物，这里不作推测。

  > The slide does not name the compound; no guess is made here.

考点速记 KEY POINTS

- 常规 COSY 21 min vs 超快 0.2 s：约快 6300 倍（计算）。
  COSY 21 min vs UF 0.2 s: ≈ 6300× faster (computed).
- 本例 UF 的 SNR 与常规相近，但并非总是如此。
  Here UF SNR is similar, but not always.
- UF 产生伪峰的可能性相当大。
  UF has considerable potential for artifacts.
- 曲面图：高度 = 强度，对角峰最高。
  Surface plot: height = intensity; diagonal peaks tallest.

S23 · 超快 NMR 的严重缺点（薄荷醇 COSY）

S23 · The severe drawbacks of ultrafast NMR (menthol COSY)

幻灯片要点 / What the slide says

超快 NMR 听起来是个绝妙的主意，但实际上有严重的缺点：SNR 通常很差，需要高浓度；它对样品中的不均匀性（例如浓度梯度）非常敏感；有些峰可能是虚假峰，甚至会缺失。

> While it sounds like an excellent idea, in practice there are severe drawbacks. SNR is usually quite poor, so high concentrations are needed. It is very sensitive to inhomogeneities in the sample (e.g., concentration gradients). Some peaks can be spurious or even missing.

机理：每个缺点从哪里来 / Mechanism: where each drawback comes from

背景 SNR：整张二维谱只扫一次，没有平均；每个"增量"只来自一层样品；梯度读出需要很宽的检测带宽，噪声更大。重复扫描平均（$\mathrm{SNR} \propto \sqrt{N_{\mathrm{scans}}}$）会吃掉时间优势，所以只能提高浓度。

> **Background** SNR: the whole 2D comes from one scan with no averaging; each "increment" comes from one slice; the gradient readout needs a wide bandwidth, which adds noise. Averaging repeats ($\mathrm{SNR} \propto \sqrt{N_{\mathrm{scans}}}$) eats the time advantage, so the remedy is a higher concentration.

背景 不均匀性：t_1 信息存放在位置 z 上。浓度沿管变化，各"增量"的信号量就不同；B_0 不均匀，各层频率就不同。两者都给 t_1 数据加上额外调制，FT 后在 F1 方向产生虚假峰、峰形畸变，或让弱峰消失。

> **Background** Inhomogeneity: t_1 is stored in position z. A concentration gradient gives the "increments" different amounts of signal; poor B_0 homogeneity gives the slices different frequencies. Either adds a false modulation to the t_1 data, so after the transform F1 shows false peaks, distorted shapes or lost weak peaks.

$$\frac{4.5\ \mathrm{min}}{100\ \mathrm{ms}} = \frac{270\ \mathrm{s}}{0.1\ \mathrm{s}} = 2700$$

由图中两个时间算出：约快 2700 倍 / computed from the two times in the figure: about 2700 times faster

图怎么读：薄荷醇的 300 MHz ^{1}H COSY / Reading the figure: 300 MHz ^{1}H COSY of menthol

- 左：Conventional COSY，采集 4.5 分钟；右：Ultrafast COSY，100 毫秒。横轴约 3.5–0.5 ppm，纵轴刻度 1、2、3 ppm。
  > Left: Conventional COSY, 4.5 minutes; right: Ultrafast COSY, 100 msec. Horizontal axes ≈ 3.5–0.5 ppm, vertical ticks 1, 2, 3 ppm.
- 常规谱等高线小而紧凑。超快谱的峰更宽、呈方块状并变形，对角峰（约 3.3、0.9 ppm）尤其大；约 3.3 与 1.8 ppm 之间的一对交叉峰在超快谱中只剩微弱痕迹，即"峰可能缺失"。
  > Conventional contours are small and compact. Ultrafast peaks are broader, box-shaped and distorted, the diagonal peaks (≈ 3.3, 0.9 ppm) especially; a cross-peak pair between ≈ 3.3 and ≈ 1.8 ppm leaves only faint traces, i.e. peaks can go missing.

二维采集 2D acquisition	扫描/增量 Scans	优点 Gain	代价 Cost
常规（线性 t_1） conventional (linear t_1)	N	最稳、SNR 最好 most robust, best SNR	最慢 slowest
NUS（S19–S20） NUS (S19–S20)	$M < N$	省 N/M 倍（例 4×） saves N/M (e.g. 4×)	SNR 降、可能伪峰 lower SNR, artifacts
超快（S21–S23） ultrafast (S21–S23)	1	亚秒级（例 2700–6300×） sub-second (2700–6300×)	SNR 差、怕不均匀、虚假/缺失峰 poor SNR, inhomogeneity, false/missing peaks

考点速记 KEY POINTS

- 缺点一：SNR 差 → 需要高浓度。
  Drawback 1: poor SNR → high concentrations needed.
- 缺点二：对样品不均匀（如浓度梯度）非常敏感。
  Drawback 2: very sensitive to sample inhomogeneity.
- 缺点三：有的峰是虚假峰，有的会缺失。
  Drawback 3: some peaks spurious, some missing.
- 薄荷醇：4.5 min → 100 ms，约快 2700 倍。
  Menthol: 4.5 min → 100 ms, ≈ 2700× faster.

S24 · 魔角旋转方法：走出溶液，转子

S24 · Magic angle spinning approaches: beyond solutions, rotors

幻灯片要点 / What the slide says

前面讲过的 NMR 全部基于溶液态。但 NMR 也能测固体、凝胶以及介于两者之间的几乎任何样品。只要样品含凝胶或固体，通常就需要魔角旋转（magic angle spinning, MAS）。MAS 在转子（rotor）里进行，最常见的转子直径为 $4\ \mathrm{mm}$。

> All the NMR discussed so far was solution-state. NMR can also be done on solids, gels and almost anything in between. Anything containing gels or solids usually needs magic angle spinning (MAS). It is done inside a rotor, most commonly $4\ \mathrm{mm}$ in diameter.

为什么固体和凝胶需要 MAS / Why solids and gels need MAS

背景：溶液中分子快速、各向同性地翻滚，把偶极耦合、化学位移各向异性（CSA）等各向异性相互作用平均为零，所以谱线很窄。固体和凝胶中分子运动受限，这些相互作用随分子相对 B_0 的取向而变；每个取向给出不同频率，叠加起来就是一条很宽的谱线。MAS 用转子的机械旋转代替缺失的分子翻滚（机理见 S25）。

> **Background**: in solution, fast isotropic tumbling averages anisotropic interactions such as dipolar coupling and chemical shift anisotropy (CSA) to zero, so lines are narrow. In solids and gels motion is restricted and these interactions depend on each molecule's orientation to B_0; every orientation gives a different frequency and the sum is one very broad line. MAS replaces the missing tumbling with mechanical rotation of the rotor (mechanism on S25).

读图：转子照片与尺寸表 / Reading the figure: rotor photo and table

照片中五个白色转子立在一枚硬币旁作比例尺，从左到右由细到粗。下表照录图中的数字：

> The photo shows five white rotors beside a coin for scale, thinnest to thickest from left to right. The table copies the figure's numbers:

直径 Diameter / mm	0.7	1.3	3.2	4	7
最高 MAS 频率 Max. MAS frequency / kHz	111	67	24	15	7
体积 Volume / µL	0.59	3	46.7*	99	359.6

* 薄壁转子（thin-wall rotor）的体积。规律：转子越细，最高转速越高，样品越少。0.7 mm 转子能转到 111 kHz，却只装 0.59 µL；7 mm 转子装 359.6 µL，只能转到 7 kHz。4 mm（15 kHz、99 µL）是转速与样品量之间的常用折中。

> * Volume of a thin-wall rotor. Trend: the thinner the rotor, the higher its top spin rate and the less sample it holds. A 0.7 mm rotor reaches 111 kHz but holds only 0.59 µL; a 7 mm rotor holds 359.6 µL but spins only to 7 kHz. The 4 mm rotor (15 kHz, 99 µL) is the usual compromise between spin rate and sample amount.

背景：为什么细转子转得快 / Background: why thin rotors spin faster

限制转速的是转子外壁的线速度 $v = \pi d \nu_{\mathrm{rot}}$（材料强度的极限）。按表中数字计算，五种转子的 v 都在约 $150\text{–}270\ \mathrm{m\,s^{-1}}$，所以直径越小，允许的 ν_{rot} 越高。

> The limit is the surface speed of the rotor wall, $v = \pi d \nu_{\mathrm{rot}}$ (set by material strength). From the table, all five rotors run at about $150\text{–}270\ \mathrm{m\,s^{-1}}$, so a smaller diameter allows a higher ν_{rot}.

考点速记 KEY POINTS

- 含凝胶或固体的样品通常需要 MAS。
  Gels or solids usually require MAS.
- 转子越细：转速越高、体积越小（0.7 mm：111 kHz，0.59 µL）。
  Thinner rotor: faster, less sample (0.7 mm: 111 kHz, 0.59 µL).
- 最常用 4 mm 转子：最高 15 kHz，99 µL。
  Most common: 4 mm rotor, up to 15 kHz, 99 µL.
- 易错：3.2 mm 的 46.7 µL 是薄壁转子。
  Trap: 46.7 µL (3.2 mm) refers to a thin-wall rotor.

S25 · 魔角：54.74° 从何而来

S25 · The magic angle: where 54.74° comes from

幻灯片要点 / What the slide says

魔角（magic angle）与磁场方向偏离约 54.74°。幻灯片称它是"一阶勒让德多项式的根"（见下方编者注）。以这个角度旋转，可以抵消大量原本会造成谱线展宽的相互作用。

> The magic angle is offset about 54.74° from the magnetic field. The slide calls it "the root of a first order Legendre polynomial" (see the editor's note below). Spinning at this angle cancels a large variety of interactions that otherwise broaden the spectrum.

为什么偏偏是这个角度 / Why exactly this angle

背景：偶极耦合、CSA 等相互作用的频率都含取向因子 $3\cos^2\theta_\mathrm{B}-1$。样品绕一根与 B_0 成 θ 角的轴快速旋转时，时间平均后这个因子变成两项之积：

> **Background**: dipolar couplings, CSA and similar interactions all scale with the orientation factor $3\cos^2\theta_\mathrm{B}-1$. When the sample spins fast about an axis at angle θ to B_0, the time average of this factor becomes a product of two terms:

$$\overline{3\cos^2\theta_\mathrm{B}-1}=\frac{1}{2}\left(3\cos^2\theta-1\right)\left(3\cos^2\beta-1\right)$$

θ_B：相互作用轴与 B_0；θ：转轴与 B_0；β：相互作用轴与转轴

/ θ_B: interaction axis to B_0; θ: rotor axis to B_0; β: interaction axis to rotor axis

β 随晶粒取向而变，但第一项只取决于转轴方向，对所有取向都相同。让第一项为零，所有取向的各向异性项就同时消失，只剩各向同性化学位移和 J 耦合，宽峰收窄为尖峰：

> β differs from crystallite to crystallite, but the first term depends only on the rotor axis and is the same for all of them. Make it zero and the anisotropic term vanishes for every orientation at once; only the isotropic shift and J couplings remain, and broad lines collapse to sharp ones:

$$P_2(\cos\theta)=\frac{1}{2}\left(3\cos^2\theta-1\right)=0\;\Rightarrow\;\cos^2\theta=\frac{1}{3}$$

$$\theta_\mathrm{m}=\arccos\frac{1}{\sqrt{3}}\approx 54.74^\circ$$

条件：转速须大于相互作用的宽度（以 Hz 计），否则平均不完全，留下旋转边带（S26–S27）；很强的 $^1\mathrm{H}$–$^1\mathrm{H}$ 偶极耦合即使高速旋转也难以完全去除（S29、S31）。

> Condition: the spin rate must exceed the interaction's width in Hz; otherwise averaging is incomplete and spinning sidebands remain (S26–S27). Very strong $^1\mathrm{H}$–$^1\mathrm{H}$ dipolar couplings are hard to remove even at high rates (S29, S31).

读图 / Reading the figure

左：MAS 探头照片，白色竖箭头为 B_0，装样品（红色）的转子倾斜安装。右：示意图，竖直的 B_0 与转轴（rotation axis）夹角 θ，标注 $\Theta=54.74^\circ$ 与 $\nu_\mathrm{rot}\approx 1$–$15\ \mathrm{kHz}$；蓝色弯箭头表示绕转轴旋转；转子两侧的红色虚线是梯度线圈（gradient coil）。

> Left: photo of a MAS probe; the white vertical arrow is B_0 and the rotor holding the sample (red) is mounted at a tilt. Right: schematic with vertical B_0 at angle θ to the rotation axis, labelled $\Theta=54.74^\circ$ and $\nu_\mathrm{rot}\approx 1$–$15\ \mathrm{kHz}$; the curved blue arrow shows spinning about the axis; the red dashed lines on both sides of the rotor are the gradient coil.

编者注 / Editor's note

幻灯片原文写"一阶勒让德多项式的根"。准确地说，魔角是**二阶**勒让德多项式 $P_2(\cos\theta)=\frac{1}{2}(3\cos^2\theta-1)$ 的根；一阶 $P_1(\cos\theta)=\cos\theta$ 的根在 90°。

> The slide says "root of a first order Legendre polynomial". Strictly, the magic angle is a root of the **second**-order Legendre polynomial $P_2(\cos\theta)=\frac{1}{2}(3\cos^2\theta-1)$; the first-order $P_1(\cos\theta)=\cos\theta$ has its root at 90°.

考点速记 KEY POINTS

- $\theta_\mathrm{m}=\arccos(1/\sqrt{3})\approx 54.74^\circ$，从 B_0 量起。
  $\theta_\mathrm{m}=\arccos(1/\sqrt{3})\approx 54.74^\circ$, measured from B_0.
- 易错：是二阶 P_2 的根，不是一阶。
  Trap: it is a root of P_2, not P_1.
- 魔角使 $3\cos^2\theta-1=0$，各向异性展宽平均为零。
  At θ_m, $3\cos^2\theta-1=0$: anisotropic broadening averages out.
- 图中 $\nu_\mathrm{rot}\approx 1$–$15\ \mathrm{kHz}$；须快于展宽宽度。
  Figure: $\nu_\mathrm{rot}\approx 1$–$15\ \mathrm{kHz}$; must exceed the linewidth.

S26 · MAS 转速（一）：从粉末谱到旋转边带

S26 · The MAS spin rate (1): from powder spectrum to sidebands

幻灯片要点 / What the slide says

转速（spin rate, ν_{rot}）是让 MAS 方法区别于溶液方法的关键参数之一。右图展示不同转速下的谱。转速为 0 时是"粉末"谱（powder spectrum），通常无法使用；转速增大，边带（sidebands）彼此离得越来越远。

> The spin rate (ν_{rot}) is one of the key parameters that make MAS different from solution NMR. The figure shows spectra at different spin rates. At 0 the result is a "powder" spectrum and usually unusable; as the rate increases, the sidebands get further apart.

机理：边带为什么出现在转速的整数倍 / Mechanism: why sidebands sit at multiples of the spin rate

背景：静止的固体里每个晶粒取向给出不同频率，所有取向叠加成宽而有特征形状的粉末线形。慢速旋转时，每个晶粒的频率随转子转动作周期性变化，周期为 $1/\nu_{\mathrm{rot}}$。周期信号的傅里叶变换是一组间隔为 ν_{rot} 的离散谱线，于是宽峰被"切成"一排旋转边带（spinning sidebands）：

> **Background**: in a static solid every crystallite orientation gives a different frequency, and all of them add up to a broad powder pattern of characteristic shape. Under slow spinning, each crystallite's frequency oscillates with the rotor period $1/\nu_{\mathrm{rot}}$. The Fourier transform of a periodic signal is a set of discrete lines spaced by ν_{rot}, so the broad line is "chopped" into a comb of spinning sidebands:

$$\nu_{\mathrm{SSB}} = \nu_{\mathrm{iso}} \pm k\,\nu_{\mathrm{rot}}, \qquad k = 1, 2, \ldots$$

背景公式，即幻灯片所说"边带相距等于转速" / background form of the slide's "spaced by the spin rate"

转速越高，边带越稀、越弱，强度向各向同性位置 ν_{iso} 的中心峰集中；当 ν_{rot} 远大于粉末线形的宽度时，几乎只剩中心峰。

> The faster the spinning, the sparser and weaker the sidebands, and intensity gathers into the centreband at the isotropic position ν_{iso}; once ν_{rot} far exceeds the powder width, nearly only the centreband is left.

读图 / Reading the figure

纵轴：样品旋转频率（kHz），自下而上 0.0、0.1、0.2、0.3、0.4、0.6、0.8、1.0、1.5、2.0；横轴：共振频率（kHz），−4 到 4。

> Vertical axis: sample spinning frequency (kHz), bottom to top 0.0, 0.1, 0.2, 0.3, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0. Horizontal axis: resonance frequency (kHz), −4 to 4.

- **0.0 kHz**：一条连续的宽粉末线形，约从 −2 到 +3 kHz，在 0 稍左处有尖顶。
  > **0.0 kHz**: one continuous broad powder pattern from about −2 to +3 kHz, peaking just left of 0.
- **0.1–0.4 kHz**：密集的边带梳，包络仍是粉末线形的形状。
  > **0.1–0.4 kHz**: a dense comb of sidebands whose envelope still traces the powder pattern.
- **0.6–1.0 kHz**：边带变稀，能数出单根谱线。
  > **0.6–1.0 kHz**: sidebands thin out; single lines can be counted.
- **1.5–2.0 kHz**：0 处一根强峰，只剩几根弱边带；2.0 kHz 时弱边带约在 ±2 与 ±4 kHz。
  > **1.5–2.0 kHz**: one strong line at 0 and a few weak sidebands; at 2.0 kHz they sit near ±2 and ±4 kHz.

考试提醒 / Exam reminder

换一个转速重测：边带跟着转速移动，中心峰不动，由此可分辨真峰与边带（**背景**，常用技巧）。

> Re-run at a different spin rate: sidebands move with the rate while the centreband stays put, which tells true peaks from sidebands (**background**, a standard trick).

考点速记 KEY POINTS

- 0 Hz：粉末谱，通常无法使用。
  0 Hz: powder spectrum, usually unusable.
- 转速越高：边带越远、越弱，强度归中心峰。
  Faster: sidebands further apart, weaker; intensity to centreband.
- 边带位置：$\nu_{\mathrm{iso}} \pm k\,\nu_{\mathrm{rot}}$。
  Sidebands at $\nu_{\mathrm{iso}} \pm k\,\nu_{\mathrm{rot}}$.
- 换转速：边带移动，中心峰不动。
  Change the rate: sidebands move, centreband does not.

S27 · MAS 转速（二）：旋转边带这种伪峰

S27 · The MAS spin rate (2): spinning sidebands as artifacts

幻灯片要点 / What the slide says

大多数 MAS 谱都有旋转边带，它是一种伪峰（artifact）。边带出现在与母峰相距等于转速的位置，在很大的峰（例如水）旁最明显。上图转速为 $500\ \mathrm{Hz}$，边带相当烦人；而在 $2500\ \mathrm{Hz}$ 时问题就小得多。

> Most MAS spectra have spinning sidebands, which are a type of artifact. They occur at a distance equal to the spin rate and are prominent next to very large peaks such as water. The top figure has a spin rate of $500\ \mathrm{Hz}$ and the sidebands are quite annoying; at $2500\ \mathrm{Hz}$ they are much less of an issue.

为什么巨大的峰（水）最糟 / Why huge peaks such as water are worst

背景：边带强度与母峰强度成正比。即使边带只占母峰的一小部分，水峰比代谢物峰大好几个数量级，它的边带仍可与真实信号一样高甚至更高。在二维谱里，每根水峰边带还会沿间接维拖出一条竖直条纹，盖住该 $^{1}\mathrm{H}$ 位移处的交叉峰。

> **Background**: sideband intensity scales with the parent peak. Even if a sideband is a small fraction of its parent, water is orders of magnitude larger than metabolite peaks, so its sidebands can be as tall as real signals or taller. In a 2D spectrum each water sideband also smears a vertical streak along the indirect dimension and hides cross peaks at that $^{1}\mathrm{H}$ shift.

为什么提高转速有用：边带与水峰的距离等于 ν_{rot}。$500\ \mathrm{Hz}$ 时一串边带都落在谱宽之内；$2500\ \mathrm{Hz}$ 时第一对边带已被推到谱的两端，中间区域变干净。

> Why a faster rate helps: sidebands sit ν_{rot} away from water. At $500\ \mathrm{Hz}$ a whole series falls inside the spectral window; at $2500\ \mathrm{Hz}$ the first pair is already pushed to the two edges, leaving the middle clean.

$$\nu_{\mathrm{SSB}} = \nu_{\mathrm{water}} \pm k\,\nu_{\mathrm{rot}}, \qquad k = 1, 2, \ldots$$

500 Hz：一串边带落在谱内；2500 Hz：只剩谱边一对 / 500 Hz: a series inside the window; 2500 Hz: one pair at the edges

读图：两张 HSQC / Reading the figure: two HSQC spectra

横轴为 $^{1}\mathrm{H}$ 化学位移（10.5 到 –0.5 ppm），纵轴为 $^{13}\mathrm{C}$ 化学位移（10 到 190 ppm，向下增大）。

> Horizontal axis: $^{1}\mathrm{H}$ chemical shift (10.5 to –0.5 ppm); vertical axis: $^{13}\mathrm{C}$ chemical shift (10 to 190 ppm, increasing downward).

- **上图（500 Hz）**：水峰（约 4.7 ppm）两侧有一排竖直条纹贯穿整个 $^{13}\mathrm{C}$ 轴，间距等于图中"500 Hz"标尺。
  > **Top (500 Hz)**: a row of vertical streaks on both sides of water (about 4.7 ppm) runs through the whole $^{13}\mathrm{C}$ axis, spaced like the "500 Hz" scale bar.
- **下图（2500 Hz）**："2500 Hz"标尺从水峰量到左侧谱边附近的一条条纹；右侧约 0 ppm 处还有一条对称的条纹，中间再无边带。
  > **Bottom (2500 Hz)**: the "2500 Hz" bar runs from water to one streak near the left edge; a matching streak sits near 0 ppm on the right, with no sidebands in between.
- 两图中真正的交叉峰相同：主要在 $^{1}\mathrm{H}$ 1–3 ppm / $^{13}\mathrm{C}$ 10–40 ppm 和 $^{1}\mathrm{H}$ 3.5–5.5 ppm / $^{13}\mathrm{C}$ 50–80 ppm。
  > The real cross peaks are the same in both: mainly $^{1}\mathrm{H}$ 1–3 / $^{13}\mathrm{C}$ 10–40 ppm and $^{1}\mathrm{H}$ 3.5–5.5 / $^{13}\mathrm{C}$ 50–80 ppm.

考点速记 KEY POINTS

- 旋转边带是伪峰，距母峰 $k\,\nu_{\mathrm{rot}}$。
  Spinning sidebands are artifacts, $k\,\nu_{\mathrm{rot}}$ from the parent.
- 500 Hz：一串条纹；2500 Hz：只剩谱边一对。
  500 Hz: a series of streaks; 2500 Hz: one pair at the edges.
- 母峰越大边带越强：水是最大来源。
  Bigger parent, bigger sidebands: water is the worst.
- 提高转速把边带推出感兴趣区。
  Faster spinning pushes sidebands out of the region of interest.

S28 · MAS 转速（三）：样品和转子决定能转多快

S28 · The MAS spin rate (3): the sample and rotor set the limit

幻灯片要点 / What the slide says

- 转速取决于样品和转子尺寸。
  > The spin rate depends on the sample and the rotor size.
- 4 mm 转子最常见，测固体时可用到约 15 kHz；液体/凝胶通常低得多。
  > 4 mm rotors are the most common and can be used to about 15 kHz for solids; for liquids/gels the rate is usually much lower.
- 细胞在约 3500 Hz 时容易破裂，组织等样品必须考虑这一点。
  > Cells tend to break apart at about 3500 Hz, which must be considered for tissue samples and the like.
- 一般来说转速越快越好，但越难实现。
  > Generally, faster spin rates are better but harder to do.

为什么细胞限制转速 / Why cells limit the spin rate

背景：转子里的样品受离心加速度 $a = (2\pi\nu_{\mathrm{rot}})^2 r$，随转速的平方增长。按下式估算，3500 Hz 时约为重力加速度的 7×10^4 倍，足以把细胞压向转子壁、撕裂膜结构，所以活细胞和组织只能慢转。凝胶和半固体中分子本身仍在运动，残余的各向异性展宽小，慢转也够用。

> **Background**: the sample feels a centrifugal acceleration $a = (2\pi\nu_{\mathrm{rot}})^2 r$, which grows with the square of the rate. The estimate below gives about 7×10^4 times gravity at 3500 Hz, enough to press cells against the rotor wall and tear membranes, so living cells and tissue must spin slowly. In gels and semi-solids the molecules still move, residual anisotropic broadening is small, and slow spinning suffices.

$$a = (2\pi\nu_{\mathrm{rot}})^2 r \approx (2\pi \times 3500\ \mathrm{s^{-1}})^2 \times 1.5 \times 10^{-3}\ \mathrm{m} \approx 7 \times 10^5\ \mathrm{m\,s^{-2}}$$

背景估算，假设 $r = 1.5$ mm / background estimate, assuming $r = 1.5$ mm

读图：Tb 配合物的 ^{1}H 谱 / Reading the figure: ^{1}H spectra of a Tb complex

右上结构：中心 Tb 与三个含 N 的吡啶环配体配位，每个配体带两个羧酸根，整体标注 3– 与 3 Na^+。横轴 $\delta(^1\mathrm{H})$ 从 +1500 到 –1500 ppm，三条谱的转速与射频功率成对标注：

> Structure (top right): a central Tb bound by three pyridine-ring ligands, each carrying two carboxylates, labelled 3– with 3 Na^+. The axis $\delta(^1\mathrm{H})$ runs from +1500 to –1500 ppm; each trace is labelled with its spin rate and RF power:

MAS	RF 功率 RF power	谱图 Trace
111.111 kHz	435 kHz	蓝：约 +700 到 –600 ppm 间一组分开的尖峰 blue: a set of separate peaks between about +700 and –600 ppm
60 kHz	200 kHz	红：峰变宽、相互重叠、强度低 red: broader, overlapping, weaker peaks
30 kHz	100 kHz	绿：几乎是一条平线 green: almost a flat line

这张图同时说明"更快更好"和"更难做"：只有 111.111 kHz 才分辨出峰，而它需要 435 kHz 的射频功率。**背景**：Tb^{3+} 是顺磁离子，使 ^{1}H 位移跨越上千 ppm、谱线极宽，所以需要极高转速。

> The figure shows both "faster is better" and "harder to do": only 111.111 kHz resolves the peaks, and that needs 435 kHz of RF power. **Background**: Tb^{3+} is paramagnetic, spreading ^{1}H shifts over thousands of ppm with very broad lines, hence the need for extreme spin rates.

考点速记 KEY POINTS

- 4 mm 测固体约 15 kHz；液体/凝胶低得多。
  4 mm: about 15 kHz for solids; much lower for liquids/gels.
- 离心加速度 $\propto \nu_{\mathrm{rot}}^2$。
  Centrifugal acceleration $\propto \nu_{\mathrm{rot}}^2$.
- 细胞约 3500 Hz 破裂：组织样品限速。
  Cells break at about 3500 Hz: tissue limits the rate.
- 图：111.111 kHz MAS 配 435 kHz RF 才分辨出峰。
  Figure: peaks resolve only at 111.111 kHz MAS, 435 kHz RF.

S29 · MAS 转速（四）：超快 MAS 与小转子

S29 · The MAS spin rate (4): very fast MAS and small rotors

幻灯片要点 / What the slide says

- 很高的转速有助于提高分辨率，这对固体尤其关键。
  > Very fast spin rates help improve resolution, which is especially critical for solids.
- 更快的转速要用更小的转子。
  > Smaller rotors are used for faster spin rates.
- 小转子既昂贵，操作起来又非常麻烦。
  > They are both expensive and very annoying to work with.
- 作者找到的最高值是 $150\ \mathrm{kHz}$："听起来不快，但在 $0.5\ \mathrm{mm}$ 转子里约为 600 万 RPM"（见编者注）。
  > The highest the author found was $150\ \mathrm{kHz}$: "while this does not sound fast it is ~6 million RPM in a $0.5\ \mathrm{mm}$ rotor" (see the editor's note).

为什么更快 → 分辨率更高，且必须用小转子 / Why faster means better resolution, and why it needs small rotors

背景：只有当 ν_{rot} 大于相互作用的宽度时，MAS 才能把它完全平均（S25）。固体中的 $^{1}\mathrm{H}$ 有很强的同核偶极耦合网络，需要几十到上百 kHz 才能明显压窄谱线；转速每提高一档，残余线宽就再缩小一些。而转子外壁线速度 $v = \pi d\,\nu_{\mathrm{rot}}$ 有上限（S24），要转得更快只能让直径更小。

> **Background**: MAS fully averages an interaction only when ν_{rot} exceeds its width (S25). $^{1}\mathrm{H}$ in solids forms a strong homonuclear dipolar network that needs tens to over a hundred kHz before lines narrow clearly; each step up in rate shrinks the residual linewidth further. Because the wall speed $v = \pi d\,\nu_{\mathrm{rot}}$ is capped (S24), faster spinning is only possible with a smaller diameter.

为什么"昂贵又麻烦"（**背景**）：亚毫米转子只装亚微升样品（S24 表：0.7 mm 仅 0.59 µL），装样困难，转子易损，信号也随样品量变少。

> Why "expensive and annoying" (**background**): sub-millimetre rotors hold sub-microlitre samples (S24 table: 0.7 mm holds only 0.59 µL), are hard to pack and easy to damage, and the signal shrinks with the sample amount.

单位换算：150 kHz 是多少 rpm / Unit check: 150 kHz in rpm

$$\nu_{\mathrm{rot}} = 150\ \mathrm{kHz} = 1.5\times10^{5}\ \mathrm{s}^{-1} \;\Rightarrow\; 1.5\times10^{5}\times 60\ \mathrm{s\,min}^{-1} = 9.0\times10^{6}\ \mathrm{rpm}$$

$$1\ \mathrm{Hz} = 60\ \mathrm{rpm}\ /\ 1\ \mathrm{Hz} = 60\ \mathrm{rpm}$$

编者注 / Editor's note

幻灯片写"约 600 万 RPM"。按 $1\ \mathrm{Hz} = 60\ \mathrm{rpm}$ 换算，150 kHz 是 9.0×10^{6} rpm（900 万）；600 万 rpm 对应的是 100 kHz。

> The slide says "~6 million RPM". With $1\ \mathrm{Hz} = 60\ \mathrm{rpm}$, 150 kHz is 9.0×10^{6} rpm (9 million); 6 million rpm would be 100 kHz.

考点速记 KEY POINTS

- 超快 MAS 提高分辨率，对固体最关键。
  Very fast MAS improves resolution; most critical for solids.
- 更快 ⇒ 更小的转子：昂贵、难操作。
  Faster needs smaller rotors: expensive and fiddly.
- 最高 150 kHz，0.5 mm 转子。
  Record cited: 150 kHz in a 0.5 mm rotor.
- 易错：150 kHz = 9.0×10^{6} rpm，不是 600 万。
  Trap: 150 kHz = 9.0×10^{6} rpm, not 6 million.

S29（续）· 读图：转子尺寸与 ^{13}C、^{1}H 谱

S29 (cont.) · Reading the figure: rotor sizes, ^{13}C and ^{1}H spectra

上部：转子与转速 / Top: rotors and spin rates

红色箭头"MAS (kHz)"上标 18、24、35、60，对应下方 4、3.2、2.5、1.3 mm 四个转子（rotor (mm)）。右上是一张 ^{1}H 谱，$\delta(^{1}H)$ 10–6 ppm，有许多分开的尖峰（图上未标转速）。

> The red arrow "MAS (kHz)" reads 18, 24, 35, 60 above four rotors of 4, 3.2, 2.5 and 1.3 mm ("rotor (mm)"). Top right is a ^{1}H spectrum, $\delta(^{1}H)$ 10–6 ppm, with many separate sharp lines (no spin rate is given for it).

转子 Rotor / mm	4	3.2	2.5	1.3
MAS / kHz（S29 图 figure）	18	24	35	60
最高 MAS / kHz（S24 表 table）	15	24	—	67

(a) ^{13}C：CO 与 Cα / (a) ^{13}C: CO and Cα

横轴 $\delta(^{13}C)$ 自 250 ppm 向右递减（刻度 250、200、150、100）；CO 峰在 200 与 150 之间，Cα 峰在 100 以右。自下而上：**static** 两个宽粉末驼峰；**1 kHz、2 kHz** 密集的边带梳；**4 kHz** 只剩几根间隔 4 kHz 的边带；**10 kHz** CO 与 Cα 都是尖峰，CO 旁只剩一两根小边带。

> Axis $\delta(^{13}C)$ decreasing from 250 ppm to the right (ticks 250, 200, 150, 100); the CO peak lies between 200 and 150, the Cα peak right of 100. Bottom to top: **static**, two broad powder humps; **1 and 2 kHz**, dense sideband combs; **4 kHz**, a few sidebands 4 kHz apart; **10 kHz**, sharp CO and Cα lines with only one or two small sidebands beside CO.

(b) ^{1}H：H^N 与 $H\alpha_1$、$H\alpha_2$ / (b) ^{1}H: H^N and $H\alpha_1$, $H\alpha_2$

横轴 $\delta(^{1}H)$ 20 到 0 ppm，自下而上 15、25、40、60、80、111 kHz。15 kHz 只有一个宽包；转速升高，H^N 峰越来越尖，Hα 区先出现肩峰；到 80 与 111 kHz，$H\alpha_1$ 与 $H\alpha_2$ 分成两个独立的峰。

> Axis $\delta(^{1}H)$ 20 to 0 ppm; bottom to top 15, 25, 40, 60, 80, 111 kHz. At 15 kHz there is only one broad hump; as the rate rises the H^N peak sharpens and a shoulder appears in the Hα region; at 80 and 111 kHz $H\alpha_1$ and $H\alpha_2$ split into two separate peaks.

对比 (a) 与 (b) / Comparing (a) and (b)

^{13}C 在 10 kHz 已经很尖，^{1}H 要到 80–111 kHz 才分辨出 $H\alpha_1$/$H\alpha_2$。**背景**：^{13}C 的展宽主要是 CSA，量级为几 kHz；^{1}H 旋磁比大、丰度高，同核偶极耦合强得多，所以对转速的要求高得多。

> ^{13}C is already sharp at 10 kHz, while ^{1}H needs 80–111 kHz to resolve $H\alpha_1$/$H\alpha_2$. **Background**: ^{13}C broadening is mainly CSA of a few kHz; ^{1}H has a large gyromagnetic ratio and high abundance, so its homonuclear dipolar coupling is far stronger and demands much faster spinning.

注意 / Caution

S29 图与 S24 表的转速不完全一致（4 mm：18 对 15 kHz；1.3 mm：60 对 67 kHz），来自不同来源的转子。答题时引用题目所指的那张图。

> The S29 figure and the S24 table disagree slightly (4 mm: 18 vs 15 kHz; 1.3 mm: 60 vs 67 kHz); the rotors come from different sources. Quote the figure the question refers to.

考点速记 KEY POINTS

- 图：4 / 3.2 / 2.5 / 1.3 mm → 18 / 24 / 35 / 60 kHz。
  Figure: 4 / 3.2 / 2.5 / 1.3 mm → 18 / 24 / 35 / 60 kHz.
- ^{1}H：80–111 kHz 才分开 $H\alpha_1$/$H\alpha_2$。
  ^{1}H: $H\alpha_1$/$H\alpha_2$ split only at 80–111 kHz.
- ^{13}C：10 kHz 时 CO、Cα 已成尖峰。
  ^{13}C: CO and Cα sharp by 10 kHz.
- ^{1}H 偶极耦合强 ⇒ 需更高转速。
  Strong ^{1}H dipolar coupling ⇒ faster spinning needed.

S30 · HR-MAS NMR：凝胶与半固体

S30 · HR-MAS NMR: gels and semi-solids

幻灯片要点 / What the slide says

- MAS 是涉及旋转的统称，但它有多种变体。
  > MAS is a general term that involves spinning, but there are several variations of MAS.
- HR-MAS（高分辨魔角旋转，high-resolution magic angle spinning）是最不"固体"的一种，主要用于凝胶。
  > HR-MAS (high-resolution magic angle spinning) is the least "solid" one and is mainly used for gels.
- 它通常用较低转速，能研究凝胶和半固体，但不能研究真正的固体。
  > It typically uses slower spin rates and can study gels and semi-solids, but not true solids.
- 通常配有梯度。
  > It is normally equipped with a gradient.
- 不旋转与旋转的差别非常大！
  > The difference between not spinning and spinning is drastic!

为什么慢转就够、为什么需要梯度 / Why slow spinning suffices, and why a gradient

背景：凝胶和半固体中分子仍有较大的局部运动，各向异性相互作用已被部分平均，剩下的展宽主要来自残余偶极耦合和样品内部不同区域的磁化率差异，几 kHz 的 MAS 就能消除，谱线接近溶液水平，这就是"高分辨"的含义。慢转也保护了组织和细胞（S28 的 3500 Hz 限制）。真固体的展宽大得多，需要高转速和大功率，这是 HR-MAS 探头做不到的（S32）。

> **Background**: in gels and semi-solids the molecules keep substantial local motion, so anisotropic interactions are already partly averaged. The remaining broadening comes mainly from residual dipolar couplings and magnetic susceptibility differences across the sample, which a few kHz of MAS removes, giving near-solution linewidths; hence "high resolution". Slow spinning also protects tissue and cells (the 3500 Hz limit of S28). True solids are broadened far more and need fast spinning and high power, which an HR-MAS probe cannot deliver (S32).

梯度（**背景**）：有了脉冲场梯度，就能在 MAS 样品上运行溶液 NMR 的梯度实验，例如梯度溶剂峰抑制和扩散编辑（S33）。

> The gradient (**background**): pulsed field gradients let solution-NMR gradient experiments run on a MAS sample, such as gradient solvent suppression and diffusion editing (S33).

读图 / Reading the figure

三条谱共用横轴 10 到 –5 ppm（图上未标核种）。**Static**（静态，绿）：约 5.3 与 1.9 ppm 处两个宽驼峰。**Static swollen**（静态溶胀，粉）：同样两个宽驼峰，几乎没有变窄，说明仅溶胀样品不够。**4 kHz HR-MAS**（蓝）：同一区域变成一组尖峰，最高峰约在 1.9 ppm，旁边约 1.2 ppm 和约 5.3 ppm 也有尖峰，6.5–7 ppm 和 2.5–4 ppm 还有小峰。

> All three traces share an axis from 10 to –5 ppm (the nucleus is not labelled). **Static** (green): two broad humps near 5.3 and 1.9 ppm. **Static swollen** (pink): the same two broad humps, barely narrower, so swelling alone is not enough. **4 kHz HR-MAS** (blue): the same region becomes a set of sharp lines, the tallest near 1.9 ppm, with sharp lines near 1.2 and 5.3 ppm and small ones at 6.5–7 and 2.5–4 ppm.

考点速记 KEY POINTS

- HR-MAS：最不"固体"的 MAS，测凝胶、半固体。
  HR-MAS: least "solid" MAS, for gels and semi-solids.
- 易错：HR-MAS 不能测真固体。
  Trap: HR-MAS cannot measure true solids.
- 转速较低（图中 4 kHz），通常带梯度。
  Slow spinning (4 kHz in the figure), usually with a gradient.
- 静态溶胀仍是宽峰：要旋转才变尖。
  Static swollen stays broad: spinning sharpens it.

S31 · 固体 NMR

S31 · Solid-state NMR

幻灯片要点 / What the slide says

- 固体 NMR（solid-state NMR）与溶液 NMR 很不相同。
  > Solid-state NMR is quite different from solution NMR.
- 通常即使在旋转，分辨率也很差。
  > Resolution is typically very poor even when spinning.
- 提高转速有帮助，固体用的转速也更高；但分辨率很少能达到溶液态的水平。
  > A higher spin rate helps, and spin rates are higher for solids; still, resolution rarely reaches that of the solution state.
- 谱往往很宽，弛豫很快，需要大功率脉冲。
  > Spectra tend to be very broad, relaxation is fast, and high-power pulses are needed.

为什么宽、为什么弛豫快、为什么要大功率 / Why broad, why fast relaxation, why high power

背景：**宽**：刚性固体中 $^1\mathrm{H}$ 之间的偶极耦合可达几十 kHz，常规转速只能部分平均；无定形固体中各分子的局部环境不同，位移本身也是一个分布。**弛豫快**：这里指横向弛豫 T_2 很短，FID 在极短时间内衰减，对应宽线 $\Delta\nu_{1/2} = 1/(\pi T_2)$。**大功率**：要在极短的脉冲内均匀激发很宽的谱，并用去耦压制强偶极耦合，射频场 B_1 必须很强。

> **Background**: **broad**, because $^1\mathrm{H}$–$^1\mathrm{H}$ dipolar couplings in a rigid solid reach tens of kHz and ordinary spin rates average them only partly, and because in amorphous solids each molecule's surroundings differ, so the shift itself is a distribution. **Fast relaxation**: here a very short transverse relaxation T_2; the FID dies almost at once, matching a broad line $\Delta\nu_{1/2} = 1/(\pi T_2)$. **High power**: exciting a very wide spectrum evenly with very short pulses, and decoupling strong dipolar couplings, both need a strong RF field B_1.

$$\Delta\nu_{1/2} = \frac{1}{\pi T_2}$$

背景关系式（洛伦兹线形）：T_2 越短，谱线越宽 / background relation (Lorentzian line): shorter T_2, broader line

读图：有机固体的 $^1\mathrm{H}$ NMR / Reading the figure: $^1\mathrm{H}$ NMR of organic solids

图题"¹H NMR of Organic Solids"，横轴 100 到 −60 ppm；结构为两个苯并稠合的六元环单元经 C=C 双键相连。

> Title "¹H NMR of Organic Solids"; axis 100 to −60 ppm; the structure is two benzo-fused six-membered-ring units joined by a C=C double bond.

- **Static Solid**（静态固体）：一个宽包几乎覆盖整个轴（约 +70 到 −55 ppm），最高点在 7 ppm 附近。
  > **Static Solid**: one broad hump covering almost the whole axis (about +70 to −55 ppm), highest near 7 ppm.
- **Solid 12 kHz MAS**：中心峰在约 7 ppm，宽度仍有若干 ppm；两侧约 +30、−20、+55、−40 ppm 处有较小的驼峰（**背景解读**：多为旋转边带）。
  > **Solid 12 kHz MAS**: a central peak near 7 ppm, still several ppm wide, with smaller humps near +30, −20, +55 and −40 ppm (**background reading**: mostly spinning sidebands).
- **Liquid**（液体）：约 7 与 2 ppm 两条极窄的线。
  > **Liquid**: two very narrow lines near 7 and 2 ppm.

结论：12 kHz MAS 把宽包压成一个主峰，但与液体谱相比仍差得很远，这正是"分辨率很少达到溶液态"。

> Conclusion: 12 kHz MAS squeezes the broad hump into one main peak, yet it is still far from the liquid spectrum; this is what "resolution rarely reaches the solution state" means.

考点速记 KEY POINTS

- 固体：即使 MAS，分辨率也很少达到溶液水平。
  Solids: even with MAS, resolution rarely matches solution.
- 固体的转速高于凝胶/溶液样品。
  Solids are spun faster than gels or solutions.
- 谱宽、弛豫快、需大功率脉冲。
  Broad spectra, fast relaxation, high-power pulses.
- 图：12 kHz MAS 仍是数 ppm 宽的峰；液体是两条尖线。
  Figure: 12 kHz MAS still gives ppm-wide peaks; liquid gives two sharp lines.

S32 · CMP NMR：一个探头测所有相

S32 · CMP NMR: one probe for every phase

幻灯片要点 / What the slide says

- CMP NMR（综合多相 NMR，comprehensive multiphase NMR）是较新的发明，大致是在 HR-MAS 基础上加上测固体的能力。
  > Comprehensive multiphase (CMP) NMR is a relatively recent invention that roughly takes HR-MAS and adds the ability to do solids.
- 主要挑战是功率承受能力（power handling），所以 CMP 探头能承受比 HR-MAS 大得多的射频功率。
  > The main challenge is power handling, so CMP probes can take much more RF power than HR-MAS probes.
- 一个探头就能研究样品的所有状态，很方便。
  > All states of a sample can be studied with one probe, which is convenient.
- 转速更接近 HR-MAS，探头带梯度。
  > Spin rates are closer to HR-MAS, and these probes have gradients.

CMP 如何把溶液、凝胶、固体放进一个探头 / How CMP combines solution, gel and solid in one probe

1. **梯度**（继承自 HR-MAS）：在同一转子里做基于扩散的 $^{1}\mathrm{H}$ 编辑，分出溶液相和凝胶相（S33）。
   > **Gradients** (inherited from HR-MAS): diffusion-based $^{1}\mathrm{H}$ editing in the same rotor separates the solution and gel phases (S33).
2. **高功率承受能力**：**背景**，交叉极化（CP）和大功率 $^{1}\mathrm{H}$ 去耦都需要强射频场，有了它才能做固相的 $^{13}\mathrm{C}$ CP-MAS（S33）。
   > **High power handling**: **background**, cross polarization (CP) and high-power $^{1}\mathrm{H}$ decoupling both need strong RF, and this is what makes $^{13}\mathrm{C}$ CP-MAS of the solid phase possible (S33).
3. **接近 HR-MAS 的转速**：样品保持完整（细胞约 3500 Hz 破裂，S28），可以不取出样品、连续跟踪各相随时间的变化。
   > **Spin rates near HR-MAS**: the sample stays intact (cells break near 3500 Hz, S28), so changes in each phase can be followed over time without removing the sample.

代价（结合 S34）：为了兼顾凝胶，CMP 的转速和功率比不上专用固体探头。

> The trade-off (with S34): to keep gels intact, CMP cannot match a dedicated solids probe in spin rate and power.

读图：萌发种子随时间的谱 / Reading the figure: a germinating seed over time

五条谱自下而上为 0、24、48、72、96 h，每条旁配种子照片：0 h 是干种子，24 h 种子变大，48 h 长出根，72 与 96 h 根和芽越来越长。横轴只标"ppm"，刻度 150、100、50（图上未标核种）。0 与 24 h：最高峰约在 30 ppm，60–80 ppm 有一组较小的峰；48 h 起，60–80 ppm 的一组峰变得最强，30 ppm 峰相对变小。幻灯片没有给出峰的归属，也没有说明这些谱是哪一相的编辑谱。

> Five traces, bottom to top, at 0, 24, 48, 72 and 96 h, each beside a photo of the seed: dry at 0 h, swollen at 24 h, a root at 48 h, and an ever longer root and shoot at 72 and 96 h. The axis says only "ppm", with ticks at 150, 100, 50 (no nucleus given). At 0 and 24 h the tallest line is near 30 ppm with a smaller group at 60–80 ppm; from 48 h the 60–80 ppm group becomes the strongest and the 30 ppm line shrinks in comparison. The slide gives no assignments and does not say which phase-edited spectrum these are.

考点速记 KEY POINTS

- CMP = HR-MAS + 测固体的能力。
  CMP = HR-MAS plus the ability to do solids.
- 转速接近 HR-MAS，带梯度；一个探头测所有相。
  HR-MAS-like spin rates, gradients, one probe for all phases.
- 关键是功率承受能力，远高于 HR-MAS 探头。
  Key: power handling, far above an HR-MAS probe.
- 示例：萌发种子 0–96 h 原位跟踪。
  Example: a germinating seed followed in situ for 0–96 h.

S33 · 谱编辑（CMP）：每个实验选出哪一相

S33 · Spectral editing (CMP): which phase each experiment selects

编者注 / Editor's note

本页幻灯片除标题外没有文字层，以下内容全部从原图读出。两个弛豫过滤的标签在原图上印作 T_2（"T_2 ^{13}C CP-MAS"与"I-T_2 ^{13}C CP-MAS"），本书照原图写。

> Apart from its title this slide has no text layer; everything below is read from the figure. The two relaxation-filter labels are printed as T_2 ("T_2 ^{13}C CP-MAS" and "I-T_2 ^{13}C CP-MAS"), and this book follows the figure.

图的结构：顶部两张"标准谱"作起点。左边 **标准 ^{1}H NMR**：除真固体外的所有相，向下分出三种 ^{1}H 编辑。右边 **标准 ^{13}C CP-MAS**：所有具有固体特征的组分，"CP 过滤先选出所有固体"，再分出两种 ^{13}C 过滤。底部色带从"真溶液"（蓝）经"固体特征增强"到"真固体"（粉）。

> Layout: two "standard" spectra on top are the starting points. Left, **standard ^{1}H NMR**: all phases except true solids, splitting into three ^{1}H edits. Right, **standard ^{13}C CP-MAS**: all components with solid character; "CP filter first selects all solids", then two ^{13}C filters. A colour bar at the bottom runs from "true solution" (blue) through "increasing solid characteristics" to "true solid" (pink).

实验 Experiment	选出什么（原图）Selects (figure)	核 Nucleus	物理意义 Physical meaning
标准 ^{1}H Standard ^{1}H	除真固体外的所有相 all phases except true solids	^{1}H	溶液、凝胶、半固体叠在一张谱上 solution, gel and semi-solid on one spectrum
反扩散编辑 IDE	可自由移动的分子：180 ms 内移动 > 1 μm → 可溶代谢物 molecules free to move: > 1 μm in 180 ms → soluble metabolites	^{1}H	真溶液中快速扩散的小分子 fast-diffusing small molecules in true solution
扩散编辑 DE	不能自由移动的分子：180 ms 内 < 1 μm → 类凝胶 molecules not free to move: < 1 μm in 180 ms → gel-like	^{1}H	扩散很慢的大分子或受限分子，谱线仍较窄 slow-diffusing large or confined molecules, lines still fairly narrow
弛豫编辑 ^{1}H RADE	扩散编辑测不到的快弛豫半固体 fast-relaxing semi-solids that diffusion editing cannot detect	^{1}H	运动更受限、弛豫快的组分 more restricted, fast-relaxing components
标准 CP-MAS（CP 过滤 CP filter）	所有具有固体特征的组分 all components with solid character	^{13}C	依赖"刚性、偶极高效的偶极相互作用" relies on a "rigid dipole-efficient dipolar interaction"
弛豫过滤 T_2 ^{13}C CP-MAS	动态固体 dynamic solids	^{13}C	固体中仍有分子运动的部分 solid parts that still have molecular motion
反弛豫过滤 I-T_2 ^{13}C CP-MAS	结晶固体 crystalline solids	^{13}C	刚性、有序的结晶部分 rigid, ordered crystalline parts

沿色带排序：IDE（可溶代谢物）→ DE（类凝胶）→ RADE（半固体）→ T_2 过滤（动态固体）→ I-T_2（结晶固体），即真溶液 → 真固体。

> Order along the colour bar: IDE (soluble metabolites) → DE (gel-like) → RADE (semi-solids) → T_2 filter (dynamic solids) → I-T_2 (crystalline solids), i.e. true solution → true solid.

考点速记 KEY POINTS

- ^{1}H 编辑：IDE 溶液、DE 凝胶、RADE 半固体。
  ^{1}H edits: IDE solution, DE gel, RADE semi-solid.
- 分界：180 ms 内移动 > 1 μm 还是 < 1 μm。
  Cut-off: moves > 1 μm or < 1 μm in 180 ms.
- ^{13}C CP 过滤：动态固体（T_2）、结晶固体（I-T_2）。
  ^{13}C CP filters: dynamic (T_2), crystalline (I-T_2).
- 易错：标准 ^{1}H 看不到真固体。
  Trap: standard ^{1}H misses true solids.

S33（续）· 各种编辑怎样挑出一相

S33 (cont.) · How each edit picks out one phase

扩散编辑与反扩散编辑（^{1}H） / Diffusion editing and inverse diffusion editing (^{1}H)

背景：第一个梯度按位置给分子"打标记"，经过扩散时间（图中 180 ms）后第二个梯度"解码"。期间移动 > 1 μm 的分子无法重聚、信号消失，几乎不动的分子保留下来，所以 DE 谱只剩慢扩散的类凝胶组分。IDE 通常是参考谱（弱梯度）减去 DE 谱，只剩快扩散的溶液组分。

> **Background**: a first gradient "labels" molecules by position and, after a diffusion delay (180 ms in the figure), a second gradient "decodes". Molecules that moved > 1 μm fail to refocus and lose their signal, nearly stationary ones survive, so DE keeps only slow-diffusing gel-like components. IDE is usually a reference spectrum (weak gradient) minus DE, leaving only fast-diffusing solution components.

$$\ell \approx \sqrt{2D\Delta}, \qquad \Delta = 0.18\ \mathrm{s}$$
$$D = 10^{-9}\ \mathrm{m^2\,s^{-1}} \ \Rightarrow\ \ell \approx 19\ \mu\mathrm{m} \gg 1\ \mu\mathrm{m}$$
$$D = 10^{-12}\ \mathrm{m^2\,s^{-1}} \ \Rightarrow\ \ell \approx 0.6\ \mu\mathrm{m} < 1\ \mu\mathrm{m}$$

背景估算：第二行为小分子，第三行为类凝胶，D 取典型数量级

/ background estimate: row 2 a small molecule, row 3 gel-like, typical orders of D

弛豫编辑 ^{1}H RADE / Relaxation editing, ^{1}H RADE

原图原话：弛豫编辑"恢复扩散编辑检测不到的快弛豫半固体的信号"。**背景**：扩散编辑序列至少 180 ms，快弛豫信号在检测前已衰减，因此改用基于弛豫的选择。幻灯片未给出 RADE 的全称。

> The figure's words: relaxation editing "recovers signals from fast-relaxing semi-solids that cannot be detected via diffusion editing". **Background**: diffusion editing lasts at least 180 ms, so fast-relaxing signals decay before detection and a relaxation-based selection is used instead. The slide does not expand RADE.

CP 过滤与 T_2 过滤（^{13}C） / CP filter and T_2 filters (^{13}C)

背景：交叉极化（CP）经 ^{1}H–^{13}C 偶极耦合把磁化转给 ^{13}C。溶液中翻滚把偶极耦合平均为零，CP 不发生，所以 CP-MAS 只见具有固体特征的组分，即图中间的"刚性、偶极高效的偶极相互作用"。两个 T_2 过滤再按弛豫把这些固体分开：原图中弛豫过滤对应动态固体，反弛豫过滤（"I-"，通常为差谱）对应结晶固体。

> **Background**: cross polarization (CP) passes magnetization from ^{1}H to ^{13}C through their dipolar coupling. Tumbling in solution averages that coupling to zero and CP fails, so CP-MAS sees only components with solid character, the "rigid dipole-efficient dipolar interaction" drawn in the middle. The two T_2 filters then split these solids by relaxation: the figure assigns the relaxation filter to dynamic solids and the inverse filter ("I-", usually a difference spectrum) to crystalline solids.

图中各编辑谱的标注 / Labels on the edited spectra

- **^{1}H 可溶代谢物**（×8，8–1 ppm）：His、Phe、Tyr、Thr、Glu/Ala、Gly、Arg、Arg/Lys、Val、Lys、Ala。
  > **^{1}H soluble metabolites** (×8, 8–1 ppm): His, Phe, Tyr, Thr, Glu/Ala, Gly, Arg, Arg/Lys, Val, Lys, Ala.
- **^{1}H 类凝胶**（×16）：$(CH_2)_n-CH_3$、$-CO-CH_2-CH_2-$、$-CH_2-CH=$、$=CH-CH_2-CH=$、甘油 α 与 γ、$-CH_2-O$、ω_3
  > **^{1}H gel-like** (×16): $(CH_2)_n-CH_3$, $-CO-CH_2-CH_2-$, $-CH_2-CH=$, $=CH-CH_2-CH=$, glycerol α and γ, $-CH_2-O$, ω_3.
- **^{1}H 半固体**（×16）：Amides、Proteins、$(CH_2)_n$、$-CH_2-CH=$、$=CH-CH_2-CH=$、ω_3。
  > **^{1}H semi-solids** (×16): amides, proteins, $(CH_2)_n$, $-CH_2-CH=$, $=CH-CH_2-CH=$, ω_3.
- **^{13}C 动态固体**（160–40 ppm，膜图标）：$-C=C-$、$-C-O-C-$、$-CH_2-$、$-CH_3$。
  > **^{13}C dynamic solids** (160–40 ppm, membrane icon): $-C=C-$, $-C-O-C-$, $-CH_2-$, $-CH_3$.
- **^{13}C 结晶固体**（虾图标）：$-C=O$、碳水化合物（甲壳素）、异头碳（甲壳素）、$-CH_2-$、$-CH_3$。
  > **^{13}C crystalline solids** (shrimp icon): $-C=O$, carbohydrates (chitin), anomeric carbons (chitin), $-CH_2-$, $-CH_3$.

考点速记 KEY POINTS

- DE：快扩散分子被梯度消掉，留下凝胶。
  DE: gradients erase fast diffusers, leaving gels.
- IDE ≈ 参考谱 – DE 谱：只剩溶液相。
  IDE ≈ reference – DE: solution phase only.
- CP 需要未被平均的偶极耦合 ⇒ 只见固体。
  CP needs unaveraged dipolar coupling, so it sees solids only.
- $\ell \approx \sqrt{2D\Delta}$：小分子 180 ms 约 19 μm。
  $\ell \approx \sqrt{2D\Delta}$: a small molecule moves about 19 μm in 180 ms.

S34 · 该用哪一种：CMP、固体探头还是 HR-MAS

S34 · When should I use what: CMP, solids probe or HR-MAS

幻灯片要点 / What the slide says

- CMP 用途广，看起来像是总体上的最优选择，但并非总是如此。
  > CMP seems like the optimal choice in general since it is versatile, but this is not always the case.
- 如果只关心固体，固体探头可能更好，因为它允许更高的功率和更快的转速。
  > If you only care about solids, a solids probe may be better, by allowing higher power and faster spin rates.
- 如果关心相与相之间发生的变化，CMP 是很好的选择。
  > If you care about changes occurring between phases, CMP is a great choice.
- HR-MAS 只对溶液和凝胶有用，但正因如此，它很适合生物组织样品之类的东西。
  > HR-MAS is only helpful for solutions and gels, but that makes it great for things like biological tissue samples.

对比表（仅依据幻灯片陈述） / Comparison (from the slides' statements only)

	CMP	固体探头 Solids probe	HR-MAS
适用 Use when	关心相之间的变化 changes between phases matter	只关心固体 only solids matter	只有溶液和凝胶，如生物组织 solutions and gels only, e.g. tissue
优势 Strength	通用；一个探头测所有相（S32） versatile; one probe for all phases (S32)	更高功率、更快转速 higher power, faster spinning	很适合生物组织样品 great for biological tissue
局限 Limit	并非总是最佳；转速接近 HR-MAS（S32） not always optimal; HR-MAS-like spin rates (S32)	幻灯片未说明 not stated on the slide	不能测真固体（S30） no true solids (S30)

决策顺序 / Decision order

1. 只关心固体？→ 固体探头（功率更高、转速更快）。
   > Only solids? → solids probe (more power, faster spinning).
2. 要看相与相之间的变化？→ CMP。
   > Changes between phases? → CMP.
3. 样品只有溶液和凝胶（如组织）？→ HR-MAS。
   > Only solutions and gels (e.g. tissue)? → HR-MAS.

考试提醒 / Exam reminder

"CMP 最通用"不等于"CMP 总是最好"：纯固体问题选固体探头。组织样品适合 HR-MAS，与 S28 一致：细胞约 3500 Hz 破裂，组织只能慢转，而 HR-MAS 本来就慢转。

> "CMP is the most versatile" does not mean "CMP is always best": pick a solids probe for a solids-only question. Tissue suits HR-MAS, consistent with S28: cells break near 3500 Hz, tissue must spin slowly, and HR-MAS spins slowly anyway.

考点速记 KEY POINTS

- 只要固体 → 固体探头：功率更高、转速更快。
  Solids only → solids probe: more power, faster spinning.
- 溶液 + 凝胶（组织）→ HR-MAS。
  Solutions + gels (tissue) → HR-MAS.
- 跨相变化 → CMP。
  Changes between phases → CMP.
- 易错：CMP 通用但并非总是最优。
  Trap: CMP is versatile but not always optimal.

S35 · CMP 示例（一）：土壤中的含氟化学品

S35 · CMP examples (1): fluorinated chemicals in soil

幻灯片要点 / What the slide says

这里用 CMP 追踪土壤中含氟化学品的分子归宿。可以看到 PFOA（全氟辛酸）更多地被固定（sequestered）进固相，而 PFP 变化不大。

> Here CMP was used to track the molecular fate of fluorinated chemicals in a soil. PFOA (perfluorooctanoic acid) was sequestered much more into the solid phase, while PFP did not really change much.

为什么这是 CMP 的典型用途 / Why this is a typical CMP job

问题本身就是"相与相之间的变化"（S34）。CMP 在同一个转子里、不做萃取，就能分别读出溶液相、凝胶相和固相中的信号，并随时间重复测量；化合物从溶液进入固相时，溶液相信号下降、固相信号上升，可以同时看到。幻灯片没有说明观测的是哪种核，也没有解释两种化合物行为不同的原因。

> The question is exactly "changes between phases" (S34). In one rotor and without extraction, CMP reads the solution, gel and solid signals separately and repeats the measurement over time; when a compound moves from solution into the solid, the solution signal falls while the solid signal rises, and both are seen together. The slide does not say which nucleus was observed, nor why the two compounds behave differently.

读图 / Reading the figure

(a) Solution and Gel Phase：纵轴"Absolute % Change"，断轴（2–5 与 30–80 两段）；横轴时间 0–120（min，与 (b) 同）。图例：PFOA 实心、PFP 空心；方块 = 溶液相，圆点 = 凝胶相。

> **(a) Solution and Gel Phase**: vertical axis "Absolute % Change", broken (segments 2–5 and 30–80); horizontal axis time 0–120 (min, as in (b)). Legend: PFOA filled, PFP open; squares = solution, circles = gel.

曲线 Series（约值 approx.）	读数 Reading
PFOA 溶液相 solution	≈67 → 20 min 时 ≈57 → 之后稳定在 ≈54–55 ≈67 → ≈57 by 20 min → steady at ≈54–55
PFP 溶液相 solution	≈35 缓慢降到 ≈30 ≈35 drifting down to ≈30
凝胶相 gel（两者 both）	只有 ≈2.5–4；PFP 从 ≈4.3 降到 ≈2.7，PFOA ≈3 only ≈2.5–4; PFP ≈4.3 → ≈2.7, PFOA ≈3
PFOA 固相 solid (b)	≈1.0（约 10 min）→ ≈1.47 → ≈1.6 → 45 min 后 ≈1.7–1.75 ≈1.0 (about 10 min) → ≈1.47 → ≈1.6 → ≈1.7–1.75 after 45 min
PFP 固相 solid (b)	≈1.32 → ≈1.38 后基本不变 ≈1.32 → ≈1.38, then flat

(b) Solid Phase：纵轴"Relative Change"（0.8–1.8），横轴 Time (min)。左下示意图分 SOLUTION、GEL、SOLID 三层：PFP 画成带 OH 和五个 F 的苯环，PFOA 画成全氟碳链末端接 COOH；在 GEL 和 SOLID 层，F 原子周围画有灰色圆球，表示与土壤基质接触。

> **(b) Solid Phase**: vertical axis "Relative Change" (0.8–1.8), horizontal axis Time (min). The cartoon at lower left has three layers, SOLUTION, GEL and SOLID: PFP is drawn as a benzene ring with OH and five F, PFOA as a perfluorinated chain ending in COOH; in the GEL and SOLID layers grey spheres surround the F atoms, showing contact with the soil matrix.

考点速记 KEY POINTS

- CMP 原位追踪同一化合物在三相间的分布。
  CMP tracks one compound across three phases in situ.
- PFP：变化小，固相 ≈1.3–1.4 后持平。
  PFP: little change; solid flat at ≈1.3–1.4.
- PFOA：溶液相 ≈67 → ≈55，固相升到 ≈1.7。
  PFOA: solution ≈67 → ≈55; solid rises to ≈1.7.
- 易错：(a) 是绝对 % 变化且断轴，(b) 是相对变化。
  Trap: (a) is absolute % with a broken axis; (b) is relative.

S36 · CMP 示例（二）：盐酸羟嗪软胶囊

S36 · CMP examples (2): a hydroxyzine hydrochloride softgel

幻灯片要点 / What the slide says

这里用 CMP 研究一种药物：盐酸羟嗪（hydroxyzine hydrochloride）的软胶囊（softgel capsule）。结果它同时含有所有的相！

> Here CMP was used to study a pharmaceutical: a softgel capsule of hydroxyzine hydrochloride. It turned out to contain all phases!

三种编辑各看哪一相 / Which phase each edit shows

图中三个箭头正好对应 S33 的编辑方法：IDE → 溶液相（$^{1}\mathrm{H}$），DE → 凝胶相（$^{1}\mathrm{H}$），CP → 固相（$^{13}\mathrm{C}$）。同一个胶囊放在一个转子里，不拆开、不溶解，三相各得一张谱，这就是 S32 所说"一个探头研究所有状态"的实际用法。

> The three arrows match the S33 edits: IDE → solution phase ($^{1}\mathrm{H}$), DE → gel phase ($^{1}\mathrm{H}$), CP → solid phase ($^{13}\mathrm{C}$). One capsule in one rotor, neither opened nor dissolved, gives one spectrum per phase; this is S32's "all states with one probe" in practice.

读图 / Reading the figure

左侧照片：白色转子旁放着一粒橙色软胶囊。右侧三条谱：

> Left photo: a white rotor beside an orange softgel capsule. Right, three spectra:

- **IDE → Solution**（红，$^{1}\mathrm{H}$ ppm，8 到 0）：尖锐的线，最高峰约 1.3 ppm，另有约 0.9、1.6、2.0–2.3、2.8、4.1–4.3、5.3 ppm 的峰；8.0–6.8 ppm 放大插图中有几条尖锐的芳香区峰（约 7.9 与 6.8 ppm 最强）。

  > **IDE → Solution** (red, $^{1}\mathrm{H}$ ppm, 8 to 0): sharp lines, tallest near 1.3 ppm, others near 0.9, 1.6, 2.0–2.3, 2.8, 4.1–4.3 and 5.3 ppm; the 8.0–6.8 ppm inset shows several sharp aromatic lines (strongest near 7.9 and 6.8 ppm).

- **DE → Gel**（蓝，$^{1}\mathrm{H}$ ppm）：峰位与溶液谱相近（最高仍约 1.3 ppm），但谱线更宽；8.0–6.8 ppm 插图几乎是平的宽包。

  > **DE → Gel** (blue, $^{1}\mathrm{H}$ ppm): peaks at similar shifts (tallest still near 1.3 ppm) but broader; the 8.0–6.8 ppm inset is an almost flat broad hump.

- **CP → Solid**（黑，$^{13}\mathrm{C}$ ppm，约 190 到 10）：约 173 ppm 与约 130 ppm 有峰，约 73 与 63 ppm 两条最强的尖峰，20–40 ppm 有一簇小峰。

  > **CP → Solid** (black, $^{13}\mathrm{C}$ ppm, about 190 to 10): peaks near 173 and 130 ppm, the two strongest sharp lines near 73 and 63 ppm, and a cluster of small peaks at 20–40 ppm.

读图推断（背景） / Inference from the figure (background)

尖锐的芳香区峰只出现在 IDE 溶液谱里。羟嗪分子含芳环，所以至少有一部分药物溶解在溶液相中。幻灯片本身没有给出各峰的归属，也没有说明三相分别来自胶囊的哪一部分。

> Sharp aromatic lines appear only in the IDE solution spectrum. Hydroxyzine contains aromatic rings, so at least part of the drug is dissolved in the solution phase. The slide itself gives no peak assignments and does not say which part of the capsule each phase comes from.

考点速记 KEY POINTS

- 盐酸羟嗪软胶囊：一个样品同时含溶液、凝胶、固体。
  Hydroxyzine HCl softgel: solution, gel and solid in one sample.
- IDE → 溶液（$^{1}\mathrm{H}$）；DE → 凝胶（$^{1}\mathrm{H}$）。
  IDE → solution ($^{1}\mathrm{H}$); DE → gel ($^{1}\mathrm{H}$).
- 易错：固相用 $^{13}\mathrm{C}$ CP 观测，不是 $^{1}\mathrm{H}$。
  Trap: the solid phase is observed by $^{13}\mathrm{C}$ CP, not $^{1}\mathrm{H}$.
- 芳香区尖峰只在 IDE 溶液谱插图中出现。
  Sharp aromatic lines appear only in the IDE solution inset.

S37 · 不同尺寸的探头

S37 · Different Probe Sizes

幻灯片要点 / What the slide says

标准的 NMR 探头（probe）是为 5 mm 核磁管设计的，但这常常并不理想。例如化合物的量非常有限时，放进更小的管子里分析可能更好，这样就不必把它稀释太多。因此还有其他尺寸的探头可用。

> A standard NMR probe is meant for 5 mm tubes, but this is often not ideal. For example, if you have a very limited amount of compound, it may be better to analyze it in a smaller tube so you don't have to dilute it much. As such, there are other sized probes available.

为什么管子越小，稀释越少 / Why a smaller tube means less dilution

背景 样品必须充满线圈的检测区（还要高出一些，才能匀场、得到好峰形），所以最少要加多少溶剂由管子和线圈的尺寸决定，而不是由你有多少化合物决定。化合物的量 n 固定时，浓度 $c = n/V$；体积越小，浓度越高，单位体积里的核越多。

> **Background** The sample must fill the coil's detection region (and a bit more, for shimming and a good lineshape), so the minimum solvent volume is set by the tube and coil size, not by how much compound you have. For a fixed amount n, the concentration is $c = n/V$: the smaller the volume, the higher the concentration and the more nuclei per unit volume.

$$c = \frac{n}{V} \qquad \frac{c_{1.7\,\mathrm{mm}}}{c_{5\,\mathrm{mm}}} = \frac{600\ \mu\mathrm{L}}{28\ \mu\mathrm{L}} \approx 21$$

同样的化合物量，按图中推荐体积计算（背景算例）

/ same amount of compound, recommended volumes from the photo (background example)

图怎么读 / Reading the figure

- 照片里三支装有有色溶液的核磁管，从左到右越来越细：5 mm 管（橙色）标「600 mcl」，3 mm 管（紫色）标「150 mcl」，1.7 mm 管（黑色）标「28 mcl」。
  > The photo shows three tubes of coloured solution, thinner from left to right: the 5 mm tube (orange) is labelled "600 mcl", the 3 mm tube (purple) "150 mcl", the 1.7 mm tube (black) "28 mcl".
- 「mcl」即微升（μL）：600、150、$28\ \mu\mathrm{L}$。相对 5 mm 管，3 mm 管只需 $1/4$ 的体积，1.7 mm 管约 $1/21$。
  > "mcl" means microlitre (μL): 600, 150, $28\ \mu\mathrm{L}$. Relative to 5 mm, the 3 mm tube needs $1/4$ of the volume and the 1.7 mm tube about $1/21$.

本节路线图（S37–S49） / Section map (S37–S49)

幻灯片 Slides	主题 Topic	核心问题 Key question
S37–39	小探头 small probes	体积小 → 少稀释、质量灵敏度更高 less volume → less dilution, better mass sensitivity
S40–46	微线圈 microcoils	质量受限样品的极限方案；B_1 不均匀等难题 the extreme for mass-limited samples; B_1 inhomogeneity and other problems
S47–49	大线圈 larger coils	浓度灵敏度好；功率、加热、质量灵敏度差 good concentration sensitivity; power, heating, poor mass sensitivity

考点速记 KEY POINTS

- 标准探头用 5 mm 管，样品少时并不理想。
  Standard probes take 5 mm tubes; not ideal for little sample.
- 量固定时 $c = n/V$：管越小，稀释越少。
  Fixed amount: $c = n/V$, smaller tube, less dilution.
- 5 / 3 / 1.7 mm 管：$600/150/28\ \mu\mathrm{L}$。
  5 / 3 / 1.7 mm tubes: $600/150/28\ \mu\mathrm{L}$.
- 图中「mcl」= 微升 μL，不是毫升。
  "mcl" on the photo = μL, not mL.

S38 · 不同尺寸的探头：1.7 mm 与 1 mm

S38 · Different Probe Sizes: 1.7 mm and 1 mm

幻灯片要点 / What the slide says

对蛋白质或珍贵样品，管子通常更小，可小到 1 mm，其中 1.7 mm 非常常见。这些也可以是超低温探头（cryoprobe，冷探头），常用于蛋白质：蛋白质纯化成本高，经同位素标记的更贵。1.7 mm 探头的推荐体积是 30 µL，约为 5 mm 管的 1/20。

> For proteins or precious samples, the tubes are typically smaller, down to 1 mm, with 1.7 mm being very common. These can also be cryoprobes, commonly used for proteins, which are expensive to purify, especially if labelled. A 1.7 mm probe has a recommended volume of 30 µL, about 20 times less than a 5 mm.

背景 超低温探头把线圈和前置放大器冷却到很低温度，热噪声大幅下降，所以同样样品的 SNR 更高。小管与超低温探头两条路可以叠加。

> **Background** A cryoprobe cools the coil and preamplifier to a very low temperature, which cuts the thermal noise, so the same sample gives a higher SNR. Small tubes and cryogenic cooling can be combined.

图怎么读：同样 172 µg 蔗糖 / Reading the figure: the same 172 µg of sucrose

- 蓝线：$172\ \mu g$ 蔗糖溶于 $5\ \mu L\ D_2O$，PRESAT，1 次扫描，1 mm TXI 微量探头（图中原文「1mm TXI MikroProbe」），400 MHz。
  > Blue: $172\ \mu g$ sucrose in $5\ \mu L\ D_2O$, PRESAT, 1 scan, 1 mm TXI microprobe (printed "1mm TXI MikroProbe"), 400 MHz.
- 红线：同样 $172\ \mu g$ 蔗糖溶于 $500\ \mu L\ D_2O$，PRESAT，1 次扫描，5 mm TXI 探头，400 MHz。只有探头和体积不同。
  > Red: the same $172\ \mu g$ in $500\ \mu L\ D_2O$, PRESAT, 1 scan, 5 mm TXI probe, 400 MHz. Only the probe and the volume differ.
- 横轴 ppm，约 5.5–3.3。蓝线的蔗糖峰（约 5.3 ppm 的端基双峰和 4.2–3.3 ppm 的环上质子）比红线高好几倍；红线上蔗糖峰很小，约 4.7 ppm 处却有一个大的残留水峰，箭头注明「residual water suppressed by PRESAT」（中间有 presat 打出的缺口）。
  > The axis runs from about 5.5 to 3.3 ppm. Blue sucrose peaks (anomeric doublet near 5.3 ppm, ring protons at 4.2–3.3 ppm) are several times taller than red; red has tiny sucrose peaks but a large residual water peak at about 4.7 ppm, marked "residual water suppressed by PRESAT" (with a notch cut by presat).

$$\frac{172\ \mu g}{342.3\ g\,mol^{-1}} \approx 0.50\ \mu mol: \quad \frac{0.50\ \mu mol}{5\ \mu L} \approx 100\ mM, \quad \frac{0.50\ \mu mol}{500\ \mu L} \approx 1.0\ mM$$

背景算例：蔗糖摩尔质量 342.3，浓度差 100 倍

/ background example: sucrose molar mass 342.3; a 100-fold concentration difference

背景 残留水（D_2O 中的 HDO）的量与溶剂体积成正比，蔗糖的量却固定，所以在 500 µL 里水峰相对蔗糖大约 100 倍，在 5 µL 里几乎看不到。

> **Background** The residual water (HDO in D_2O) scales with solvent volume while the sucrose amount is fixed, so in 500 µL the water is about 100 times larger relative to sucrose, and in 5 µL it is barely visible.

编者注 / Editor's note

体积说法：S37 照片写 1.7 mm 管 28 µL，本页文字写推荐 30 µL、「约少 20 倍」（$600/30 = 20$，$600/28 \approx 21$），两者一致到约数。本页的 5 mm 对照用的是 500 µL，不是 600 µL。横轴最后两个刻度印作「3.2」「3.4」，按 0.2 ppm 间距第一个应为 3.6。

> Volumes: the S37 photo gives 28 µL for 1.7 mm; this slide's text says 30 µL recommended, "~20 times less" ($600/30 = 20$, $600/28 \approx 21$); they agree as round numbers. The 5 mm comparison here uses 500 µL, not 600 µL. The last two axis ticks are printed "3.2", "3.4"; by the 0.2 ppm spacing the first should read 3.6.

考点速记 KEY POINTS

- 蛋白质 / 珍贵样品：1.7 mm 常用，可小到 1 mm。
  Proteins / precious samples: 1.7 mm common, down to 1 mm.
- 1.7 mm 推荐 $30\ \mu L$，约为 5 mm 的 $1/20$。
  1.7 mm: $30\ \mu L$ recommended, ~$1/20$ of 5 mm.
- 同 172 µg：5 µL（1 mm）峰远高于 500 µL（5 mm）。
  Same 172 µg: 5 µL (1 mm) far beats 500 µL (5 mm).
- 陷阱：两者质量相同，浓度差 100 倍（100 vs 1 mM）。
  Trap: same mass, 100-fold concentration (100 vs 1 mM).

S39 · 小探头的好处

S39 · Benefits of smaller probes

幻灯片要点 / What the slide says

小探头的第一大好处是需要的样品体积更少，所以不会把样品稀释太多。此外它在质量灵敏度（mass sensitivity）上也有优势：一般来说，对给定质量的样品，小探头表现更好。如果样品质量有限或非常昂贵，小探头可能是理想选择。

> The first major benefit of a smaller probe is that you need less sample volume, so you aren't diluting your sample too much. However, there is also an advantage in mass sensitivity: in general, a smaller probe performs better for a given mass of sample. If you are limited in sample mass or your sample is very expensive, a smaller probe may be ideal.

两种灵敏度 / Two kinds of sensitivity

背景 质量灵敏度 S_m：每单位样品量（质量或摩尔数）得到的 SNR，回答「我只有这么多，能测到多少」。浓度灵敏度（concentration sensitivity）S_c：每单位浓度得到的 SNR，回答「浓度一定、样品管够，能测到多少」。由 $n = cV$ 可把两者联系起来：

> **Background** Mass sensitivity S_m is the SNR per unit amount of sample (mass or moles): "I only have this much; how much signal do I get?" Concentration sensitivity S_c is the SNR per unit concentration: "at this concentration, with the tube full, how much signal?" Since $n = cV$, the two are linked:

$$\mathrm{SNR} = S_\mathrm{m}\, n = S_\mathrm{m}\, c\, V \quad \Rightarrow \quad S_\mathrm{c} = S_\mathrm{m}\, V$$

背景关系：小线圈 S_m 高但 V 小；大线圈相反（S47）

/ background relation: a small coil has high S_m but small V; a large coil the reverse (S47)

为什么小线圈的质量灵敏度更高 / Why a smaller coil has better mass sensitivity

背景 按互易原理，一个核在线圈里感应的信号正比于线圈在该处单位电流产生的 B_1（B_1/i）。线圈越小，核离导线越近，B_1/i 越大，每个核贡献的信号就越大；线圈的噪声却没有同样快地增大。所以样品量固定时，把它装进刚好容纳它的小线圈，SNR 最高。

> **Background** By the reciprocity principle, the signal a nucleus induces in a coil is proportional to the B_1 the coil makes there per unit current (B_1/i). In a smaller coil the nuclei sit closer to the wire, B_1/i is larger and each nucleus contributes more signal, while the coil's noise does not grow as fast. So for a fixed amount, a coil just large enough to hold it gives the best SNR.

考试提醒 / Exam reminder

「小探头更灵敏」只对**质量有限**的样品成立。样品量不受限时，大体积装下更多的核，5 mm（或更大）探头给出的总信号更多（S45 牛奶、S47）。

> "Smaller probes are more sensitive" holds only for **mass-limited** samples. When sample is plentiful, a larger volume holds more nuclei, so a 5 mm (or larger) probe gives more total signal (S45 milk, S47).

考点速记 KEY POINTS

- 小探头：少稀释 + 质量灵敏度高。
  Small probe: less dilution + higher mass sensitivity.
- S_m = 每单位量的 SNR；S_c = 每单位浓度的 SNR。
  S_m: SNR per amount; S_c: SNR per concentration.
- $S_\mathrm{c} = S_\mathrm{m} V$（背景）：两者此消彼长。
  $S_\mathrm{c} = S_\mathrm{m} V$ (background): a trade-off.
- 样品少或贵 → 选小探头。
  Little or expensive sample → choose a small probe.

S39（续）· 读图：线圈尺寸与质量灵敏度

S39 (cont.) · Reading the plots: coil size vs mass sensitivity

右侧两张图：a)「Coil Size (I.D.) vs. Mass Sensitivity」，横轴为线圈内径（Coil size μm (I.D.)），0–1000 μm；b)「Coil size (O.D.) vs. Mass sensitivity」，横轴为线圈外径（O.D.），0–1000 μm。纵轴都是「Mass Sensitivity」，线性刻度，最低刻度印作 1.0E+01，a) 到 7.0E+05，b) 到 1.8E+04；幻灯片没有给出单位。

> Two plots: a) "Coil Size (I.D.) vs. Mass Sensitivity", x = coil inner diameter, 0–1000 μm; b) "Coil size (O.D.) vs. Mass sensitivity", x = outer diameter, 0–1000 μm. Both y-axes read "Mass Sensitivity" on a linear scale whose lowest tick is printed 1.0E+01, up to 7.0E+05 in a) and 1.8E+04 in b); the slide gives no unit.

每张图有蓝、红、绿三组数据点和虚线幂函数拟合。幻灯片没有说明三种颜色分别代表什么，下表只抄录图上的拟合式（x 为线圈尺寸，μm）：

> Each plot has blue, red and green data points with dotted power-law fits. The slide does not say what the three colours stand for; the table copies only the fits printed on the plots (x = coil size in μm):

颜色 Colour	a) I.D. 拟合 fit	R^2	b) O.D. 拟合 fit	R^2
蓝 blue	$y = 9 \times 10^{8}\, x^{-2.306}$	0.979	$y = 4 \times 10^{9}\, x^{-2.509}$	0.9674
红 red	$y = 2 \times 10^{8}\, x^{-2.177}$	0.9901	$y = 5 \times 10^{8}\, x^{-2.293}$	0.9828
绿 green	$y = 7 \times 10^{7}\, x^{-2.079}$	0.9958	$y = 1 \times 10^{8}\, x^{-2.13}$	0.9915

- a) 的数据点约在 20、50、100 μm（绿色另有 500、1000 μm）；20 μm 处蓝点约 6.7×10^5，红点约 2.7×10^5，绿点约 1.3×10^5，到 100 μm 已接近零。
  > a) Points sit at about 20, 50, 100 μm (green also at 500, 1000 μm); at 20 μm blue ≈ 6.7×10^5, red ≈ 2.7×10^5, green ≈ 1.3×10^5; by 100 μm they are near zero.
- b) 的数据点约在 130、190、260、790 μm；130 μm 处蓝点约 1.6×10^4，红约 6.3×10^3，绿约 3.1×10^3。曲线都呈「L」形：几百 μm 以下陡升，以上几乎贴着横轴。
  > b) Points at about 130, 190, 260, 790 μm; at 130 μm blue ≈ 1.6×10^4, red ≈ 6.3×10^3, green ≈ 3.1×10^3. Every curve is L-shaped: steep below a few hundred μm, almost flat on the axis above.

$$S_\mathrm{m} = a\, d^{-b}, \quad b \approx 2.08\text{–}2.51 \;\Rightarrow\; \frac{S_\mathrm{m}(d/2)}{S_\mathrm{m}(d)} = 2^b \approx 4.2\text{–}5.7$$

由图中拟合式推算：线圈直径减半，质量灵敏度约升 4–6 倍

/ from the printed fits: halving the coil diameter raises mass sensitivity about 4–6 times

考试提醒 / Exam reminder

关键是指数为负且绝对值约 2 以上：尺寸越小，质量灵敏度增长越快。纵轴是线性的，所以大尺寸处的点看起来是零，其实只是小得多；R^2 都在 0.97–1.00 之间，幂律拟合很好。

> The key is a negative exponent of magnitude above about 2: the smaller the coil, the faster mass sensitivity grows. The y-axis is linear, so large-coil points look like zero but are only much smaller; every R^2 lies between 0.97 and 1.00, so the power laws fit well.

考点速记 KEY POINTS

- I.D. 与 O.D. 两图都显示：线圈越小，S_m 越高。
  Both I.D. and O.D. plots: smaller coil, higher S_m.
- 直径减半 → S_m 约 ×4–6（推算）。
  Halve d → S_m about ×4–6 (derived).
- 拟合 $S_\mathrm{m} = a\, d^{-b}$，$b \approx 2.1$–$2.5$。
  Fits $S_\mathrm{m} = a\, d^{-b}$ with $b \approx 2.1$–2.5.
- 陷阱：颜色含义与纵轴单位幻灯片未给出。
  Trap: the slide gives neither colour meanings nor y units.

S40 · 微线圈 NMR

S40 · Microcoil NMR

幻灯片要点 / What the slide says

把「小探头」推向极端，就得到微线圈（microcoil）的想法：本质上就是超小的 NMR 线圈，质量灵敏度更好。微线圈有相当多的难题，所以只用于小众场合；它们非常适合「质量受限」（mass-limited）的样品。

> Taking this to an extreme gives the idea of microcoils: essentially super small NMR coils that give better mass sensitivity. There are considerable challenges with microcoils, so their use is niche. They are great for "mass limited" samples.

什么叫「质量受限」 / What "mass limited" means

限制来自样品的**总量**，而不是溶解度：总共只有纳升级的体积或微克级的物质，加溶剂只会把它稀释。本节的例子是单个卵（S41）、大型溞卵（S45）和活体大鼠脑内的一小块组织（S46）。S39 的结论在这里推到极致：样品有多小，线圈就做多小。

> The limit is the **total amount** of sample, not its solubility: there are only nanolitres or micrograms in total, and adding solvent just dilutes it. This section's examples are single ova (S41), *D. magna* eggs (S45) and a small region of a living rat brain (S46). The S39 result is pushed to its limit: make the coil as small as the sample.

图怎么读：两种微线圈 / Reading the figure: two kinds of microcoil

- 上图：螺线管式微线圈（solenoid）。细铜线密绕在一根细管上，比例尺箭头 $500\ \mu\mathrm{m}$，约等于整个绕组的长度。
  > Top: a solenoid microcoil. Fine copper wire is tightly wound on a thin tube; the scale arrow, $500\ \mu\mathrm{m}$, spans about the length of the winding.
- 下图：平面式微线圈（planar）。金属导线在平板基底上绕成同心螺旋，中心留空，右边引出两块接线焊盘；左下比例尺 $500\ \mu\mathrm{m}$。
  > Bottom: a planar microcoil. A metal track is laid out as a flat spiral on a substrate with an open centre and two contact pads leading off to the right; scale bar $500\ \mu\mathrm{m}$ at the lower left.

背景 两者的差别决定了后面的难题。螺线管把样品包在线圈*里面*，内部 B_1 比较均匀；平面线圈用光刻做在芯片上，便于做小、做成阵列（S42），但样品放在线圈*上方*，B_1 随离开平面的高度迅速衰减，所以很不均匀（S43 说的正是「(planar) microcoils」）。

> **Background** The difference between the two sets up the later problems. A solenoid wraps the sample *inside* the coil, where B_1 is fairly uniform. A planar coil is made lithographically on a chip, so it is easy to shrink and to put into arrays (S42), but the sample sits *above* the coil and B_1 falls quickly with height above the plane, so it is very non-uniform (S43 refers to exactly "(planar) microcoils").

类型 Type	样品位置 Sample	B_1	优点 Strength
螺线管 **solenoid**	线圈内 inside	较均匀 fairly uniform	脉冲较好（背景） better pulses (background)
平面 **planar**	线圈上方 on top	很不均匀 very non-uniform	易微型化、易做阵列（背景） easy to shrink and array (background)

考点速记 KEY POINTS

- 微线圈 = 小探头的极限：质量灵敏度最好。
  Microcoil = small probe taken to the extreme: best mass sensitivity.
- 适用于质量受限样品；难题多，用途小众。
  For mass-limited samples; many challenges, niche use.
- 两种：螺线管式、平面式（比例尺 500 μm）。
  Two kinds: solenoid and planar (scale bars 500 μm).
- 平面线圈 B_1 随高度衰减 → 不均匀（背景）。
  Planar: B_1 decays with height → non-uniform (background).

S41 · 微线圈 NMR：单个卵

S41 · Microcoil NMR: single ova

幻灯片要点 / What the slide says

这里是一个超小微线圈的例子。这类线圈已被用来研究单个卵（ovum，复数 ova）。

> Here is an example of a super small microcoil. Such coils have been used to study individual ova.

为什么非用微线圈不可：图 a 中所有卵都不到 $1.5\ \mathrm{nL}$，而 5 mm 管装 $600\ \mu\mathrm{L} = 6 \times 10^5\ \mathrm{nL}$。一个约 $0.5\ \mathrm{nL}$ 的卵只占其中约百万分之一，在 5 mm 探头里根本测不到；线圈缩到几十微米、刚好罩住一个卵，才能有足够信号。

> Why a microcoil is the only option: every ovum in panel a is below $1.5\ \mathrm{nL}$, while a 5 mm tube holds $600\ \mu\mathrm{L} = 6 \times 10^5\ \mathrm{nL}$. A $0.5\ \mathrm{nL}$ ovum is about one millionth of that and is undetectable in a 5 mm probe; only a coil shrunk to tens of micrometres, just covering one ovum, gives enough signal.

$$\frac{600\ \mu\mathrm{L}}{0.5\ \mathrm{nL}} = \frac{6 \times 10^5\ \mathrm{nL}}{0.5\ \mathrm{nL}} \approx 1.2 \times 10^6$$

背景算例：一个卵与一支 5 mm 管的体积比 / background example: one ovum vs one 5 mm tube

图怎么读 / Reading the figure

- a：纵轴「Ovum volume [nL]」0–1.5。按引线读（约值）：猪 ≈ 1.43，鲸 ≈ 1.15，文昌鱼（lancelet）≈ 0.7，人与豚鼠 ≈ 0.52，海胆与大鼠 ≈ 0.18，蛔虫（roundworm）$\approx 0.05\ \mathrm{nL}$。

  > a: y-axis "Ovum volume [nL]", 0–1.5. Read along the leader lines (approx.): pig ≈ 1.43, whale ≈ 1.15, lancelet ≈ 0.7, human and guinea pig ≈ 0.52, sea urchin and rat ≈ 0.18, roundworm ≈ 0.05 nL.

- 两条红线把 b、c 两张照片接到轴上：b *Richtersius coronifer* 的卵约 $0.52\ \mathrm{nL}$（带刺的球形），c *Heligmosomoides polygyrus bakeri* 的卵约 $0.1\ \mathrm{nL}$（椭圆形）；两张的比例尺都是 $50\ \mu\mathrm{m}$。

  > Two red lines tie photos b and c to the axis: b, the ovum of *Richtersius coronifer*, ≈ $0.52\ \mathrm{nL}$ (a spiny sphere); c, *Heligmosomoides polygyrus bakeri*, ≈ $0.1\ \mathrm{nL}$ (an oval egg); both scale bars are $50\ \mu\mathrm{m}$.

- d：宽 $0.6\ \mathrm{mm}$ 的微芯片，左下角虚线框是线圈；放大图是八边形平面螺旋线圈，比例尺 $50\ \mu\mathrm{m}$。

  > d: a microchip $0.6\ \mathrm{mm}$ wide; the dashed box at the lower left is the coil, enlarged as an octagonal planar spiral with a $50\ \mu\mathrm{m}$ scale bar.

- e：截面图。印刷电路板上是微芯片，微线圈在芯片表面，卵（绿色）就放在微线圈上；键合线由硅胶保护，上面罩聚苯乙烯杯，杯内填 1.5% 琼脂糖凝胶，边缘用蜡密封。

  > e: cross-section. The microchip sits on a printed circuit board with the microcoil on its surface and the ovum (green) placed on the microcoil; the bonding wires are covered with silicone glue, a polystyrene cup on top is filled with 1.5% agarose gel, and wax seals the edges.

背景 凝胶把卵固定在线圈正上方：平面线圈只在紧贴表面的薄层里灵敏，卵稍一移动信号就丢（参见 S43、S45）。

> **Background** The gel holds the ovum right above the coil: a planar coil is sensitive only in a thin layer next to its surface, so any movement of the ovum loses the signal (see S43, S45).

考点速记 KEY POINTS

- 微线圈可测单个卵：卵体积 $< 1.5\ \mathrm{nL}$。
  Microcoils can measure one ovum: volumes $< 1.5\ \mathrm{nL}$.
- 人卵约 $0.5\ \mathrm{nL}$，约为 5 mm 管的百万分之一。
  Human ovum ≈ $0.5\ \mathrm{nL}$, ~10^{-6} of a 5 mm tube.
- 芯片宽 $0.6\ \mathrm{mm}$，线圈比例尺 $50\ \mu\mathrm{m}$。
  Chip $0.6\ \mathrm{mm}$ wide; coil scale bar $50\ \mu\mathrm{m}$.
- 卵放在线圈上，用 1.5% 琼脂糖凝胶固定。
  Ovum sits on the coil, held in 1.5% agarose gel.

S42 · 微线圈阵列

S42 · Microcoil Arrays

幻灯片要点 / What the slide says

线圈很小，所以可以扩展成多个线圈，即微线圈阵列（microcoil array）。这样可以同时研究多个样品，非常有用。例如可以同时测暴露组和对照组的生物，从而减小差异（variability）。

> Since the coils are small, it is possible to expand to multiple coils. This allows multiple samples to be studied at once and is quite powerful. For example, you can study exposed and control organisms at the same time, which reduces variability.

为什么「同时测」能减小差异 / Why measuring at the same time reduces variability

背景 分别测量时，暴露组与对照组之间混进了很多与处理无关的差别：不同的时间、匀场、温度、仪器漂移，活体样品还有不同的生理状态。阵列中所有线圈处在同一磁体、同一时刻、同一条件下，这些因素对两组的影响相同，比较时相互抵消，剩下的差别才更可能来自暴露本身。每组放几个线圈还能给出重复样本，可以做统计。

> **Background** Measured one after another, the exposed and control groups pick up many differences unrelated to the treatment: different times, shims, temperatures, instrument drift and, for live samples, different physiological states. In an array every coil sits in the same magnet, at the same moment, under the same conditions, so these factors affect both groups alike and cancel in the comparison; what remains is more likely due to the exposure. Several coils per group also give replicates for statistics.

背景 另一个好处是通量：N 个线圈在一次实验的时间里得到 N 个样品的谱。这依赖平面线圈易于在同一基底上批量制作（S40）。

> **Background** Another gain is throughput: N coils give N spectra in the time of one experiment. This relies on planar coils being easy to make side by side on one substrate (S40).

图怎么读 / Reading the figure

- 示意图：3 × 3 个橙色平面螺旋线圈，被细线分成三列，每个线圈上放一只小型水生生物（外形像大型溞，图中没有注明）。

  > Illustration: a 3 × 3 grid of orange planar spiral coils, divided by thin lines into three columns, with one small aquatic organism on each coil (it looks like *D. magna*; the figure does not say).

- 三列生物颜色不同（左列深褐、中列绿色、右列灰白），可理解为三个处理组、每组三个重复。这是读图推断，幻灯片没有标注各组。

  > The three columns have different colours (dark brown, green, grey-white), which reads as three treatment groups with three replicates each. This is an inference; the slide does not label the groups.

考试提醒 / Exam reminder

阵列能做成是**因为线圈小**；它的价值是多个样品**同时**测量，暴露组与对照组共享同样的仪器条件，从而减小差异。

> Arrays are possible **because the coils are small**; their value is measuring several samples **at once**, so exposed and control groups share the same instrument conditions and variability drops.

考点速记 KEY POINTS

- 线圈小 → 可做成多线圈阵列。
  Small coils → multi-coil arrays.
- 暴露组与对照组同时测 → 减小差异。
  Exposed and control measured together → less variability.
- 多个样品同时测，功能强大。
  Several samples at once: quite powerful.
- 图：3 × 3 平面线圈，每个上放一只生物。
  Figure: 3 × 3 planar coils, one organism on each.

S43 · 微线圈 NMR 的难题：B_1 均匀性

S43 · Microcoil NMR – Challenges: B_1 homogeneity

幻灯片要点 / What the slide says

微线圈的主要难题是 B_1（射频场）均匀性差：在（平面）微线圈上，脉冲的效果不太好。下面是标准探头（左）和微线圈（右）的章动曲线（nutation curve）。样品永远不会被反转，所以二维实验很难做。

> The main challenge with microcoils is poor B_1 field homogeneity: pulses are not very good on (planar) microcoils. Below is a nutation curve for a standard probe (left) and a microcoil (right). The sample never inverts, so 2D experiments are tough.

机理：章动曲线 / Mechanism: the nutation curve

背景 章动曲线是在固定功率下逐步加长脉冲宽度 t_p，每次记一张一维谱。翻转角 $\theta = \gamma B_1 t_\mathrm{p}$，信号 $\propto \sin\theta$：$90°$ 最大，$180°$ 为零（磁化被反转到 $-z$），$270°$ 最负，$360°$ 再为零。B_1 均匀时，样品各处翻转角相同，曲线是一条干净的正弦。

> **Background** A nutation curve steps the pulse length t_p at fixed power and records a 1D spectrum each time. The flip angle is $\theta = \gamma B_1 t_\mathrm{p}$ and the signal $\propto \sin\theta$: maximum at $90°$, zero at $180°$ (magnetization inverted to $-z$), most negative at $270°$, zero again at $360°$. With a uniform B_1 every part of the sample turns by the same angle and the curve is a clean sine.

均匀 B_1（背景） / uniform B_1 (background)

$$S(t_\mathrm{p}) \propto \sin\left(\gamma B_1 t_\mathrm{p}\right)$$

不均匀 B_1（背景） / non-uniform B_1 (background)

$$S(t_\mathrm{p}) \propto \left\langle \sin\left(\gamma B_1(\mathbf{r})\, t_\mathrm{p}\right)\right\rangle_\mathrm{sample}$$

背景 平面线圈的 B_1 在导线附近强、离开平面迅速变弱，样品不同位置 $\mathbf{r}$ 的翻转角差别很大。第一个最大值之后，各处的 $\sin\theta$ 有正有负、彼此抵消，总信号只降到一个正的平台，而不是经过零点变负；离线圈远、B_1 很弱的区域还在慢慢转向 $90°$，曲线后段因而缓慢回升。没有哪一个脉冲宽度能让整个样品都翻转 $180°$。

> **Background** A planar coil's B_1 is strong near the wire and drops quickly away from the plane, so the flip angle varies widely with position $\mathbf{r}$. After the first maximum, the $\sin\theta$ values from different regions have both signs and cancel, so the total falls only to a positive plateau instead of passing through zero; regions far from the coil, with weak B_1, are still turning towards $90°$, so the curve rises slowly at the end. No single pulse length turns the whole sample by $180°$.

考点速记 KEY POINTS

- 微线圈主要难题：B_1 均匀性差。
  Main microcoil problem: poor B_1 homogeneity.
- B_1 各处不同 → $\sin\theta$ 平均后不过零。
  B_1 varies → averaged $\sin\theta$ never crosses zero.
- $\theta = \gamma B_1 t_\mathrm{p}$，信号 $\propto \sin\theta$（背景）。
  $\theta = \gamma B_1 t_\mathrm{p}$, signal $\propto \sin\theta$ (background).
- 样品从不反转 → 二维实验困难。
  The sample never inverts → 2D experiments are tough.

S43（续）· 读图：标准探头与微线圈的章动曲线

S43 (cont.) · Reading the figure: standard probe vs microcoil

两张图横轴都是「Pulse Length (μs)」，刻度 50、150、250，小刻度每 25 μs。每根竖线是一张一维谱的峰，脉冲宽度从约 10 μs 每次加 10 μs，到约 300 μs，共约 30 张。以下时间都是读图约值。

> Both x-axes read "Pulse Length (μs)", ticks at 50, 150, 250, minor ticks every 25 μs. Each vertical line is the peak of one 1D spectrum; the pulse length goes from about 10 μs in 10 μs steps to about 300 μs, about 30 spectra. All times below are approximate readings.

特征 Feature	左：标准探头（红）Left: standard probe (red)	右：微线圈（蓝）Right: microcoil (blue)
最大值 maximum	约 60–70 μs（90°） ≈ 60–70 μs (90°)	约 70–80 μs ≈ 70–80 μs
第一个零点 first zero	约 140 μs（180°，反转） ≈ 140 μs (180°, inversion)	没有零点 none
负信号 negative signal	约 200–210 μs 最负（270°） most negative ≈ 200–210 μs (270°)	从不为负 never negative
之后 afterwards	约 270 μs 再过零（360°），随后转正 zero again ≈ 270 μs (360°), then positive	约 190–200 μs 降到最大值的约 60 %，然后缓慢回升 falls to ≈ 60 % of maximum at ≈ 190–200 μs, then rises slowly

左图正是一条正弦：90°、180°、270°、360° 的位置之比约为 1 : 2 : 3 : 4，所以从曲线就能定出 90° 脉冲宽度。右图只有一个宽的正峰，之后停在正的平台上：既读不出准确的 90°，也找不到 180°。

> The left curve is a sine: the 90°, 180°, 270° and 360° points are in the ratio of about 1 : 2 : 3 : 4, so the 90° pulse length can be read off it. The right curve has one broad positive hump and then a positive plateau: no exact 90° can be read, and no 180° exists.

考试提醒：为什么二维实验难 / Exam reminder: why 2D experiments suffer

背景 多脉冲实验（COSY、HSQC 等）的相干转移、反转和重聚都依赖准确的 90° 和 180° 脉冲。每个脉冲在样品各处的误差都不同，误差逐个脉冲累积，信号大部分损失，下一页的微线圈 COSY 就是结果。一维谱只需一个脉冲，所以仍然能做。

> **Background** Coherence transfer, inversion and refocusing in multipulse experiments (COSY, HSQC …) all need accurate 90° and 180° pulses. Each pulse errs differently at each point in the sample and the errors compound pulse by pulse, so most of the signal is lost; the microcoil COSY on the next page is the result. A 1D spectrum needs only one pulse, so it still works.

考点速记 KEY POINTS

- 标准探头：90° ≈ 65 μs，180° ≈ 140 μs（读图）。
  Standard probe: 90° ≈ 65 μs, 180° ≈ 140 μs (read).
- 微线圈：一个正峰后停在约 60 % 的平台，从不变负。
  Microcoil: one hump, then a ~60 % plateau; never negative.
- 均匀 B_1：零点比例 180° : 360° = 1 : 2。
  Uniform B_1: zeros at 180° and 360°, ratio 1 : 2.
- 陷阱：一维谱照样能测，难的是多脉冲二维实验。
  Trap: 1D still works; multipulse 2D is what fails.

S44 · 微线圈 NMR 的难题：COSY 无信号

S44 · Microcoil NMR – Challenges: COSY with no signal

幻灯片要点 / What the slide says

左边是用 5 mm 探头做的 COSY（¹H–¹H 化学位移相关谱）；右边是微线圈 COSY，由于章动差（S43），没有信号。

> The left is a COSY pulsed with a 5 mm probe; the right is a microcoil COSY, which has no signal due to poor nutation (S43).

为什么 COSY 特别受影响 / Why COSY is hit so hard

背景 最简单的 COSY 是 $90^\circ - t_1 - 90^\circ - t_2$（采集）。交叉峰来自第二个 90° 脉冲把相干从一个质子转移到与它耦合的质子；翻转角偏离 90° 越多，转移越少。在微线圈里两个脉冲在样品各处都不准，而且各处不准的程度不同，各处的贡献相位和幅度各异，叠加后大部分抵消。相位循环是按理想脉冲设计的，也不能再干净地选出所需的相干路径。剩下的一点信号淹没在噪声里。

> **Background** The simplest COSY is $90^\circ - t_1 - 90^\circ - t_2$ (acquire). Cross peaks arise when the second 90° pulse transfers coherence from one proton to its coupling partner; the further the flip angle is from 90°, the less is transferred. In a microcoil both pulses are wrong everywhere, by different amounts at different points, so the contributions differ in phase and amplitude and largely cancel when summed. Phase cycling, designed for ideal pulses, can no longer select the wanted coherence pathway cleanly either. What little remains is lost in the noise.

$$90^\circ \; - \; t_1 \; - \; 90^\circ \; - \; t_2 \; (\text{acquire})$$

背景：基本 COSY 序列；两个脉冲都须接近 90°

/ background: the basic COSY sequence; both pulses must be close to 90°

图怎么读 / Reading the figure

- 两张图两个轴都是「¹H Chemical Shift」（横轴约 6–0.6，右侧纵轴约 0.6–6 ppm）。幻灯片没有说明样品。
  > Both axes of both plots read "¹H Chemical Shift" (x about 6–0.6, right-hand y about 0.6–6 ppm). The slide does not name the sample.
- 左（5 mm，红色）：对角线从右上（约 0.7 ppm）延伸到左下（约 5.8 ppm），0.8–2.6 ppm 一段最密，4.8–5.8 ppm 另有一组；对角线两侧有成对、关于对角线对称的交叉峰，例如 $3.7 \leftrightarrow 1.2$、$5.15 \leftrightarrow 2.35$、$5.2 \leftrightarrow 1.65$ ppm（读图约值）。
  > Left (5 mm, red): the diagonal runs from the upper right (≈ 0.7 ppm) to the lower left (≈ 5.8 ppm), densest at 0.8–2.6 ppm with another group at 4.8–5.8 ppm; cross peaks come in pairs mirrored about the diagonal, e.g. $3.7 \leftrightarrow 1.2$, $5.15 \leftrightarrow 2.35$, $5.2 \leftrightarrow 1.65$ ppm (approximate readings).
- 右（微线圈，蓝色）：同样的坐标范围里只有零散的小点，看不到对角线，也没有成对的交叉峰。
  > Right (microcoil, blue): over the same ranges there are only scattered small specks, no diagonal and no paired cross peaks.

考点速记 KEY POINTS

- 5 mm COSY：对角峰 + 对称的交叉峰。
  5 mm COSY: diagonal + symmetric cross peaks.
- COSY 交叉峰靠第二个 90° 转移相干（背景）。
  COSY cross peaks need the second 90° transfer (background).
- 微线圈 COSY：无信号，原因是章动差。
  Microcoil COSY: no signal, due to poor nutation.
- 陷阱：失败的根源是 B_1 不均匀，不是磁场 B_0。
  Trap: the cause is B_1 inhomogeneity, not B_0.

S45 · 微线圈 NMR 的难题：放样与适用样品

S45 · Microcoil NMR – Challenges: placement and which samples

幻灯片要点 / What the slide says

把样品放到微线圈上可能困难得多。此外，对大多数样品，微线圈并没有真正的好处：如果分析的是牛奶之类的东西，放进 5 mm 探头很可能得到多得多的信号。另一方面，对大型溞卵（*D. magna* 卵）这类样品，它会非常有用。

> It can be much more challenging to place a sample on a microcoil. Additionally, there is no real benefit for most samples: if you are analyzing something like milk, you will likely get way more signal if you put it into a 5 mm probe. On the other hand, for things like *D. magna* eggs it can be very useful.

为什么牛奶该用 5 mm / Why milk belongs in a 5 mm probe

牛奶不缺：浓度固定，想装多少就装多少，决定信号的是浓度灵敏度 $S_c = S_m V$（S39）。微线圈每个核的灵敏度高，但检测体积小几个数量级。下面用 S46 的数字粗算，只作示意（两处是不同的线圈）：

> Milk is not scarce: its concentration is fixed and you can fill as much as you like, so what counts is concentration sensitivity, $S_c = S_m V$ (S39). A microcoil gains per nucleus but detects a volume orders of magnitude smaller. A rough estimate with the S46 numbers, for illustration only (they belong to a different coil):

$$\frac{\mathrm{SNR}_{\mathrm{micro}}}{\mathrm{SNR}_{5\,\mathrm{mm}}} = \frac{S_{\mathrm{m,micro}}}{S_{\mathrm{m,5\,mm}}} \cdot \frac{V_{\mathrm{micro}}}{V_{5\,\mathrm{mm}}} \approx 76 \times \frac{450\ \mathrm{nL}}{600\ \mu\mathrm{L}} \approx 0.06$$

背景粗算：同浓度、样品充足时，5 mm 信号约多 17 倍

/ background estimate: same concentration, plenty of sample → 5 mm gives ~17× more signal

大型溞卵正好相反：一个卵的总量极少，属于质量受限样品，放进 5 mm 管只会被稀释到测不出；微线圈把灵敏度集中在卵所在的那一点。**背景** 平面线圈只在紧贴表面的薄层里灵敏（S43），所以样品必须准确放在线圈正上方，这就是「放样难」。

> *D. magna* eggs are the opposite: one egg is a tiny total amount, a mass-limited sample that a 5 mm tube would dilute below detection, while a microcoil concentrates its sensitivity right where the egg is. **Background** A planar coil is sensitive only in a thin layer just above its surface (S43), so the sample must sit exactly over the coil; that is why placement is hard.

图怎么读 / Reading the figure

- 标题「50 µm planar microcoil」。照片：绿色基底上两块金色引线相对，交汇处顶着一个球形样品。幻灯片没有说明是什么样品，结合文字应是一个大型溞卵（推断）。

  > Title "50 µm planar microcoil". Photo: two gold leads meet on a green substrate, with a spherical sample sitting where they join. The slide does not name it; from the text it is presumably a *D. magna* egg (inference).

- 谱：横轴 6.0–1.0 ppm。约 1.3 ppm 最高，0.97 与 0.90 ppm 一对，1.6、2.05、2.25（2.35 处有肩峰）、2.8 ppm 中等，4.3、4.1 ppm 很小，5.35 ppm 一个小峰（读图约值）。

  > Spectrum: axis 6.0–1.0 ppm. Tallest at ≈ 1.3 ppm; a pair at 0.97 and 0.90; medium peaks at 1.6, 2.05, 2.25 (shoulder at 2.35) and 2.8; very small ones at 4.3 and 4.1; a small peak at 5.35 ppm (approximate readings).

- **背景** 这组位置与脂质的典型峰一致（1.3 亚甲基链、0.9 甲基、2.0 烯丙位、2.25 羰基 α 位、2.8 双烯丙位、4.1/4.3 甘油骨架、5.35 烯烃 CH），属推断，幻灯片未标注。

  > **Background** These positions match a typical lipid profile (1.3 methylene chain, 0.9 methyl, 2.0 allylic, 2.25 α to carbonyl, 2.8 bis-allylic, 4.1/4.3 glycerol backbone, 5.35 olefinic CH); an inference, not labelled on the slide.

考点速记 KEY POINTS

- 难题：样品难以准确放到微线圈上。
  Challenge: placing the sample on the microcoil.
- 质量受限（如大型溞卵）→ 微线圈很有用。
  Mass-limited (*D. magna* eggs) → microcoil is very useful.
- 样品充足（如牛奶）→ 5 mm 信号多得多。
  Plentiful sample (milk) → far more signal in 5 mm.
- 判据：$S_c = S_m V$，看样品是否受限。
  Criterion: $S_c = S_m V$; is the sample limited?

S46 · 微线圈 NMR 的应用：植入式针式微线圈

S46 · Microcoil NMR – Applications: an implantable needle coil

幻灯片要点 / What the slide says

质量灵敏度的提升可以非常大！这里把一个微线圈做在可植入的针上（植入式针式微线圈），再用它检测大鼠的脑组织。灵敏度约为 5 mm NMR 探头的 100 倍。

> The mass sensitivity improvement can be massive! Here a microcoil was built on an implantable needle, which was then used to examine the brain tissue of a rat. The sensitivity was ~100 times larger than a 5 mm NMR probe.

要点是线圈到了样品里：不取样、不稀释，线圈只「看」针尖周围 $450\ \mathrm{nL}$ 的活组织。它也是质量受限的情形：被测的就是那一小块组织，按 S39 线圈越小，每单位量的信号越大。

> The point is that the coil goes into the sample: no extraction, no dilution, and the coil only "sees" $450\ \mathrm{nL}$ of living tissue around the needle tip. It is a mass-limited case too: what is measured is just that small piece of tissue, and by S39 the smaller coil gives more signal per unit amount.

图怎么读 / Reading the figure

- 标题「Implantable NMR microcoil」；两个红框：「Volume detection : 450 nL」（检测体积）和「Gain in sensitivity (x 76)」（灵敏度增益 ×76）。
  > Title "Implantable NMR microcoil"; two red boxes: "Volume detection : 450 nL" and "Gain in sensitivity (x 76)".
- 中间：「Stereotactic implantation of the NMR microcoil in:」，即用立体定位把微线圈植入「Healthy rat」（健康大鼠）和「C6 glioma-bearing rat」（C6 胶质瘤大鼠）；画有大鼠头部、插入的针和接头照片。底部红框：「Observation of different metabolic profiles at sub-microliter scale」（亚微升尺度观察到不同的代谢谱）。
  > Centre: "Stereotactic implantation of the NMR microcoil in:" a "Healthy rat" and a "C6 glioma-bearing rat"; a drawing of a rat head with the needle and a photo of the connector. Bottom red box: "Observation of different metabolic profiles at sub-microliter scale".
- 两张谱横轴「Chemical Shift (PPM)」，刻度 4、3、2、1。峰标签按图原样，后附约略位置（ppm）：Tau；tCho 约 3.2；tCr 约 3.0；Glu 约 2.35，仅健康谱有标注；NAA 约 2.0；Lac + Lip 约 1.3；Lip 约 0.9。
  > Both spectra: axis "Chemical Shift (PPM)", ticks $4, 3, 2, 1$. Peak labels as printed, with approximate positions (ppm): Tau; tCho ≈ 3.2; tCr ≈ 3.0; Glu ≈ 2.35, labelled in the healthy spectrum only; NAA ≈ 2.0; Lac + Lip ≈ 1.3; Lip ≈ 0.9.
- 健康大鼠：NAA 最高，tCr 第二。胶质瘤大鼠：Lac + Lip 最高，Lip 也很高，NAA 相对变低。**背景** NAA 下降与乳酸、脂质升高是脑肿瘤谱的常见特征。
  > Healthy rat: NAA tallest, tCr second. Glioma rat: Lac + Lip tallest, Lip also high, NAA relatively lower. **Background** Lower NAA with higher lactate and lipid is a common pattern in brain-tumour spectra.

编者注 / Editor's note

文字说灵敏度「约 100 倍」（~100 times），图中写「Gain in sensitivity (x 76)」。两者都按原文保留；要具体数字时引用图中的 ×76，「约 100 倍」是取整的说法。峰标签按图写作 tCr、tCho，另有 Tau。

> The text says "~100 times" larger sensitivity; the figure says "Gain in sensitivity (x 76)". Both are kept as printed; quote the figure's ×76 when a number is needed, "~100 times" is the rounded wording. Peak labels follow the figure: tCr and tCho, plus Tau.

考点速记 KEY POINTS

- 植入式针式微线圈：立体定位植入活体大鼠脑。
  Implantable needle coil: stereotactically placed in a live rat brain.
- 检测体积 $450\ \mathrm{nL}$（亚微升）。
  Detection volume $450\ \mathrm{nL}$ (sub-microlitre).
- 增益：图 ×76，文字「约 100 倍」。
  Gain: ×76 in the figure, "~100 times" in the text.
- 胶质瘤：Lac + Lip 升高，NAA 相对降低。
  Glioma: Lac + Lip up, NAA relatively down.

S47 · 大线圈

S47 · Larger Coils

幻灯片要点 / What the slide says

微线圈的质量灵敏度好，但浓度灵敏度差；直径较大的探头正好相反。探头过去有 8 mm、10 mm 和 20 mm 的版本；由于灵敏度普遍提高，这些如今已不常见。

> A microcoil has a good mass sensitivity but a poor concentration sensitivity. The converse is true for a larger diameter probe. Probes used to be made in 8 mm, 10 mm and 20 mm versions; these are uncommon nowadays as sensitivity has generally improved.

机理：同一条关系的两端 / Mechanism: two ends of one relation

用 S39 的 $S_c = S_m V$ 就能看懂：微线圈 S_m 很高，但 V 只有纳升级，浓度一定时装进去的核很少，所以浓度灵敏度差；大线圈每个核的灵敏度低，但 V 大，装得下更多的核，所以浓度灵敏度好。**背景** 灵敏度普遍提高来自更高的场强、超低温探头（S38）等；常规探头更灵敏之后，需要大直径探头的场合就少了。

> The S39 relation $S_c = S_m V$ explains it: a microcoil has a very high S_m but only a nanolitre-scale V, so at a given concentration it holds few nuclei and its concentration sensitivity is poor; a large coil has a lower sensitivity per nucleus but a large V that holds many more nuclei, so its concentration sensitivity is good. **Background** The general gain in sensitivity comes from higher fields, cryoprobes (S38) and the like; once ordinary probes became more sensitive, fewer jobs needed a large-diameter probe.

线圈 Coil	质量灵敏度 S_m	浓度灵敏度 S_c	适合 Best for
微线圈 **microcoil**	好 good	差 poor	质量受限样品：单个卵、大型溞卵（S41、S45） mass-limited samples: single ova, *D. magna* eggs (S41, S45)
1–3 mm 小管 **1–3 mm tubes**	较好 better	较差 lower	蛋白质、珍贵样品（S38） proteins, precious samples (S38)
5 mm	标准 standard	标准 standard	大多数样品，如牛奶（S45） most samples, e.g. milk (S45)
8–20 mm 大线圈 **8–20 mm coils**	差 poor	好 good	量大但很稀的样品，如雨水（S48） plentiful, dilute samples, e.g. rainwater (S48)

图怎么读 / Reading the figure

- 两张照片，没有比例尺和数据。左：一支细长的核磁管，红色管帽。右：一支较粗的样品管，黄色管帽，管内有一根细的内管，下面接白色的底座。照片只示意不同的管径，读不出具体尺寸。

  > Two photos, no scale bar and no data. Left: a long thin NMR tube with a red cap. Right: a wider tube with a yellow cap and a thin inner tube, on a white base. The photos only illustrate different diameters; no size can be read from them.

考点速记 KEY POINTS

- 微线圈：质量灵敏度好，浓度灵敏度差。
  Microcoil: good mass, poor concentration sensitivity.
- 曾有 8、10、20 mm 探头，如今少见。
  8, 10, 20 mm probes existed; uncommon now.
- 大线圈：正好相反。
  Larger coil: the converse.
- 原因：灵敏度普遍提高（场强、超低温探头，背景）。
  Why: sensitivity improved (fields, cryoprobes; background).

S48 · 大线圈：样品量与直径的平方成正比

S48 · Larger Coils: sample amount scales with diameter squared

幻灯片要点 / What the slide says

一般来说，较大探头能装的样品量与直径的平方成正比。这意味着如果样品很多，用较大的探头会有帮助。这对雨水这类稀样品很有用。

> In general, the amount of sample you can put in a larger probe scales with the square of the diameter. This means that if you have a lot of sample, using a larger probe can help. This can be useful for dilute samples such as rainwater.

为什么是直径的平方 / Why the square of the diameter

检测区可以看成一段圆柱：截面积 $\pi(d/2)^2$ 乘以线圈高度 h。线圈高度基本不变时，体积只随截面积增长，即与 d^2 成正比。直径翻倍，体积变为 4 倍。

> The detection region is roughly a cylinder: cross-section $\pi(d/2)^2$ times the coil height h. With the coil height roughly fixed, the volume grows only with the cross-section, i.e. with d^2. Doubling the diameter gives 4 times the volume.

$$V = \pi \left(\frac{d}{2}\right)^2 h \;\propto\; d^2 \qquad (h \text{ fixed})$$

线圈高度固定时成立 / holds at fixed coil height

$$\frac{V_8}{V_5} = \left(\frac{8}{5}\right)^2 \approx 2.6, \qquad \frac{V_{10}}{V_5} = \left(\frac{10}{5}\right)^2 = 4, \qquad \frac{V_{20}}{V_5} = \left(\frac{20}{5}\right)^2 = 16$$

背景算例：相对 5 mm 探头可多装的样品量 / background example: how much more sample than a 5 mm probe

雨水这类样品量大但溶质极稀，浓度无法提高，只能多装：$S_c = S_m V$，V 变大，浓度灵敏度变好。**背景** 若 SNR 真的翻倍，由 $\mathrm{SNR} \propto \sqrt{N_{\mathrm{scans}}}$，达到同样 SNR 所需扫描次数降为 1/4。但 S49 指出，样品加倍并不会让信号加倍，所以实际收益小于 V 的增长。

> Rainwater is plentiful but its solutes are very dilute; the concentration cannot be raised, so the only option is more volume: $S_c = S_m V$ improves as V grows. **Background** If the SNR really doubled, $\mathrm{SNR} \propto \sqrt{N_{\mathrm{scans}}}$ would cut the scans needed for the same SNR to 1/4. But S49 points out that doubling the sample does not double the signal, so the real gain is smaller than the growth in V.

图怎么读 / Reading the figure

- 照片：一个大口径探头。长长的金属圆筒是探头主体，顶端露出样品管；底部是方形底座，侧面印有「DOTY SCIENTIFIC」，带接头和调谐旋钮。只是照片，没有数据。

  > Photo: a large-bore probe. The long metal cylinder is the probe body with a sample tube at the top; the square base, printed "DOTY SCIENTIFIC", carries connectors and tuning knobs. A photo only; no data.

考点速记 KEY POINTS

- 样品量 $V \propto d^2$（线圈高度固定）。
  Sample amount $V \propto d^2$ (fixed coil height).
- 样品多且稀（如雨水）→ 大探头有帮助。
  Plentiful, dilute (rainwater) → a larger probe helps.
- 10 mm 约装 5 mm 的 4 倍，20 mm 约 16 倍。
  10 mm holds ~4× a 5 mm; 20 mm ~16×.
- 陷阱：是 d^2 不是 d^3；体积翻倍 ≠ 信号翻倍（S49）。
  Trap: d^2, not d^3; double volume ≠ double signal (S49).

S49 · 大线圈的难题：功率、均匀性与射频加热

S49 · Larger Coils – Challenges: power, homogeneity, RF heating

幻灯片要点 / What the slide says

大线圈通常需要大得多的功率，而且场没有那么均匀，这可能导致相当严重的样品加热（射频加热）。此外它们的质量灵敏度更差：放入两倍的样品并不会得到两倍的信号。例如，8 mm 探头因性能差已基本被淘汰。

> Larger coils typically need much more power and do not have as homogeneous fields. This can lead to considerable sample heating. Further, they have poorer mass sensitivity: putting double the amount of sample in does not result in double the signal. As an example, 8 mm probes have been largely phased out due to poor performance.

机理 / Mechanism

背景 要在更大的线圈里得到同样的 B_1（同样的 90° 脉冲宽度），射频能量要充满更大的体积，所需功率大致与线圈体积成正比。射频的电场在导电（含盐）样品中产生损耗而发热；去耦这种长时间连续照射的平均功率最大，加热最严重。场不均匀，加热也不均匀，样品中出现温差。

> **Background** To get the same B_1 (the same 90° pulse length) in a larger coil, the RF energy must fill a larger volume, so the power needed grows roughly with coil volume. The RF electric field deposits energy as heat in conductive (salty) samples; decoupling, a long continuous irradiation, has the highest average power and heats the most. An uneven field heats unevenly, so temperature differences appear across the sample.

$$P \propto B_1^2 V_{\text{coil}} \qquad (\text{same } Q \text{ and frequency})$$

背景粗略关系：同样的 B_1，线圈越大所需功率越大 / rough background relation: same B_1, larger coil, more power

信号为什么不随样品量线性增长：按 S39，线圈变大时每单位量的灵敏度 S_{m} 下降；**背景** 边缘区域的 B_1、B_0 均匀性也更差，那里的样品贡献得少。所以 V 加倍，SNR 增加不到一倍。

> Why the signal does not grow in proportion to the amount: by S39 the sensitivity per unit amount S_{m} falls as the coil grows; **Background** the B_1 and B_0 homogeneity is also worse near the edges, so sample there contributes less. Doubling V therefore less than doubles the SNR.

图怎么读：射频加热造成两个温度群体 / Reading the figure: two temperature populations

- 图标题「Two temperature population from RF heating」，样品：蛋白脂质体（Proteoliposome sample）。横轴「ΔTemperature, K」，刻度 30、10、-10，越往左温升越大。右侧橙色楔形箭头「increase decoupling power」：自上而下去耦功率增大，共 6 条谱。

  > Figure title "Two temperature population from RF heating"; sample: proteoliposome. The x-axis is "ΔTemperature, K" with ticks 30, 10, -10; further left means more heating. An orange wedge on the right, "increase decoupling power": decoupling power rises from top to bottom over 6 traces.

- 顶部（功率最低）：一个峰，约 $\Delta T \approx 3\ \text{K}$。往下：主峰（蓝虚线）逐渐左移，到底部约 $12\ \text{K}$；同时在更热的一侧分出一个小峰（红虚线），底部约 $20\ \text{K}$（读图约值）。

  > Top (lowest power): one peak at about $\Delta T \approx 3\ \text{K}$. Going down, the main peak (blue dashed line) moves left to about $12\ \text{K}$ at the bottom, and a small peak splits off on the hotter side (red dashed line), about $20\ \text{K}$ at the bottom (approximate readings).

- 含义：功率越大，样品整体越热，而且分成两个温度不同的部分（两个温度群体），说明射频加热并不均匀。

  > Meaning: more power heats the whole sample and splits it into two parts at different temperatures (two temperature populations), so RF heating is not uniform.

考点速记 KEY POINTS

- 大线圈：功率大、场不均匀 → 样品加热。
  Large coil: more power, less uniform field → sample heating.
- 去耦功率增大 → 温升增大并分成两个温度群体。
  More decoupling power → more heating, two temperature populations.
- 质量灵敏度更差：样品加倍，信号不到两倍。
  Poorer mass sensitivity: double sample, less than double signal.
- 8 mm 探头因性能差已基本淘汰。
  8 mm probes largely phased out for poor performance.

S50 · 混合物分析技术

S50 · Techniques for Mixture Analysis

讲解 / Explanation

前面讨论的主要是单一化合物，即合成后经纯化得到的样品。但我们常常需要研究混合物：这严格说略超出"解析结构"的范围，却往往更贴近实际；有机化学样品本身也常含杂质。

> So far we have mainly discussed single compounds, as obtained after a synthesis with purification. Often, however, we want to study mixtures. This is arguably a bit outside just figuring out structures, but it is often more realistic, and organic chemistry samples often contain impurities.

难点在于所有组分的峰叠加在同一张谱图上：峰数成倍增加、彼此重叠，强峰掩盖弱峰，而且单看一张谱无法知道哪个峰属于哪个分子。本节（S51–S66）的方法都回答同一个问题：怎样只"看见"想要的化合物，以及为此付出多少信号。

> The difficulty is that every component's peaks fall in the same spectrum: the peak count multiplies, peaks overlap, strong peaks hide weak ones, and one spectrum does not say which peak belongs to which molecule. Every method in this section (S51–S66) answers one question: how to see only the compound you want, and how much signal that costs.

图读：混合物的 ^{13}C 谱 / Reading the figure: ^{13}C spectrum of a mixture

横轴 f1（ppm）在多处截断，只保留有峰的区段（约 138–134、130–124、113、64–62、52、34–18、15–12 ppm）。红色箭头把每个结构连到它的一个峰（按图中所画结构）：≈ 138 ppm 为 1-己烯，≈ 128 ppm 为苯（最高峰之一），≈ 125 ppm 为甲苯，≈ 63 ppm 为异丙醇；另有 2,2,3-三甲基丁烷；26–27 ppm 的两根高峰为环辛烷与环己烷，≈ 22 ppm 为正己烷与正戊烷。

> The f1 (ppm) axis is broken in several places to keep only regions with peaks (about 138–134, 130–124, 113, 64–62, 52, 34–18, 15–12 ppm). Red arrows link each drawn structure to *one* of its peaks: 1-hexene (≈ 138 ppm), benzene (≈ 128 ppm, one of the tallest), toluene (≈ 125 ppm), isopropanol (≈ 63 ppm), 2,2,3-trimethylbutane, cyclooctane and cyclohexane (two tall peaks at 26–27 ppm), *n*-hexane and *n*-pentane (≈ 22 ppm).

每个化合物其实有多个碳峰，未标注的峰（尤其 20–35 ppm 最密的一段）同样来自这些组分。即使在位移范围很宽的 ^{13}C 谱中，归属混合物也要先知道里面有什么。**背景** ^{1}H 只有约 0–12 ppm，重叠更严重，这正是后面各种选择性实验的出发点。

> Each compound has several carbon peaks; the unmarked ones (especially in the crowded 20–35 ppm stretch) also belong to these components. Even in ^{13}C, with its wide shift range, assigning a mixture requires knowing what is in it. **Background** ^{1}H spans only about 0–12 ppm, so overlap is worse; this motivates the selective experiments that follow.

本节路线图 / Section map

幻灯片 Slides	方法 Method	选出什么 What it selects
S51–52	选择性 TOCSY selective TOCSY	一个可分辨峰所在的整条耦合链 the whole coupled chain of one resolved peak
S53–54	单重态 NMR singlet-state NMR	化学位移非常接近的自旋对 spin pairs with (very) similar shifts
S55–58	仅氨基酸 NMR amino-acid-only NMR	含 $^{\beta}HC-^{\alpha}HC-C=O$ 单元的小分子 small molecules with the $^{\beta}HC-^{\alpha}HC-C=O$ motif
S59–60	GEMSTONE	即使重叠也能挑出的单个多重峰 one multiplet, even when overlapped
S61–66	DREAMTIME	位移、耦合与 J 都匹配的指定化合物 chosen compounds matching shift, coupling and J

考点速记 KEY POINTS

- 真实样品多为混合物：所有组分的峰叠在同一谱中。
  Real samples are mixtures: all components share one spectrum.
- ^{1}H 约 0–12 ppm，比 ^{13}C 更易重叠（背景）。
  ^{1}H spans ~0–12 ppm: more overlap than ^{13}C (background).
- 每个分子有多个峰，图中箭头只标其中一个。
  Each molecule has several peaks; arrows mark only one.
- 本节主线：选出目标化合物，并权衡信号损失。
  Theme: select the target, weigh the signal lost.

S51 · 混合物分析：选择性 TOCSY

S51 · Mixture Analysis – selective TOCSY

讲解 / Explanation

TOCSY（全相关谱）告诉我们谱图中一整条耦合链：混合期内磁化沿 J 耦合一步步传遍同一自旋体系的所有质子。选择性 TOCSY（selective TOCSY，一维 TOCSY）可以用来考察混合物中的某些组分：先用选择性脉冲只激发一个特定共振峰，TOCSY 传递就只发生在与该共振峰相连（同一耦合链）的质子上。

> TOCSY tells us about an entire coupled chain: during mixing, magnetization spreads along J couplings to every proton of the same spin system. A selective TOCSY can examine certain components of a mixture: one specific resonance is excited with a selective pulse, and the TOCSY transfer then reaches only protons connected to that resonance (its coupled chain).

前提：被激发的峰必须在混合物谱中可分辨，否则选择性脉冲会同时激发别的化合物（S59 的 GEMSTONE 就是为重叠峰设计的）。**背景** 代价：只得到一个自旋体系；离激发点越远、耦合越弱的质子，分到的信号越少。

> Prerequisite: the excited peak must be resolved in the mixture spectrum, otherwise the selective pulse also excites other compounds (GEMSTONE, S59, is built for overlapped peaks). **Background** Cost: you get one spin system only, and protons further from the excited one, or more weakly coupled, receive less signal.

图读：1D TOCSY for Mixture Analysis / Reading the figure: 1D TOCSY for Mixture Analysis

最下面一条是混合物（Mixture）的普通 ^{1}H 谱：三个化合物的峰在 7.2–7.8 ppm 交错，另有 2.4 ppm 的单峰（横轴在 7.2 与 2.4 ppm 之间截断）。上面三条是选择性 TOCSY，红色箭头标出被激发的峰：

> The bottom trace is the normal ^{1}H spectrum of the mixture: peaks of three compounds interleave at 7.2–7.8 ppm, plus a singlet at 2.4 ppm (the axis is broken between 7.2 and 2.4 ppm). The three traces above are selective TOCSYs; red arrows mark the excited peak:

- 激发 8.6 ppm → 只剩吡啶（含 N 的六元环）的 8.6、7.7、7.3 ppm 三组峰。
  > Excite 8.6 ppm → only pyridine (the ring with N): 8.6, 7.7, 7.3 ppm.
- 激发 7.4 ppm → 只剩邻二氯苯的 7.4 与 7.2 ppm 两组峰。
  > Excite 7.4 ppm → only 1,2-dichlorobenzene: 7.4 and 7.2 ppm.
- 激发 2.4 ppm 的甲苯甲基 → 甲苯的环质子（约 7.2 ppm）出现，但很弱，放大 ×8 才看清。
  > Excite toluene's methyl at 2.4 ppm → toluene's ring protons (≈ 7.2 ppm) appear, but weakly, shown ×8.

考试提醒 / Exam reminder

"选择"靠化学位移（一个可分辨的峰），结果却是整条耦合链。**背景** 甲苯的 CH_3 与环质子之间只有很小的远程耦合，传递效率低，所以要 ×8。

> The selection uses chemical shift (one resolved peak), but the output is the whole coupled chain. **Background** Toluene's CH_3 has only small long-range couplings to the ring protons, so transfer is weak, hence ×8.

考点速记 KEY POINTS

- 选择性 TOCSY = 选择性激发一个峰 + TOCSY 混合。
  Selective TOCSY = excite one peak + TOCSY mixing.
- 前提：被激发的峰不能与他峰重叠。
  Prerequisite: the excited peak must be resolved.
- 输出：该峰所属自旋体系的全部质子。
  Output: every proton of that spin system.
- 甲苯 CH_3→环质子传递弱，需 ×8。
  Toluene CH_3→ring transfer is weak: shown ×8.

S52 · 混合物分析：选择性 TOCSY（2-戊烯醛）

S52 · Mixture Analysis – selective TOCSY (2-pentenal)

讲解 / Explanation

这是另一个谱更宽的例子。普通质子谱显示混合物中所有组分的峰；蓝色谱是选择性 TOCSY（全相关谱），只显示 2-戊烯醛。注意它与纯化合物谱之间的强度差别。

> Another example with a larger spectrum. The normal proton spectrum shows peaks from everything in the mixture; the blue spectrum, a selective TOCSY, shows only 2-pentenal. Note the difference in intensities compared with the pure compound.

为什么强度不同？**背景** 选择性 TOCSY 的峰强不按质子数分配，而取决于从被激发质子出发的传递效率：被激发的峰和与它强耦合的近邻最强，链远端的质子较弱，比例还随混合时间改变。所以选择性 TOCSY 用来**识别和归属**组分，而不能像普通一维谱那样靠积分定量。

> Why do intensities differ? **Background** In a selective TOCSY, intensity does not follow proton count; it follows transfer efficiency from the excited proton: the excited peak and its strongly coupled neighbours are strongest, protons further down the chain are weaker, and the ratios change with mixing time. A selective TOCSY therefore **identifies and assigns** a component; it cannot quantify by integrals like a normal 1D spectrum.

图读：三条谱，横轴 12.0–0 ppm / Reading the figure: three traces, axis 12.0–0 ppm

- 底部红色 "1D proton of mixture"：混合物普通谱。除 2-戊烯醛的峰外，5.0–5.5 ppm 和 1.5–3.0 ppm 还有大量其他组分的峰。
  > Bottom, red, "1D proton of mixture": besides 2-pentenal, many peaks of other components at 5.0–5.5 and 1.5–3.0 ppm.
- 中间绿色 "1D proton of 2-pentenal"：纯物参比。峰在约 9.5（醛基，双峰）、6.0–7.0、2.3 和 1.1 ppm，其中 1.1 ppm 一组最高。
  > Middle, green, "1D proton of 2-pentenal": the pure reference, with peaks at ≈ 9.5 (aldehyde doublet), 6.0–7.0, 2.3 and 1.1 ppm; the 1.1 ppm group is tallest.
- 顶部蓝色 "1D TOCSY of mixture"：其他组分全部消失，只剩 2-戊烯醛的峰；但约 9.5 ppm 的醛基峰远高于其余，2.3 与 1.1 ppm 的峰比纯物谱弱得多。
  > Top, blue, "1D TOCSY of mixture": all other components are gone, only 2-pentenal remains; but the aldehyde peak at ≈ 9.5 ppm towers over the rest, and the 2.3 and 1.1 ppm peaks are much weaker than in the pure spectrum.

编者注 / Editor's note

幻灯片没有标出被激发的是哪个峰。从强度分布看最可能是醛基峰（它远离其他峰，也最高），这是推断，不是幻灯片的说法。

> The slide does not mark which peak was excited. The intensity pattern suggests the aldehyde peak (isolated from other peaks and tallest); this is an inference, not a statement of the slide.

考点速记 KEY POINTS

- 普通谱 = 全部组分；选择性 TOCSY = 只有 2-戊烯醛。
  Normal spectrum: everything; selective TOCSY: 2-pentenal only.
- 用于识别与归属，不用于积分定量。
  Use it to identify and assign, not to integrate.
- TOCSY 峰强 ≠ 质子数，取决于传递效率（背景）。
  TOCSY intensity ≠ proton count; it follows transfer efficiency.
- 醛基峰 ≈ 9.5 ppm，远离混合物其他峰。
  The aldehyde at ≈ 9.5 ppm is clear of other peaks.

S53 · 单重态 NMR：用作过滤器

S53 · Singlet State NMR as a filter

讲解 / Explanation

单重态（singlet state）NMR 的理论相当复杂，但要点是：如果两个自旋的化学位移（非常）接近，就可以把它们制备到单重态。这可以当作过滤器：选出想要的化合物，去掉其余的。本例还加了一个 TOCSY（全相关谱）模块来帮助峰归属。

> The theory behind singlet-state NMR is quite complex, but essentially, if two spins have (very) similar chemical shifts, they can be put into the singlet state. This can be used as a filter: we choose the compounds we want and get rid of the others. Here a TOCSY block is added to help assign peaks.

背景 两个耦合的自旋-½ 核组成一个单重态（总自旋 0）和三个三重态。单重态不能被直接观测，也不受这两个核之间偶极弛豫的影响。序列先把磁化转入单重态，用过滤步骤消除其余所有信号，再转回可观测磁化。只有具备"近等价自旋对"的分子能走完这条路径，因此它成为化合物选择器；加上 TOCSY 后，信号再传给同一分子的其他质子。

> **Background** Two coupled spin-½ nuclei form one singlet (total spin 0) and three triplet states. The singlet is not directly observable and is immune to the pair's mutual dipolar relaxation. The sequence moves magnetization into the singlet, filters out everything else, then converts it back to observable magnetization. Only molecules with a near-equivalent spin pair complete this path, so it selects compounds; the TOCSY then passes the signal to the molecule's other protons.

图读：(a)、(b) 两组对照 / Reading the figure: panels (a) and (b)

两组上方黑色都是同一样品的标准 ^{1}H 谱（Standard ^{1}H NMR），横轴化学位移约 8.5–0.5 ppm；下方蓝色是单重态选择谱，彩色竖条标出属于各氨基酸的峰，最下面的彩色竖线标出各峰位置（幻灯片未说明其来源）。

> In both panels the black top trace is the standard ^{1}H spectrum of the same sample (shift axis about 8.5–0.5 ppm); the blue trace is the singlet selection, coloured bands mark peaks belonging to each amino acid, and coloured sticks at the bottom mark the peak positions (their source is not stated).

- (a) Singlet selection (Met, Lys)：红色 = 赖氨酸（Lys，约 3.8、3.0、1.4–1.9 ppm）；绿色 = 甲硫氨酸（Met，约 3.9、2.6、2.1 ppm）。5 ppm 附近仍有一小段残余信号。
  > (a) Met, Lys: red = lysine (≈ 3.8, 3.0, 1.4–1.9 ppm); green = methionine (≈ 3.9, 2.6, 2.1 ppm). A small residual remains near 5 ppm.
- (b) Singlet selection (Phe, Glu, Leu)：绿色 = 苯丙氨酸（Phe，约 7.3 ppm 芳香峰）；红色 = 谷氨酸（Glu，约 3.8 与 2.0–2.4 ppm）；橙色 = 亮氨酸（Leu，约 1.7 与 1.0 ppm）。
  > (b) Phe, Glu, Leu: green = phenylalanine (≈ 7.3 ppm, aromatic); red = glutamic acid (≈ 3.8 and 2.0–2.4 ppm); orange = leucine (≈ 1.7 and 1.0 ppm).

在标准谱中这些峰淹没在大量重叠峰里；选择谱基线平坦，只留下选中的几种氨基酸，而且一次实验可以同时选多种。

> In the standard spectrum these peaks are buried among many overlapping ones; the selection gives a flat baseline with only the chosen amino acids, and one experiment can select several at once.

考点速记 KEY POINTS

- 条件：两自旋化学位移（非常）接近 → 可制备单重态。
  Condition: two spins with (very) similar shifts.
- 加 TOCSY 模块：信号传到同一分子其他质子，助归属。
  An added TOCSY block spreads signal to aid assignment.
- 单重态 = 过滤器：留下目标化合物，去掉其余。
  Singlet state = filter: keep targets, remove the rest.
- 一次可选多种：(Met, Lys)、(Phe, Glu, Leu)。
  Several per run: (Met, Lys), (Phe, Glu, Leu).

S54 · 单重态 NMR：缺氧胁迫的时间序列

S54 · Singlet State NMR: an anoxic-stress time course

讲解 / Explanation

这是生物体缺氧胁迫的例子：随着时间推移，葡萄糖和氨基酸的强度上升，因为生物体在分解储能分子获取能量。方法并不完美：仍有泄漏（残余信号穿透过滤器）。优点是不需要成形脉冲，在很老的硬件上也能运行。

> An example of anoxic stress in an organism: over time glucose and amino acids increase in intensity as the organism breaks down storage molecules for energy. It is not perfect: there is still breakthrough. No shaped pulses are needed, so it works even on very old hardware.

背景 成形（选择性）脉冲要求谱仪精确调制射频的幅度和相位，老式谱仪往往做不到；只用硬脉冲和延迟的序列几乎在任何谱仪上都能跑。泄漏说明过滤不是 $100\,\%$：强的非目标峰会留下残余，读图时要区分真实信号与残余。

> **Background** Shaped (selective) pulses need precise RF amplitude and phase modulation, which old spectrometers often lack; a sequence of hard pulses and delays runs almost anywhere. Breakthrough means the filter is not $100\,\%$: strong unwanted peaks leave residues, so tell real signals from leftovers when reading the spectrum.

图读 / Reading the figure

- a) 堆叠谱：最前面是普通 $^{1}\mathrm{H}$ 谱（Normal $^{1}\mathrm{H}$ NMR），其后为时间（hours）$0, 2.3, 4.6, \ldots, 20.7\ \mathrm{h}$ 的单重态过滤谱，约每 $2.3\ \mathrm{h}$ 一条。横轴 8.0–$0.5\ \mathrm{ppm}$，在 6.5 与 $4.5\ \mathrm{ppm}$ 之间截断。3.0–$4.0\ \mathrm{ppm}$ 一带的信号随时间明显升高。

  > a) Stacked spectra: in front the normal $^{1}\mathrm{H}$ spectrum, behind it singlet-filtered spectra at $0, 2.3, 4.6, \ldots, 20.7\ \mathrm{h}$ (one every $\approx 2.3\ \mathrm{h}$). Axis 8.0–$0.5\ \mathrm{ppm}$, broken between 6.5 and $4.5\ \mathrm{ppm}$. Signals at 3.0–$4.0\ \mathrm{ppm}$ grow clearly with time.

- 普通谱上两个方框标 $128\times$（约 7.6–$6.9\ \mathrm{ppm}$）和 $16\times$（约 4.0–$3.0\ \mathrm{ppm}$），即这些区域按该倍数放大显示：目标信号在普通谱里本来很弱。

  > Two boxes on the normal spectrum read $128\times$ (≈ 7.6–$6.9\ \mathrm{ppm}$) and $16\times$ (≈ 4.0–$3.0\ \mathrm{ppm}$): those regions are drawn magnified by that factor, so the target signals are weak in the normal spectrum.

- b) 7.5–$7.3\ \mathrm{ppm}$ 与 c) 4.0–$3.25\ \mathrm{ppm}$ 的放大图：各有一条蓝色谱在上、一条黑色线状谱在下。幻灯片没有标注二者，可理解为测得谱与参比谱的对照（推断）。

  > Insets b) 7.5–$7.3\ \mathrm{ppm}$ and c) 4.0–$3.25\ \mathrm{ppm}$: a blue trace above a black line spectrum. The slide does not label them; read them as measured vs reference (inference).

考试提醒 / Exam reminder

本页的三个关键词：缺氧 → 分解储能分子 → 葡萄糖、氨基酸上升；缺点是泄漏；优点是无需成形脉冲、老硬件可用。幻灯片没有说明是哪种生物，不要自行补充。

> Three keys: anoxia → storage molecules broken down → glucose and amino acids rise; drawback: breakthrough; advantage: no shaped pulses, old hardware is fine. The slide does not name the organism; do not add one.

考点速记 KEY POINTS

- 缺氧：分解储能分子 → 葡萄糖、氨基酸随时间升高。
  Anoxia: storage molecules broken down → glucose, amino acids rise.
- 时间序列 0–$20.7\ \mathrm{h}$，约每 $2.3\ \mathrm{h}$ 一谱。
  Time series 0–$20.7\ \mathrm{h}$, one spectrum per $\approx 2.3\ \mathrm{h}$.
- 缺点：泄漏，过滤不完全。
  Drawback: breakthrough; filtering is incomplete.
- 优点：无需成形脉冲，老硬件也能用。
  Advantage: no shaped pulses; runs on old hardware.

S55 · 仅氨基酸 NMR：相干转移与结构单元

S55 · Amino Acid Only NMR: coherence transfer and the motif

讲解 / Explanation

仅氨基酸 NMR（amino-acid-only NMR）依靠向不同原子的相干转移工作。它源自一个名为 $(\mathrm{H})\mathrm{C}_\beta\mathrm{C}_\alpha\mathrm{CO}(\mathrm{C}_\alpha)\mathrm{H}_\alpha$ 的三维实验，本质上检测蛋白质或氨基酸中连在 α 碳上的质子 H_α。它只选出具有结构单元 $^{\beta}\mathrm{HC}-{}^{\alpha}\mathrm{HC}-\mathrm{C}=\mathrm{O}$ 的分子，选择通过频带选择性脉冲实现。

> The amino-acid-only experiment works by coherence transfer to different atoms. It is based on a 3D experiment called $(\mathrm{H})\mathrm{C}_\beta\mathrm{C}_\alpha\mathrm{CO}(\mathrm{C}_\alpha)\mathrm{H}_\alpha$ and essentially detects the proton on the alpha carbon, H_α, of a protein or amino acid. It selects only molecules with the structural motif $^{\beta}\mathrm{HC}-{}^{\alpha}\mathrm{HC}-\mathrm{C}=\mathrm{O}$, through band-selective pulses.

背景 名称读法：括号内的原子只作转移中继、不记录其化学位移。频带选择性脉冲分别只作用于 C_β、C_α、C' 的位移范围，每一步都要求下一个碳落在对应频带内并与上一个碳直接相连。缺任何一环（β 碳上没有 H、α 碳旁没有 $\mathrm{C}=\mathrm{O}$），相干就到不了 H_α，该分子不出信号。$\mathrm{C}-\mathrm{C}$ 之间的转移要求相邻碳都是 $^{13}\mathrm{C}$，这类实验通常用于 $^{13}\mathrm{C}$ 富集的样品；幻灯片没有说明样品的标记情况。

> **Background** In the name, atoms in parentheses relay the transfer but their shifts are not recorded. Band-selective pulses act only on the C_β, C_α and C' shift ranges, and each step needs the next carbon in its band and directly bonded to the previous one. If any link is missing (no H on the β carbon, no $\mathrm{C}=\mathrm{O}$ next to the α carbon), coherence never reaches H_α and the molecule gives no signal. $\mathrm{C}-\mathrm{C}$ transfer needs adjacent $^{13}\mathrm{C}$ nuclei, so such experiments are usually run on $^{13}\mathrm{C}$-enriched samples; the slide does not state the labelling.

图读：转移路径（红色箭头 1–5） / Reading the figure: transfer path (red arrows 1–5)

$$\mathrm{H}_\beta \xrightarrow{1} \mathrm{C}_\beta \xrightarrow{2} \mathrm{C}_\alpha \xrightarrow{3} \mathrm{C}' \xrightarrow{4} \mathrm{C}_\alpha \xrightarrow{5} \mathrm{H}_\alpha$$

第 1 步还画了 $\mathrm{H}_\alpha \to \mathrm{C}_\alpha$；第 5 步在 H_α 上检测 / step 1 also shows $\mathrm{H}_\alpha \to \mathrm{C}_\alpha$; step 5 is detected on H_α

上图是一个氨基酸骨架片段（N、NH、H_α、C_α、$\mathrm{C}'=\mathrm{O}$、C_β 和两个 H_β），箭头和数字给出上式的顺序；磁化从质子出发经 C_β、C_α 到羰基碳 C' 再返回。粗箭头下方的图用红色虚线框住 C_β–C_α：这就是被选中的结构单元。

> The upper drawing is an amino-acid backbone fragment (N, NH, H_α, C_α, $\mathrm{C}'=\mathrm{O}$, C_β and two H_β); arrows and numbers give the order above, from the protons through C_β and C_α to the carbonyl carbon C' and back. Below the bold arrow, a red dashed box encloses C_β–C_α: the selected motif.

考点速记 KEY POINTS

- 基础实验：三维 $(\mathrm{H})\mathrm{C}_\beta\mathrm{C}_\alpha\mathrm{CO}(\mathrm{C}_\alpha)\mathrm{H}_\alpha$。
  Parent: the 3D $(\mathrm{H})\mathrm{C}_\beta\mathrm{C}_\alpha\mathrm{CO}(\mathrm{C}_\alpha)\mathrm{H}_\alpha$.
- 手段：频带选择性脉冲 + 相干转移。
  Tools: band-selective pulses + coherence transfer.
- 检测 H_α；只选 $^{\beta}\mathrm{HC}-{}^{\alpha}\mathrm{HC}-\mathrm{C}=\mathrm{O}$ 单元。
  Detects H_α; selects only $^{\beta}\mathrm{HC}-{}^{\alpha}\mathrm{HC}-\mathrm{C}=\mathrm{O}$.
- 顺序：$\mathrm{H}_\beta \to \mathrm{C}_\beta \to \mathrm{C}_\alpha \to \mathrm{C}' \to \mathrm{C}_\alpha \to \mathrm{H}_\alpha$。
  Order: $\mathrm{H}_\beta \to \mathrm{C}_\beta \to \mathrm{C}_\alpha \to \mathrm{C}' \to \mathrm{C}_\alpha \to \mathrm{H}_\alpha$.

S56 · 仅氨基酸 NMR：用切趾去除蛋白质

S56 · Amino Acid Only NMR: apodization removes proteins

讲解 / Explanation

严格说，仅凭这个实验区分不了蛋白质和氨基酸：蛋白质的每个残基同样带有 $^{\beta}\mathrm{HC}-^{\alpha}\mathrm{HC}-\mathrm{C}=\mathrm{O}$ 单元。于是用切趾（apodization，窗函数）去掉蛋白质，只关注氨基酸。这很有用，因为氨基酸常是生物体健康状况的指标。

> Strictly speaking, this experiment alone would not differentiate proteins from amino acids: every protein residue also carries the $^{\beta}\mathrm{HC}-^{\alpha}\mathrm{HC}-\mathrm{C}=\mathrm{O}$ motif. Apodization (a window function) is used to remove the proteins and focus only on the amino acids. This is very useful, as amino acids are often indicators of health in an organism.

背景 为什么窗函数能分开二者：蛋白质分子大、翻滚慢、T_2 短，FID 很快衰减，信号集中在 FID 开头；游离氨基酸分子小、T_2 长，FID 持续很久。开头为零、中段最大的窗函数会把快衰减的蛋白质信号几乎全部压掉，而保留长寿命的氨基酸信号。代价：FID 开头本是信号最强的部分，压低后 SNR 下降。

> **Background** Why a window separates them: proteins are large, tumble slowly and have short T_2, so their FID dies quickly and their signal sits at the start of the FID; free amino acids are small with long T_2, so their FID lasts. A window that is zero at the start and largest mid-FID suppresses the fast-decaying protein signal and keeps the long-lived amino-acid signal. Cost: the start of the FID carries the most signal, so SNR drops.

图读：A) 与 B) 两张二维谱 / Reading the figure: 2D spectra A) and B)

两张都是 $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ 二维谱：横轴 $^{1}\mathrm{H}$ 4.4–3.0 ppm，纵轴 $^{13}\mathrm{C}$ 20–70 ppm。图例：黑色 = 氨基酸（Amino Acids），红色 = 蛋白质（Proteins），方框 = 乳酸（Lactic Acid）。

> Both are $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ 2D spectra: $^{1}\mathrm{H}$ 4.4–3.0 ppm across, $^{13}\mathrm{C}$ 20–70 ppm down. Legend: black = amino acids, red = proteins, boxes = lactic acid.

- A)：右上插图是快速衰减的 FID 和一条单调下降的权重曲线；谱中红色蛋白质峰（多为横向拉长的宽峰）与黑色氨基酸峰混在一起。
  > A): the inset shows a fast-decaying FID with a monotonically falling weighting curve; red protein peaks (mostly broad, smeared along $^{1}\mathrm{H}$) mix with black amino-acid peaks.
- B)：插图中的窗函数变成钟形（开头为零、中间最大），加权后的 FID 开头被压低；谱中红色蛋白质峰全部消失，只剩黑色氨基酸峰和方框里的乳酸峰。
  > B): the window is now bell-shaped (zero at the start, largest in the middle) and the weighted FID is suppressed at the start; every red protein peak is gone, leaving black amino-acid peaks and the boxed lactic-acid peaks.

考试提醒 / Exam reminder

去除蛋白质靠的是**数据处理**（切趾），不是脉冲序列本身；脉冲序列负责按结构单元选择，切趾负责按弛豫快慢（分子大小）区分。

> Proteins are removed by **processing** (apodization), not by the pulse sequence: the sequence selects by motif, the apodization separates by relaxation rate (molecular size).

考点速记 KEY POINTS

- 序列本身分不清蛋白质与氨基酸（骨架也含该单元）。
  The sequence alone cannot tell proteins from amino acids.
- 蛋白质 T_2 短、FID 早衰；开头为零的窗压掉它（背景）。
  Short-T_2 proteins decay early; a window zero at the start removes them.
- 用切趾（窗函数）去掉蛋白质信号。
  Apodization (window function) removes the protein signals.
- 代价：丢掉 FID 开头的强信号，SNR 降低（背景）。
  Cost: the strong early FID is discarded, lowering SNR.

S57 · 仅氨基酸 NMR：显出被掩盖的峰

S57 · Amino Acid Only NMR: revealing buried peaks

讲解 / Explanation

在这个例子中，仅氨基酸序列显示出原本会被更强共振峰掩盖的峰，差别相当惊人！乳酸也能被看到，因为它具有类似的结构单元。

> In this case the amino-acid-only sequence shows peaks that would otherwise be buried underneath the more intense resonances. The difference is quite staggering! Lactic acid can also be seen, as it has a similar structural motif.

乳酸为什么能通过：乳酸 $CH_3-CH(OH)-COOH$ 中，甲基碳带 H（相当于 $^{\beta}HC$），连 OH 的碳带 H（相当于 $^{\alpha}HC$），并紧接羧基 $C{=}O$，正好符合 $^{\beta}HC-^{\alpha}HC-C{=}O$。过滤器检验的是这种连接方式，并不要求有氨基，所以"仅氨基酸"实际是"仅含此结构单元的分子"（蛋白质再由 S56 的切趾去掉）。

> Why lactic acid passes: in $CH_3-CH(OH)-COOH$ the methyl carbon carries H (the $^{\beta}HC$), the $CH(OH)$ carbon carries H (the $^{\alpha}HC$), and it is bonded to the carboxyl $C{=}O$, so it fits $^{\beta}HC-^{\alpha}HC-C{=}O$ exactly. The filter tests this connectivity, not the amino group, so "amino acid only" really means "only molecules with this motif" (proteins are then removed by the S56 apodization).

图读：A) 常规谱 vs C) 仅氨基酸谱 / Reading the figure: A) standard vs C) amino-acid-only

- A)：常规 ^{1}H–^{13}C 二维谱（^{1}H 6–1 ppm，^{13}C 20–70 ppm），一大片连续等高线，几乎没有可分辨的单个峰。
  > A): a standard ^{1}H–^{13}C 2D spectrum (^{1}H 6–1 ppm, ^{13}C 20–70 ppm): one large continuous contour mass with almost no resolved peaks.
- C)：仅氨基酸谱（^{1}H 4.4–3.4 ppm）：一个个分立的小峰；图例黑 = 氨基酸、红 = 蛋白质、方框 = 乳酸，仍可见少量浅红色等高线。
  > C): the amino-acid-only spectrum (^{1}H 4.4–3.4 ppm): separate small peaks; legend black = amino acids, red = proteins, box = lactic acid; a few faint red contours remain.
- 两个方框（乳酸）位于 $^{1}H \approx 4.1$ ppm，$^{13}C \approx 20$ 与 ≈ 70 ppm：同一个 H_α 对应两个碳峰。
  > The two boxes (lactic acid) sit at $^{1}H \approx 4.1$ ppm, $^{13}C \approx 20$ and ≈ 70 ppm: one H_α, two carbon peaks.
- C) 的 ^{1}H 窗口只有 1 ppm 宽，即 H_α 所在区域（背景）。幻灯片只有 A)、C) 两图，右上角孤立的 "B" 没有对应图。
  > C)'s ^{1}H window is only 1 ppm wide, the H_α region (background). The slide has panels A) and C) only; a stray "B" at top right has no panel.

考点速记 KEY POINTS

- 常规 2D 中被强峰掩盖的峰在仅氨基酸谱中分立可见。
  Peaks buried in the standard 2D become resolved.
- 过滤看 $^{\beta}HC-^{\alpha}HC-C{=}O$ 连接，不看氨基。
  The filter tests connectivity, not the amino group.
- 乳酸 $CH_3CH(OH)COOH$ 含同一单元，也会通过。
  Lactic acid has the same motif, so it passes.
- 乳酸峰：$\delta_H \approx 4.1$；$\delta_C \approx 20$、70 ppm。
  Lactic acid: $\delta_H \approx 4.1$; $\delta_C \approx 20$, 70 ppm.

S58 · 仅氨基酸 NMR：全部氨基酸归属与 BPA 暴露

S58 · Amino Acid Only NMR: every amino acid assigned; BPA exposure

讲解 / Explanation

在一个代谢物提取液中，所有氨基酸都可以被归属。该方法被用来追踪双酚 A（BPA）暴露的影响。谱图简单、峰分立，就能比较不同样品中每种氨基酸的峰，看暴露是否改变了代谢（S56：氨基酸是健康指标）。

> All of the amino acids were assignable in a metabolite extract. The method was used to track the impacts of exposure to BPA (bisphenol A). With a simple spectrum of separate peaks, each amino acid can be compared between samples to see whether exposure changed metabolism (S56: amino acids are health indicators).

图读：两张已归属的二维谱 A)、B) / Reading the figure: assigned 2D spectra A) and B)

两图都是仅氨基酸谱，横轴 $^{1}\mathrm{H}$ 4.4–3.4 ppm，纵轴 $^{13}\mathrm{C}$ 20–70 ppm，红色箭头逐个标注：丙氨酸、缬氨酸、亮氨酸、异亮氨酸、苏氨酸、丝氨酸、天冬氨酸、天冬酰胺、谷氨酸、谷氨酰胺、赖氨酸、精氨酸、组氨酸、苯丙氨酸、酪氨酸、色氨酸、甲硫氨酸、半胱氨酸、脯氨酸、甘氨酸，以及乳酸。

> Both are amino-acid-only spectra ($^{1}\mathrm{H}$ 4.4–3.4 ppm, $^{13}\mathrm{C}$ 20–70 ppm) with red arrows labelling alanine, valine, leucine, isoleucine, threonine, serine, aspartic acid, asparagine, glutamic acid, glutamine, lysine, arginine, histidine, phenylalanine, tyrosine, tryptophan, methionine, cysteine, proline, glycine, and lactic acid.

- 多数氨基酸被标两次：一个峰在 $^{13}\mathrm{C} \approx 20$–$45$ ppm，另一个在 ≈ 50–70 ppm，两峰的 $^{1}\mathrm{H}$ 位移相同。**背景** 即同一个 H_α 分别与 C_β、C_α 的相关峰（苏氨酸、丝氨酸的 C_β 带 OH，两个峰都在下半区）。

  > Most amino acids are labelled twice: one peak at $^{13}\mathrm{C} \approx 20$–$45$ ppm, one at ≈ 50–70 ppm, at the same $^{1}\mathrm{H}$ shift. **Background** These are the same H_α correlated with C_β and with C_α (threonine and serine carry OH on C_β, so both their peaks lie in the lower band).

- 甘氨酸只有一个标注峰（$^{13}\mathrm{C} \approx 42$ ppm）；乳酸在 $^{1}\mathrm{H} \approx 4.1$ ppm 有上下两个峰。

  > Glycine has a single labelled peak ($^{13}\mathrm{C} \approx 42$ ppm); lactic acid has an upper and a lower peak at $^{1}\mathrm{H} \approx 4.1$ ppm.

- B) 还标出谷胱甘肽（glutathione）；左侧一条窄框是 4.50–4.55 ppm 区域的插入片段，在主窗口之外。

  > B) also labels glutathione; a narrow strip on the left is an inset of the 4.50–4.55 ppm region, outside the main window.

注意 / Caution

幻灯片没有说明 A)、B) 各是哪个样品（对照还是 BPA 暴露），也没有给出暴露的结果；不要从两图的差别自行推断 BPA 的影响。

> The slide does not say which panel is which sample (control or BPA-exposed) and gives no exposure result; do not infer BPA effects from differences between the panels.

考点速记 KEY POINTS

- 代谢物提取液中全部氨基酸可归属。
  All amino acids assignable in a metabolite extract.
- 多数氨基酸：同一 δ_{H} 上两个 $^{13}\mathrm{C}$ 峰（背景：C_β、C_α）。
  Most amino acids: two $^{13}\mathrm{C}$ peaks at one δ_{H} (C_β, C_α).
- 应用：追踪双酚 A（BPA）暴露的影响。
  Application: tracking the impact of BPA exposure.
- 陷阱：幻灯片未说明 A)/B) 哪个是暴露组。
  Trap: the slide does not say which panel is exposed.

S59 · GEMSTONE：定义与原理

S59 · GEMSTONE: definition and principle

讲解 / Explanation

GEMSTONE（梯度增强多重峰选择性靶向观测 NMR 实验，Gradient-Enhanced Multiplet-Selective Targeted-Observation NMR Experiment）可以选出想要的峰，即使它们彼此重叠，这一点与选择性 TOCSY（全相关谱）相反。它像超快 NMR（UF）一样，用梯度对样品管的不同部分进行空间编码。

> GEMSTONE, the Gradient-Enhanced Multiplet-Selective Targeted-Observation NMR Experiment, allows selection of desired peaks even if they overlap (contrary to selective TOCSY). It encodes different parts of the tube using gradients, similarly to ultrafast (UF) NMR.

背景 空间编码的意思：在梯度开着时施加选择性脉冲，管中不同高度的切片感受到不同的频率偏置，所以各切片被激发的频率略有不同。经过后续步骤，序列只保留目标质子的完整多重峰，而位移只差一点、与之重叠的多重峰被去掉。和 S21 的超快 NMR 一样，"位置 ↔ 频率"的对应是由梯度建立的。

> **Background** Spatial encoding means a selective pulse is applied while a gradient is on, so slices at different heights in the tube see different frequency offsets and are excited at slightly different frequencies. After the later steps, only the full multiplet of the target proton survives, and overlapping multiplets with slightly different shifts are removed. As in ultrafast NMR (S21), the gradient sets up the position ↔ frequency mapping.

图读（S60 使用同一张图） / Reading the figure (S60 shows the same figure)

- 横轴约 5.38–5.24 ppm，只有约 0.14 ppm 宽。最下方黑色 "Conventional ^{1}H Acquisition"（常规采集）：5.32–5.24 ppm 内一串连成片的谱线，看不出有几个多重峰。

  > The axis runs about 5.38–5.24 ppm, only ≈ 0.14 ppm wide. Bottom, black, "Conventional ^{1}H Acquisition": a continuous run of lines over 5.32–5.24 ppm; you cannot tell how many multiplets there are.

- 上方三条 GEMSTONE 谱分别以 5.255（蓝）、5.285（绿）、5.305 ppm（红）为目标，每条只剩一个干净的多重峰，位置正好在目标位移附近；三个中心只相差 0.02–0.03 ppm。

  > The three GEMSTONE traces target 5.255 (blue), 5.285 (green) and 5.305 ppm (red); each keeps one clean multiplet at its target shift, and the three centres differ by only 0.02–0.03 ppm.

- 右侧三个结构（**背景**：quinine 类金鸡纳生物碱）：彩色 H 是 $\mathrm{CH(OH)}$ 碳上被选中的质子，颜色与谱线对应。两个结构带甲氧基 $\mathrm{OCH_3}$、一个不带；两个带甲氧基的结构在该碳上的立体化学画法相反。

  > Three structures on the right (**background**: quinine-type cinchona alkaloids): the coloured H on the $\mathrm{CH(OH)}$ carbon is the selected proton, colour-matched to its trace. Two carry a methoxy $\mathrm{OCH_3}$, one does not; the two methoxy structures are drawn with opposite stereochemistry at that carbon.

考点速记 KEY POINTS

- GEMSTONE：从重叠峰中选出单个多重峰。
  GEMSTONE picks one multiplet out of overlap.
- 选择性 TOCSY 做不到：它要求激发峰可分辨。
  Selective TOCSY cannot: it needs a resolved peak.
- 手段：梯度对管内不同部分空间编码，类似 UF NMR。
  Gradient spatial encoding of the tube, as in UF NMR.
- 目标 5.255、5.285、5.305 ppm，仅差 0.02–0.03 ppm。
  Targets 5.255, 5.285, 5.305 ppm: only 0.02–0.03 ppm apart.

S60 · GEMSTONE：结果与灵敏度损失

S60 · GEMSTONE: the result and its sensitivity penalty

讲解 / Explanation

这是 GEMSTONE（梯度增强多重峰选择性靶向观测 NMR 实验）结果的例子：尽管峰重叠，GEMSTONE 谱仍能拉出想要的多重峰。但存在灵敏度损失，对快弛豫的样品可能相当大。

> Here is an example of a GEMSTONE result: despite the overlap of the peaks, the GEMSTONE spectrum can pull out the desired multiplet. There is a sensitivity penalty, which can be considerable for fast-relaxing samples.

背景 损失的来源：选择步骤（梯度空间编码加选择性脉冲）本身就丢弃了部分信号；序列中的选择性元件较长，期间横向磁化按 T_2 衰减，T_2 越短（快弛豫）损失越大。收获：每条谱只含一个多重峰，可以直接读出它的裂分和耦合常数，这在常规谱里做不到。

> **Background** Where the loss comes from: the selection itself (gradient spatial encoding plus selective pulses) discards part of the signal, and the selective elements are long, so transverse magnetization decays with T_2 during them; the shorter T_2 (fast relaxing), the larger the loss. The gain: each trace holds one multiplet, so its splitting and coupling constants can be read directly, which the conventional spectrum does not allow.

编者注 / Editor's note

S60 与 S59 用的是同一张图，完整的读图见上一页（S59）。从图上还能看出：三条 GEMSTONE 谱的基线噪声明显高于常规谱（图中没有纵坐标比例，只能定性看）。

> S60 reuses the S59 figure; the full figure reading is on the previous page (S59). One more thing the figure shows: the three GEMSTONE traces have visibly noisier baselines than the conventional trace (no vertical scale is given, so this is qualitative only).

对比：选择性 TOCSY 与 GEMSTONE / Comparison: selective TOCSY vs GEMSTONE

	选择性 TOCSY (S51–52)	GEMSTONE (S59–60)
目标峰可重叠? Target may overlap?	否 No	是 Yes
输出 Output	整条耦合链 the whole coupled chain	被选中的那个多重峰 the selected multiplet
关键手段 Key tool	选择性脉冲 + TOCSY 混合 selective pulse + TOCSY mixing	梯度空间编码（类似 UF） gradient spatial encoding (like UF)
代价 Cost	远端质子信号弱（背景） weak distant protons (background)	灵敏度损失，快弛豫样品尤甚 sensitivity penalty, worst when fast relaxing

考点速记 KEY POINTS

- 结果：在重叠中拉出目标多重峰。
  Result: the target multiplet pulled out of overlap.
- 快弛豫 = T_2 短：长选择元件期间信号衰减（背景）。
  Fast relaxing = short T_2: decay during long selective elements.
- 代价：灵敏度损失，快弛豫样品可能很大。
  Cost: sensitivity penalty, large for fast relaxers.
- S59 与 S60 是同一张图。
  S59 and S60 show the same figure.

S61 · DREAMTIME NMR：定义

S61 · DREAMTIME NMR: definition

讲解 / Explanation

DREAMTIME（为活体靶标与混合物解析设计的重聚激发及可选混合实验，Designed Refocused Excitation And optional Mixing for Targets In vivo and Mixture Elucidation）是一种相对较新的方法，本质上让你在混合物中精确选择想看的化合物，堪称选择性方法的巅峰。

> DREAMTIME, Designed Refocused Excitation And optional Mixing for Targets In vivo and Mixture Elucidation, is a relatively new approach that essentially allows you to select exactly which compounds you want to see in a mixture: essentially the pinnacle of selective approaches.

名称中的每个部分对应后面哪一页 / What each part of the name refers to

名称 Name	含义 Meaning	见 See
Designed	按目标化合物设计：位移、耦合、J 都要匹配 designed for the target: shifts, coupling and J must match	S62
Refocused Excitation	选择性激发并重聚，输出同相磁化 selective excitation with refocusing; in-phase output	S65
optional Mixing	可选的 TOCSY 模块，带出其他峰 optional TOCSY block that brings out the other peaks	S64
Targets In vivo	活体内的目标代谢物 target metabolites inside living organisms	S61–62
Mixture Elucidation	解析混合物（血液、尿液） resolving mixtures (blood, urine)	S63, S66

图读 / Reading the figure

左边是大型溞（*D. magna*）示意图，体内画着许多代谢物结构和水分子（$\mathrm{H-O-H}$）。箭头从生物体指向 "DREAMTIME NMR"，再指向右下的 $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ 二维谱框（$^{1}\mathrm{H}$ 8.0–0.0 ppm，$^{13}\mathrm{C}$ 10–130 ppm）。谱框里只画了三个青绿色结构：丙氨酸、葡萄糖、酪氨酸，意思是在活体众多分子中只看到你选定的几个。这是示意图，没有画出具体峰。

> On the left is a cartoon of *D. magna* with many metabolite structures and water molecules ($\mathrm{H-O-H}$) inside. Arrows lead from the organism to "DREAMTIME NMR" and on to a $^{1}\mathrm{H}$–$^{13}\mathrm{C}$ 2D frame ($^{1}\mathrm{H}$ 8.0–0.0 ppm, $^{13}\mathrm{C}$ 10–130 ppm). The frame shows only three teal structures (alanine, glucose, tyrosine): out of all the molecules in the organism you see only the ones you chose. It is a schematic; no actual peaks are drawn.

考点速记 KEY POINTS

- DREAMTIME：在混合物中精确选择想看的化合物。
  DREAMTIME: select exactly the compounds you want.
- optional Mixing = 可选 TOCSY 模块（S64）。
  "optional Mixing" = the optional TOCSY block (S64).
- 幻灯片原话："选择性方法的巅峰"。
  Slide's words: "the pinnacle of selective approaches".
- 面向活体靶标与混合物解析。
  Built for in vivo targets and mixture elucidation.

S62 · DREAMTIME：三重过滤条件

S62 · DREAMTIME: a three-part filter

讲解 / Explanation

这种方法本质上是一个非常好的过滤器，只让目标化合物通过。要通过过滤器，化合物必须在指定化学位移处有共振峰，这些共振峰必须彼此耦合，并且耦合常数要大致正确。结果是每个化合物给出两个峰。

> This approach is essentially a very good filter that only lets the desired compounds through. To pass, a compound must have resonances at the indicated chemical shifts, they must be coupled, and the coupling constant must be about right. The result is two peaks per compound.

$$\text{pass} \iff \left(\delta_A,\ \delta_B \text{ as designed}\right) \wedge \left(A\text{–}B \text{ coupled}\right) \wedge \left(J_{AB} \approx J_{\text{design}}\right)$$

三个条件同时满足才通过；输出 = A、B 两个峰 / all three must hold; output = the two peaks A and B

背景 为什么是两个峰：过滤器检验的是一对耦合的共振峰，通过的磁化只落在这一对峰上；不加混合时，同一分子的其他质子不出现（要看全部峰见 S64）。只符合位移、碰巧重叠的峰不满足耦合和 J 条件，进不来，所以它比只靠位移选择的方法更干净。

> **Background** Why two peaks: the filter tests one coupled pair of resonances, and the magnetization that passes ends up on that pair; without mixing, the molecule's other protons do not appear (see S64 for all peaks). A peak that merely overlaps at the right shift fails the coupling and J tests, so the result is cleaner than selection by shift alone.

图读：横轴 ^{1}H 4.0–1.0 ppm / Reading the figure: ^{1}H axis 4.0–1.0 ppm

- 顶部黑色 "In-Vivo"：活体普通谱，峰又宽又重叠。其下 "Multiple Selection"（多重选择）：四个目标同时选择，基本是下面四条之和。
  > Top, black, "In-Vivo": the ordinary in vivo spectrum, broad and overlapped. Below it "Multiple Selection": all four targets at once, essentially the sum of the four traces beneath.
- 单独选择：乳酸（约 4.1、1.3）、丙氨酸（约 3.8、1.5）、甲硫氨酸（约 2.65、2.2；图上拼作 "Methioine"）、β-葡萄糖（窗口内只见约 3.25 一个峰），单位 ppm。
  > Single selections: lactic acid (≈ 4.1, 1.3), alanine (≈ 3.8, 1.5), methionine (≈ 2.65, 2.2; spelled "Methioine" on the slide), β-glucose (only one peak, ≈ 3.25, inside the window), all in ppm.
- β-葡萄糖的第二个峰不在画出的范围内（推断，幻灯片未标）。选出的峰都窄而分立，活体谱中却只是宽包络。
  > β-glucose's second peak lies outside the plotted range (inference; not marked). The selected peaks are sharp and separate, whereas the in vivo spectrum is a broad envelope.

考点速记 KEY POINTS

- 通过条件：指定位移 + 彼此耦合 + J 大致正确。
  Pass: right shifts + coupled + J about right.
- 多重选择谱 ≈ 各单独选择谱之和。
  Multiple selection ≈ sum of the single selections.
- 结果：每个化合物两个峰（被选中的耦合对）。
  Result: two peaks per compound (the selected pair).
- 只碰巧位移重叠的峰过不了 J 条件（背景）。
  A peak that only overlaps in shift fails the J test.

S63 · DREAMTIME：血液中的多重选择

S63 · DREAMTIME: multiple selection in blood

讲解 / Explanation

还可以做多重选择，一次选出多个化合物。这里是在血液中有选择地挑出代谢物的例子。方法非常灵活，生物标志物可以随时间追踪！

> DREAMTIME can also do multiple selection, where several compounds are chosen at once. This is an example of metabolites in blood being selectively chosen. There is considerable versatility, and biomarkers can be traced over time!

背景 全血的普通谱里，大分子（蛋白质、脂类等）给出宽包络，小分子代谢物几乎被淹没。DREAMTIME 只让目标代谢物通过，峰清楚到可以逐个读强度，于是可以在不同时间点重复测量，追踪标志物的变化。

> **Background** In an ordinary spectrum of whole blood, large molecules (proteins, lipids and others) give a broad envelope that swamps the small metabolites. DREAMTIME passes only the target metabolites, clean enough to read each one's intensity, so measurements can be repeated over time to trace biomarkers.

图读：B 图，横轴 ^{1}H 5–0 ppm / Reading the figure: panel B, ^{1}H axis 5–0 ppm

- 顶部 "Whole Blood"（全血）：宽峰包络，只有约 5.0 ppm 处一根尖峰突出。"Multiple Selection"：多重选择谱，3.0–1.9 ppm 一段另画了放大 ×4 的副本。

  > Top, "Whole Blood": a broad envelope, with one sharp peak near 5.0 ppm. "Multiple Selection": the combined selection, with the 3.0–1.9 ppm stretch redrawn at ×4.

- 单独选择（ppm）：丙氨酸约 3.8、1.5；谷氨酸约 3.75、2.35、2.1；天冬氨酸约 3.9、2.7–2.8；乳酸约 4.1、1.3；β-葡萄糖 3.2–3.9 一组峰；缬氨酸约 3.6、2.3、1.0。

  > Single selections (ppm): alanine ≈ 3.8, 1.5; glutamic acid ≈ 3.75, 2.35, 2.1; aspartic acid ≈ 3.9, 2.7–2.8; lactic acid ≈ 4.1, 1.3; β-glucose a group at 3.2–3.9; valine ≈ 3.6, 2.3, 1.0.

编者注 / Editor's note

本图中谷氨酸、β-葡萄糖、缬氨酸等显示的峰组多于两个，与 S62 "每个化合物两个峰"不同。幻灯片没有说明原因；S64 提到加 TOCSY 模块会带出其他峰，但本页没有说明是否用了它。这里只按图描述。

> Here glutamic acid, β-glucose, valine and others show more than two peak groups, unlike "two peaks per compound" on S62. The slide gives no reason; S64 says a TOCSY block brings out other peaks, but this slide does not say whether one was used. The description above follows the figure only.

考点速记 KEY POINTS

- 多重选择：一次选多个化合物。
  Multiple selection: several compounds in one run.
- 全血谱是宽包络；选择后代谢物峰可逐个读出。
  Whole blood is a broad envelope; selected metabolites read one by one.
- 用途：随时间追踪生物标志物。
  Use: tracing biomarkers over time.
- 本图六种：Ala、Glu、Asp、乳酸、β-葡萄糖、Val。
  Six here: Ala, Glu, Asp, lactate, β-glucose, Val.

S64 · DREAMTIME：加入 TOCSY 模块

S64 · DREAMTIME with an added TOCSY block

讲解 / Explanation

还可以在实验中加入一个 TOCSY（全相关谱）模块，把（同一分子的）其他所有峰都带出来。这对分子的初步鉴定非常有用，但会损失大量信号。

> A TOCSY block can also be added to the experiment, which brings out all the other peaks (of the same molecule). This is great for initial identification of the molecules, but loses a lot of signal.

背景 为什么损失信号：TOCSY 混合把被选中的磁化分摊到整个自旋体系的所有质子上，每个峰只分到一部分；混合期本身较长，还有弛豫损失。因此常用 TOCSY 版本确认"选中的确实是这个分子"（完整的峰型像指纹），之后追踪或比较时不加混合，以保住信号。

> **Background** Why signal is lost: TOCSY mixing shares the selected magnetization over every proton of the spin system, so each peak gets only part of it, and the mixing period itself is long enough to add relaxation losses. A TOCSY version is therefore useful to confirm that the selected signal is the right molecule (the full pattern acts as a fingerprint); later tracking or comparison can run without mixing to keep the signal.

图读：B 图，横轴 ^{1}H 8–0 ppm / Reading the figure: panel B, ^{1}H axis 8–0 ppm

顶部 "8 Component Multiple Selection"（八组分多重选择），下面是八条单独选择（位置约值，ppm）：

> Top, "8 Component Multiple Selection"; below, the eight single selections (approximate positions, ppm):

化合物 Compound	峰位 Peaks (ppm)	化合物 Compound	峰位 Peaks (ppm)
胆碱 Choline	4.1, 3.55	尿嘧啶 Uracil	7.6, 5.85
甘油 Glycerol	3.55–3.8	苏氨酸 Threonine	4.25, 3.6, 1.35
丙氨酸 Alanine	3.8, 1.5	酪氨酸 Tyrosine	7.2, 6.9, 3.8, 2.9–3.05
缬氨酸 Valine	3.65, 2.3, 1.0	赖氨酸 Lysine	3.8, 3.05, 1.5–1.95

- 加 TOCSY 后多数化合物出现两组以上的峰（赖氨酸最明显）；尿嘧啶、丙氨酸只有两组耦合质子，仍是两组。
  > With TOCSY most compounds show more than two peak groups (lysine most clearly); uracil and alanine have only two coupled proton groups, so they still show two.
- 胆碱只显示两组峰，没有 $\mathrm{N(CH_3)_3}$ 单峰。**背景** 该单峰与其他质子无耦合，TOCSY 传不到它。几条谱在约 5 ppm（水峰区）有残余波动。
  > Choline shows two groups and no $\mathrm{N(CH_3)_3}$ singlet. **Background** That singlet is not coupled to other protons, so TOCSY cannot reach it. Several traces show residual wiggles near 5 ppm (the water region).

考点速记 KEY POINTS

- 加 TOCSY 模块 → 带出同一分子的其他峰。
  TOCSY block → the molecule's other peaks appear.
- 磁化被分摊到整个自旋体系（背景）。
  Magnetization is shared over the whole spin system.
- 适合初步鉴定，但损失大量信号。
  Good for initial identification, but costs much signal.
- 无耦合的单峰（胆碱 $\mathrm{N(CH_3)_3}$）带不出来（背景）。
  Uncoupled singlets (choline $\mathrm{N(CH_3)_3}$) never appear.

S65 · DREAMTIME：同相磁化与选择性 HSQC

S65 · DREAMTIME: in-phase output and a selective HSQC

讲解 / Explanation

DREAMTIME 产生同相磁化，所以可以接入（几乎）任何其他 NMR 实验。这里是针对八种代谢物的选择性 HSQC（异核单量子相干谱）。谱图简化对峰归属帮助很大。

> DREAMTIME produces in-phase magnetization, so it can be added into (nearly) any other NMR experiment. Here is a selective HSQC for eight different metabolites. The spectral simplification helps a lot with peak assignments.

背景 同相磁化就像普通 90° 脉冲后得到的横向磁化（多重峰各谱线同相、没有反相成分），后面的实验可以把它直接当起点；若输出是反相磁化，就需要专门设计衔接步骤。所以 DREAMTIME 可以作为前置的"选择模块"，后面接 HSQC、TOCSY 等。

> **Background** In-phase magnetization is like the transverse magnetization after an ordinary 90° pulse (all lines of a multiplet in phase, no antiphase part), so a following experiment can use it directly as its starting point; an antiphase output would need a dedicated conversion step. DREAMTIME can therefore act as a front-end selection block for HSQC, TOCSY and others.

图读：B) 完整 HSQC vs F) 选择性 HSQC / Reading the figure: B) full HSQC vs F) selective HSQC

- B：完整 HSQC，横轴 $^{1}\mathrm{H}$ 8–1 ppm，纵轴 $^{13}\mathrm{C}$ 10–130 ppm，上方与左侧为一维投影。大量交叉峰挤在 $^{1}\mathrm{H}$ 1–4.5、$^{13}\mathrm{C}$ 10–80 ppm 的斜带上，很难逐一归属。
  > B: the full HSQC, $^{1}\mathrm{H}$ 8–1 ppm across, $^{13}\mathrm{C}$ 10–130 ppm down, with 1D projections on top and left. Many cross peaks crowd a diagonal band at $^{1}\mathrm{H}$ 1–4.5, $^{13}\mathrm{C}$ 10–80 ppm; assigning them one by one is hard.
- F：八种代谢物的选择性 HSQC，只剩少数交叉峰，每个用字母标注：胆碱 (C)、甘油 (G)、丙氨酸 (A)、缬氨酸 (V)、尿嘧啶 (U)、苏氨酸 (Th)、酪氨酸 (Ty)、赖氨酸 (L)，与 S64 是同一组八种。
  > F: the selective HSQC of eight metabolites; only a few cross peaks remain, each lettered: choline (C), glycerol (G), alanine (A), valine (V), uracil (U), threonine (Th), tyrosine (Ty), lysine (L), the same eight as S64.
- U 与 Ty 的峰在 $^{1}\mathrm{H}$ 5.8–7.6、$^{13}\mathrm{C} \approx 100$–145 ppm（芳香 / 烯烃区）；其余在 $^{1}\mathrm{H}$ 1–4.3、$^{13}\mathrm{C}$ 10–75 ppm。
  > U and Ty peaks sit at $^{1}\mathrm{H}$ 5.8–7.6, $^{13}\mathrm{C} \approx 100$–145 ppm (aromatic/olefinic); the rest at $^{1}\mathrm{H}$ 1–4.3, $^{13}\mathrm{C}$ 10–75 ppm.

考点速记 KEY POINTS

- DREAMTIME 输出同相磁化 → 可接（几乎）任何实验。
  In-phase output → plugs into (nearly) any experiment.
- 谱图简化 → 峰归属容易得多。
  Spectral simplification makes assignment much easier.
- 例：八种代谢物的选择性 HSQC。
  Example: a selective HSQC of eight metabolites.
- 字母代号：C、G、A、V、U、Th、Ty、L。
  Letter codes: C, G, A, V, U, Th, Ty, L.

S66 · DREAMTIME 聚焦

S66 · DREAMTIME Focusing

讲解 / Explanation

与大多数选择性技术一样，选择过程中会损失信号。不过可以施加聚焦（DREAMTIME 聚焦）来大幅提高 SNR：本质上把峰聚焦成一根高得多的单一尖峰。

> Like most selective techniques, you lose signal during the selection. However, focusing can be applied to raise the SNR considerably: essentially the peak is focused into a single spikelet that is much taller.

背景 一个多重峰的总信号分散在多条裂分谱线上；把它集中到一根谱线上，峰高成倍增加而噪声不变，峰高意义上的 SNR 随之大增。代价是看不到裂分（耦合常数），峰位仍可作标识。

> **Background** A multiplet's total signal is spread over several split lines; collecting it into one line multiplies the peak height while the noise stays the same, so the height-based SNR rises sharply. The price is that the splitting (coupling constants) is no longer visible; the peak position still identifies the compound.

图读：尿液 A–C，血液 D–F / Reading the figure: urine A–C, blood D–F

A、D（黑）像普通谱，显示全部峰；B、E（蓝，画为 $\times 2$）只剩粉色条标出的两个目标峰，比普通谱还弱；C、F 是聚焦后：每处只剩一根细高的尖峰。尿液横轴 8.0–6.4 ppm，血液 3.5–0.5 ppm。参比 SNR 带星号（SNR*，标在 A、D 上方），幻灯片未解释星号。

> A and D (black) look like ordinary spectra with all peaks; B and E (blue, drawn $\times 2$) keep only the two pink-marked targets and are weaker than the ordinary spectrum; C and F are focused: one thin, tall spikelet at each target. Urine axis 8.0–6.4 ppm, blood 3.5–0.5 ppm. The reference SNR carries an asterisk (SNR*, above A and D), which the slide does not explain.

样品 Sample	化合物 Compound	δ (ppm)	SNR*	聚焦 Focused	幻灯片 Slide	比值 Ratio
尿液 Urine	马尿酸 hippuric acid	≈ 7.8	234	1647	↑~700 %	7.0
尿液 Urine	糠酰甘氨酸 furoylglycine	≈ 6.6	102	918	↑~900 %	9.0
血液 Blood	β-葡萄糖 β-glucose	≈ 3.25	114	1055	↑~900 %	9.3
血液 Blood	缬氨酸 valine	≈ 1.0	21	265	↑~1200 %	12.6

编者注 / Editor's note

幻灯片的百分数等于"聚焦后 / 参比"的比值 $\times 100\,\%$，如 $1647/234 \approx 7.0$ 标为 "↑~700 %"。若按增加量计算，应为约 $+600\,\%$、$+800\,\%$、$+825\,\%$、$+1160\,\%$。答题时照幻灯片的数字写，但要知道它表示的是倍数。

> The slide's percentages equal the ratio focused / reference $\times 100\,\%$: $1647/234 \approx 7.0$ is printed "↑~700 %". As increases they would be about $+600\,\%$, $+800\,\%$, $+825\,\%$, $+1160\,\%$. Quote the slide's numbers, knowing they express the ratio.

考点速记 KEY POINTS

- 选择会丢信号；聚焦把峰集中成一根高尖峰。
  Selection loses signal; focusing makes one tall spikelet.
- 尿液：马尿酸 $234 \to 1647$，糠酰甘氨酸 $102 \to 918$。
  Urine: hippuric acid $234 \to 1647$; furoylglycine $102 \to 918$.
- 血液：β-葡萄糖 $114 \to 1055$，缬氨酸 $21 \to 265$。
  Blood: β-glucose $114 \to 1055$; valine $21 \to 265$.
- "↑~700 %"实为倍数 $\approx 7\times$（增量约 $+600\,\%$）。
  "↑~700 %" is the ratio $\approx 7\times$ (increase $\approx +600\,\%$).

S67 · 活体 NMR：无损，所以能测活的生物

S67 · In vivo NMR: nondestructive, so organisms stay alive

幻灯片要点 / What the slide says

NMR 一个很有意思的优点是它是无损（非破坏性，nondestructive）的。这意味着可以用它来研究活体生物（living organisms），即活体 NMR（in vivo NMR）。

> An interesting advantage of NMR is that it is nondestructive. This means that we can use it to study living organisms: in vivo NMR.

为什么无损 / Why it is nondestructive

背景 NMR 用的是射频，光子能量极小，远不足以断键或电离；测量也不消耗样品，测完的生物还活着，可以接着再测。以 600 MHz 为例估算一个光子的能量：

> **Background** NMR uses radio waves whose photons carry far too little energy to break bonds or ionize; nothing is consumed, so the organism is still alive afterwards and can be measured again. The photon energy at 600 MHz, as an example:

$$E = h\nu = (6.626 \times 10^{-34}\ \mathrm{J\,s}) \times (6.00 \times 10^{8}\ \mathrm{s^{-1}}) \approx 4.0 \times 10^{-25}\ \mathrm{J}$$

$$N_A E \approx 0.24\ \mathrm{J\,mol^{-1}}$$

背景估算：化学键能约 $10^5\ \mathrm{J\,mol^{-1}}$ 量级，高出约六个数量级

/ background estimate: bond energies are of order $10^5\ \mathrm{J\,mol^{-1}}$, about six orders of magnitude more

背景 好处是能对同一个个体做时间序列：先测基线，再施加化学或生物扰动，然后反复测量，每只生物都是自己的对照。破坏性方法（如提取后再测）每个时间点都要牺牲一批生物，个体差异会混进结果（幻灯片 72 再谈）。

> **Background** The payoff is a time series on the same individual: record a baseline, apply a chemical or biological perturbation, then measure repeatedly, so each organism is its own control. A destructive method (e.g. extraction, then measurement) sacrifices a new batch of organisms at every time point, and individual differences enter the result (slide 72 returns to this).

图怎么读 / Reading the figure

- 左：磁体剖面示意，中间是样品管和橙色的射频线圈，生物就放在线圈区域里。
  > Left: a cut-away magnet with the sample tube and the orange RF coil in the middle; the organisms sit inside the coil region.
- 右上「Chemical or biological perturbation begins」（化学或生物扰动开始）：三个圆中是同一种小型水生生物（图上未标名称），上方虚线圈里的分子由全红逐渐变成以黄色为主，表示分子组成随时间改变。
  > Top right, "Chemical or biological perturbation begins": three circles show the same small aquatic organism (not named); the molecules in the dashed circles above change from all red to mostly yellow, i.e. the molecular make-up changes with time.
- 下方箭头「Molecular response of living organisms is monitored over time」（随时间监测活体的分子响应）：三张横轴为 Chemical shift 的小谱中，红峰由高变矮，黄峰从无到有并逐渐变高。
  > Bottom arrow, "Molecular response of living organisms is monitored over time": in three small spectra (x-axis Chemical shift) the red peak shrinks while a yellow peak appears and grows.

考点速记 KEY POINTS

- NMR 无损 → 可研究活体生物（in vivo NMR）。
  NMR is nondestructive → living organisms can be studied.
- 背景：600 MHz 光子 ≈ 0.24 J/mol，远低于键能。
  Background: a 600 MHz photon ≈ 0.24 J/mol, far below bond energies.
- 同一个体可反复测量，形成时间序列。
  The same individual can be measured again and again.
- 图：扰动后红峰降、黄峰升 = 分子响应随时间变化。
  Figure: red falls, yellow rises = molecular response over time.

S68 · 活体 NMR 的难点：峰形差

S68 · Why in vivo NMR is hard: poor lineshape

幻灯片要点 / What the slide says

这相当有挑战性：生物会动，需要在 NMR 中维持生命，而且通常弛豫很快。因此活体谱图的峰形（线形，lineshape）往往很差。

> This is quite challenging: the organisms can move, need to be sustained in the NMR, and relaxation is typically fast. In vivo spectra therefore tend to have poor lineshape.

三个原因各怎样损坏峰形 / How each cause spoils the lineshape

- **背景** 会动：生物在线圈中的位置和姿态在扫描之间、甚至一次采集之内都在变，磁场均匀性（匀场）随之改变，各次扫描的相位和幅度不一致，叠加后峰变宽、变形。

  > **Background** Movement: the organism's position and posture in the coil change between scans and even within one acquisition, so field homogeneity (the shim) keeps changing; scans no longer add with the same phase and amplitude, and peaks broaden and distort.

- **背景** 维持生命：生物必须泡在含水的培养液里（还要供氧、换水），巨大的水峰必须抑制，残留水峰和峰脚留在谱里；组织、水和空气之间的磁化率差异也使局部 B_0 不均匀。

  > **Background** Life support: the organism lives in water-based medium (with oxygen and water exchange), so a huge water peak must be suppressed and its residue and wings remain; susceptibility differences between tissue, water and air also make the local B_0 inhomogeneous.

- **背景** 弛豫快：活组织中分子运动受限、常与大分子结合，T_2 短；线宽与 T_2^* 成反比，峰天然就宽，相邻峰互相重叠。

  > **Background** Fast relaxation: in living tissue molecular motion is restricted and molecules often bind to large ones, so T_2 is short; linewidth is inversely proportional to T_2^*, so peaks are intrinsically broad and neighbours overlap.

$$\Delta\nu_{1/2} = \frac{1}{\pi T_2^*} \qquad T_2^* = 10\ \text{ms} \ \Rightarrow\ \Delta\nu_{1/2} \approx 32\ \text{Hz}$$

背景：T_2^* 同时包含真实 T_2 和磁场不均匀的贡献 / background: T_2^* includes both the true T_2 and field inhomogeneity

图怎么读 / Reading the figure

- a（理想谱）：所画分子为 CH_3–CH(OH)–$CH(NH_2)$–COOH（**背景** 即苏氨酸；幻灯片没写名称）。三个彩色位点各对应一个窄峰：红（CH–OH）约 4.2 ppm、黄（CH–NH_2）约 3.6 ppm、蓝（CH_3）约 1.3 ppm，蓝峰最高（3 个 H）。横轴 ^{1}H Chemical shift (ppm)。

  > a (idealised): the molecule drawn is CH_3–CH(OH)–$CH(NH_2)$–COOH (**Background** threonine; the slide gives no name). Each coloured site gives one narrow peak: red (CH–OH) ≈ 4.2 ppm, yellow (CH–NH_2) ≈ 3.6 ppm, blue (CH_3) ≈ 1.3 ppm, blue tallest (3 H). x-axis ^{1}H chemical shift (ppm).

- b（活体谱）：横轴 8–1 ppm。约 5 ppm 处箭头标出「Residual water」（残留水峰），形状扭曲、上下抖动；约 4.5–3 ppm 是宽而低的鼓包；约 3–0.9 ppm 是一串标「L」的峰，约 1.3 ppm 的最高。幻灯片没有说明 L 的含义（**背景** 位置与脂质链信号相符）。

  > b (in vivo): x-axis 8–1 ppm. At ≈ 5 ppm the arrowed "Residual water" peak is distorted and jagged; a broad low hump spans ≈ 4.5–3 ppm; peaks marked "L" lie between ≈ 3 and 0.9 ppm, tallest at ≈ 1.3 ppm. The slide does not say what L means (**Background** the positions fit lipid chain signals).

- 对比：a 中清楚分开的窄峰，在 b 中变成宽峰和鼓包，单个代谢物很难认出。

  > Contrast: the clean, narrow peaks of a become broad peaks and humps in b, where single metabolites are hard to pick out.

考点速记 KEY POINTS

- 三大难点：会动、要维持生命、弛豫快。
  Three problems: movement, life support, fast relaxation.
- 背景：$\Delta\nu_{1/2} = 1/(\pi T_2^*)$，$T_2^*$ 短 → 峰宽。
  Background: $\Delta\nu_{1/2} = 1/(\pi T_2^*)$; short T_2^* → broad.
- 结果：活体谱峰形差（宽、扭曲、重叠）。
  Result: poor lineshape (broad, distorted, overlapping).
- 活体谱里常留有残留水峰（约 5 ppm）。
  In vivo spectra keep a residual water peak (≈ 5 ppm).

S69 · 流动系统：让生物活着

S69 · A flow system keeps the organisms alive

幻灯片要点 / What the slide says

生物通常被维持在一个流动（灌流）系统（flow system）中。由于它们可以自由游动，扩散实验（diffusion experiment）效果不太好。

> The organisms are normally sustained in a flow system. Because they are free to swim around, diffusion experiments don't work very well.

为什么要流动 / Why flow

背景 管中的水量很小，生物几小时内就会耗尽氧气、积累代谢废物。流动系统持续送入新鲜、含氧的水并带走旧水，生物才能在磁体里活过长时间的实验；水源还可以中途切换成含污染物的水或清水（幻灯片 72）。

> **Background** The tube holds very little water, so the organisms would use up the oxygen and accumulate waste within hours. A flow system keeps bringing fresh, oxygenated water and removing the old, so they survive long experiments in the magnet; the supply can also be switched mid-run to polluted or clean water (slide 72).

为什么扩散实验不灵 / Why diffusion experiments fail

背景 扩散实验用一对梯度按位置给分子「打标签」，再看信号在扩散时间 Δ 内衰减多少：分子移动越远，衰减越多，由此得到扩散系数 D。前提是位移只来自分子扩散。生物自己在游，整体位移远大于分子在这段时间内的扩散距离，衰减中混入游动和水流，测得的 D 失去意义；CMP 的扩散编辑（幻灯片 33，以 180 ms 内移动 1 μm 为界）同样受影响。

> **Background** A diffusion experiment labels molecules by position with a gradient pair and measures how much the signal decays over the diffusion time Δ: the further molecules move, the larger the decay, which gives the diffusion coefficient D. This assumes the displacement comes from molecular diffusion only. A swimming organism moves far more than molecules diffuse in that time, so swimming and flow enter the decay and the measured D is meaningless; CMP diffusion editing (slide 33, threshold 1 μm in 180 ms) suffers the same way.

$$\frac{S}{S_0} = \exp\left[-\gamma^2\delta^2G^2D\left(\Delta - \frac{\delta}{3}\right)\right]$$

背景：Stejskal–Tanner 式；此处 δ 是梯度时长（不是化学位移），G 梯度强度，Δ 扩散时间

/ background: Stejskal–Tanner equation; here δ is the gradient duration (not chemical shift), G the gradient strength, Δ the diffusion time

图怎么读 / Reading the figure

- 照片：竖直的玻璃管（顶部蓝色盖子），一些小型生物（图上未标名称）分散在管中不同高度游动；管底有一团白色物质；绿色背景。只作示意，没有数据。

  > Photo: a vertical glass tube (blue cap on top) with small organisms (not named) swimming at different heights; a white mass at the bottom; green background. Illustrative only, no data.

- 要点：生物分散在整根管里且位置不断变化，这正是峰形差（幻灯片 68）和扩散实验失效的原因。

  > Point: the organisms are spread through the tube and keep moving, which is why lineshape suffers (slide 68) and diffusion experiments fail.

考点速记 KEY POINTS

- 活体样品通常养在流动（灌流）系统中。
  In vivo samples normally live in a flow system.
- 生物自由游动 → 扩散实验效果差。
  Free swimming → diffusion experiments work poorly.
- 背景：扩散实验假定位移只来自分子扩散。
  Background: diffusion NMR assumes all displacement is diffusion.
- 背景：流动 = 供氧、去废物、可切换暴露与清水。
  Background: flow = oxygen in, waste out, switchable exposure.

S70 · 现代流动系统：锁场、塞子、注入与抽吸

S70 · The modern flow system: lock, plugs, injection and suction

幻灯片要点 / What the slide says

- 较新的流动系统带有氘锁场（deuterium lock）。
  > The more modern flow system has a deuterium lock.
- 有塞子（特氟龙塞），使生物不能离开线圈区域。
  > There are plugs so the organisms can't leave the coil region.
- 有一根注入管和一根抽吸管，抽吸管把多余的水抽走。
  > There is an injection tube and a suction tube, which removes excess water.

每个部件为什么重要 / Why each part matters

- **背景** 氘锁场：谱仪持续跟踪 ²H 信号的频率并修正 B_0 漂移。活体实验要连续采集数小时，没有锁场，漂移会让各次扫描的峰位错开，叠加后变宽。生物要活在普通水里（高比例 D_2O 对生物有害），所以 D_2O 装在单独的毛细管里，提供锁场信号又不接触生物。
  > **Background** Deuterium lock: the spectrometer tracks the ²H signal and corrects drift of B_0. In vivo runs last hours; without a lock, drift shifts peaks between scans and broadens the sum. The organisms must live in ordinary water (a high fraction of D_2O harms them), so the D_2O sits in its own capillary, giving a lock signal without touching them.
- 塞子：上、下两个特氟龙塞把生物关在线圈的灵敏区内。游出线圈的生物测不到，游进游出会让信号忽大忽小（**背景**）。
  > Plugs: the Teflon top and bottom plugs keep the organisms inside the coil's sensitive region. An organism outside the coil gives no signal, and swimming in and out makes the signal fluctuate (**Background**).
- 注入 + 抽吸：注入管送入新鲜的水，抽吸管抽走多余的水。**背景** 这样水面和样品形状保持不变，匀场不必反复重做。
  > Injection + suction: the injection tube brings fresh water and the suction tube removes the excess. **Background** The water level and sample geometry then stay constant, so the shim need not be redone.

图怎么读 / Reading the figure

- 左（虚线框放大）：自上而下标注 Injection capillary tube（注入毛细管）、Suction capillary tube（抽吸毛细管）、Teflon top plug（特氟龙上塞）、Glass tip（玻璃尖端）、D_2O capillary tube（D_2O 毛细管）、Teflon bottom plug（特氟龙下塞）。两塞之间 3 cm 的水层里是生物，下塞高 1 cm。
  > Left (dashed close-up), top to bottom: Injection capillary tube, Suction capillary tube, Teflon top plug, Glass tip, D_2O capillary tube, Teflon bottom plug. The organisms live in the 3 cm water column between the plugs; the bottom plug is 1 cm tall.
- 右：NMR tube 套在 Outer plastic tube（外层塑料管）里；PEEK tubing（0.76 mm ID）和 PTFE tubing（1.5 mm ID）把它连到 Injection pump（注入泵）和 Suction pump（抽吸泵），两泵都接到一个通「Air」的储水瓶，构成循环。图上没有说明哪根管线用哪种管材。
  > Right: the NMR tube sits in an outer plastic tube; PEEK tubing (0.76 mm ID) and PTFE tubing (1.5 mm ID) connect it to the injection pump and the suction pump, both linked to a reservoir open to "Air", forming a loop. The figure does not say which line uses which tubing.

考点速记 KEY POINTS

- 氘锁场：D_2O 在独立毛细管中，不接触生物。
  Lock: D_2O in its own capillary, away from the organisms.
- 注入管送新水，抽吸管抽走多余的水。
  Injection brings fresh water; suction removes the excess.
- 塞子把生物限制在线圈区域内。
  Plugs keep the organisms in the coil region.
- 生物区 3 cm，下塞 1 cm；PEEK 0.76 mm、PTFE 1.5 mm ID。
  Organism zone 3 cm, bottom plug 1 cm; PEEK 0.76, PTFE 1.5 mm ID.

S71 · 活体 CMP NMR：转子帽开孔与超慢速 MAS

S71 · In vivo CMP NMR: a hole in the rotor cap and ultra-slow MAS

幻灯片要点 / What the slide says

CMP NMR（综合多相 NMR）多了一个难题：样品要旋转。第一，转子帽（rotor cap）上必须开孔让氧气进入，否则生物会死。第二，转速不能太高，这有时被称为超慢速 MAS（ultra-slow MAS）；慢速旋转对凝胶仍有不错的好处。旋转边带可能是个问题。

> CMP NMR has the additional challenge of spinning. First, the cap must have a hole to allow oxygen in, or the organism will die. Second, the spin rate cannot be too high; this is sometimes called ultra-slow MAS, and slow spinning still gives a nice advantage for gels. Sidebands can be problematic.

机理 / Mechanism

背景 离心加速度随转速平方增长。同一转子里 2.5 kHz 的加速度是 50 Hz 的 $(2500/50)^2 = 2500$ 倍；设半径 $r \approx 1.5\ \mathrm{mm}$（假设为 4 mm 级转子），2.5 kHz 约 $4 \times 10^4\, g$，50 Hz 约 $15\, g$。代价是边带：边带间距等于转速（幻灯片 27），50 Hz 时边带紧挨真实峰、落进代谢物峰之间，巨大的残留水峰尤其明显。慢转对凝胶仍有用，因为凝胶状组织中残余的各向异性作用较弱，慢转也能部分平均，峰比静态时窄。

> **Background** Centrifugal acceleration grows with the square of the spin rate. In the same rotor 2.5 kHz gives $(2500/50)^2 = 2500$ times the acceleration of 50 Hz; with $r \approx 1.5\ \mathrm{mm}$ (assuming a 4 mm-class rotor) that is $\approx 4 \times 10^4\, g$ vs $\approx 15\, g$. The cost is sidebands: they sit one spin rate apart (slide 27), so at 50 Hz they crowd next to the real peaks and fall among the metabolite signals, worst for the huge residual water peak. Slow spinning still helps gels because the remaining anisotropic interactions in gel-like tissue are weak, so even slow spinning partly averages them and lines are narrower than static.

$$a = (2\pi\nu_{\mathrm{rot}})^2\, r \qquad\qquad \nu_{\mathrm{SSB}} = \nu_{\mathrm{iso}} \pm k\,\nu_{\mathrm{rot}},\ k = 1, 2, \ldots$$

背景：转得越慢越安全，但边带越密 / background: slower is safer, but the sidebands crowd closer

图怎么读 / Reading the figure

- 上：转子轴与 B_0 成 54.74°（魔角），转子里有一只虾形生物，标注「Hole in rotor cap」（转子帽上的孔）。下：纵轴 % Survival Rate（存活率），横轴 Time (hours) 0–48 h，带误差棒。

  > Top: the rotor axis at 54.74° (magic angle) to B_0 with a shrimp-like organism inside, labelled "Hole in rotor cap". Bottom: y-axis % survival rate, x-axis time (hours) 0–48, with error bars.

- 50 Hz：H. azteca（阿兹特克钩虾，蓝菱形）48 h 一直 100 %；D. magna（大型溞，灰圆）约 24 h 内 100 %，之后逐步下降，约 29–33 h 约 83 %、约 43 h 约 55 %、48 h 约 50 %。

  > 50 Hz: H. azteca (blue diamonds) stays at 100 % for 48 h; D. magna (grey circles) is 100 % to ≈ 24 h, then falls: ≈ 83 % at ≈ 29–33 h, ≈ 55 % at ≈ 43 h, ≈ 50 % at 48 h.

- 2.5 kHz：D. magna（红圆）头几小时约 95 %，约 6–8 h 降到 0；H. azteca（绿菱形）约 5 h 内 100 %，约 8 h 约 50 %、约 10 h 约 11 %，约 12 h 起为 0。

  > 2.5 kHz: D. magna (red circles) ≈ 95 % for the first hours, 0 by ≈ 6–8 h; H. azteca (green diamonds) 100 % to ≈ 5 h, ≈ 50 % at ≈ 8 h, ≈ 11 % at ≈ 10 h, 0 from ≈ 12 h.

考点速记 KEY POINTS

- 转子帽开孔供氧，否则生物会死。
  A hole in the rotor cap lets oxygen in, or the organism dies.
- 2.5 kHz：两种生物约 12 h 内全死；50 Hz：H. azteca 48 h 全活。
  2.5 kHz: both dead within ≈ 12 h; 50 Hz: H. azteca 100 % at 48 h.
- 转速不能高：超慢速 MAS（如 50 Hz）。
  Spin slowly: ultra-slow MAS (e.g. 50 Hz).
- 背景：$a \propto \nu_{\mathrm{rot}}^2$；代价：边带间距 = 转速。
  Background: $a \propto \nu_{\mathrm{rot}}^2$; cost: sidebands one spin rate apart.

S72 · 活体实例：鲍鱼的暴露与恢复

S72 · In vivo example: abalone exposure and recovery

幻灯片要点 / What the slide says

- 活体研究的例子很多，研究最多的生物是大型溞（D. magna）。
  > There are many in vivo examples; the most commonly studied organism is *D. magna*.
- NMR 的一项关键能力是对活体生物做过程监测（process monitoring）。
  > A key ability of NMR is process monitoring of living organisms.
- 这里一只鲍鱼（abalone）暴露于一种含氯农药（chlorinated pesticide），随后加入新鲜培养液，追踪其恢复。
  > Here an abalone was exposed to a chlorinated pesticide; fresh media was then added and recovery was tracked.
- 用破坏性技术很难做到这一点。
  > This is very hard to do with destructive techniques.

为什么破坏性方法难做 / Why destructive methods struggle

背景 破坏性方法每取一个时间点就要处死一只动物，暴露期和恢复期的每个点都来自不同个体，个体差异与时间变化混在一起。活体 NMR 让同一只鲍鱼走完「暴露 → 恢复」全过程，变化只来自时间和处理。

> **Background** A destructive method kills one animal per time point, so every point of exposure and recovery comes from a different individual, and individual variation mixes with the time course. In vivo NMR follows the same abalone through exposure → recovery, so changes come only from time and treatment.

图怎么读 / Reading the figure

- 下方流程：沿斜箭头编号 1–6，前段标「Flow in pollutant」（通入污染物），后段标「Flow in fresh water」（通入清水）；上方括号为「Study toxic response」（研究毒性响应）和「Monitor recovery」（监测恢复）。水箱颜色由蓝变绿，再回到蓝。
  > Scheme below: steps 1–6 along a diagonal arrow, the early part "Flow in pollutant", the later part "Flow in fresh water"; the brackets above read "Study toxic response" and "Monitor recovery". The tank colour goes blue → green → blue.
- 结构式：苯环上一个 OH、五个 Cl，即五氯苯酚（pentachlorophenol），也就是幻灯片所说的含氯农药。
  > Structure: a benzene ring with one OH and five Cl, pentachlorophenol, the slide's chlorinated pesticide.
- 上方谱图：6 条堆叠谱，右侧编号 1–6（与流程同样的编号）；横轴 Chemical shift (ppm)，−10 至 −40 ppm；峰标 1–5，峰 2 最高。各峰强度随谱号改变，对应暴露和恢复过程。
  > Spectra above: 6 stacked traces numbered 1–6 on the right (the same numbers as the scheme); x-axis chemical shift (ppm), −10 to −40 ppm; peaks marked 1–5, peak 2 tallest. Peak intensities change from trace to trace through exposure and recovery.

编者注 / Editor's note

谱图横轴只写「Chemical shift (ppm)」，没有标出是哪种核，幻灯片文字也没说。这里不猜测核种；考试时只描述为「−10 至 −40 ppm 的活体谱随暴露与恢复而变化」。

> The axis reads only "Chemical shift (ppm)"; neither it nor the slide text names the nucleus. It is not guessed here; in an exam describe it only as an in vivo spectrum over −10 to −40 ppm that changes with exposure and recovery.

考点速记 KEY POINTS

- 活体研究最多的生物：大型溞（D. magna）。
  Most studied in vivo organism: *D. magna*.
- 核心能力：对活体做过程监测（暴露 → 恢复）。
  Key ability: process monitoring (exposure → recovery).
- 例：鲍鱼 + 含氯农药（五氯苯酚），再换清水看恢复。
  Example: abalone + chlorinated pesticide, then fresh water.
- 陷阱：谱图未标核种，不要猜。
  Trap: the nucleus is not labelled; don't guess it.

S73 · 活体实例：结合主成分分析

S73 · In vivo example: combined with PCA

幻灯片要点 / What the slide says

活体 NMR 可以与统计方法结合，例如主成分分析（principal component analysis, PCA）。这里可以清楚地区分暴露于废水（wastewater）的生物和对照组。下方的载荷图（loadings plot）告诉我们哪些代谢物在响应中最重要。

> This can be combined with statistical techniques such as principal component analysis (PCA). Here you can clearly differentiate between the organisms exposed to wastewater and the controls. The bottom loadings plot shows which metabolites are most important in the response.

PCA 怎么读 / How PCA works

背景 每张谱切成许多积分区间，变成一行数字（变量 = 位移区间或代谢物）。PCA 找出方差最大且互相正交的方向，即主成分 PC1、PC2……，括号中的百分数是该成分解释的方差。得分（scores）是每个样品在各主成分上的坐标，组间分离看得分图；载荷（loadings）是每个变量对该成分的权重，沿哪个成分分离，就看那个成分上载荷绝对值大的变量。

> **Background** Each spectrum is cut into integration bins and becomes one row of numbers (variables = shift bins or metabolites). PCA finds mutually orthogonal directions of largest variance, the principal components PC1, PC2…; the percentage is the variance each explains. Scores are each sample's coordinates on the components, so separation is read from the scores plot; loadings are each variable's weight on a component, so the variables with large absolute loadings on the separating component explain the separation.

$$\mathbf{X} = \mathbf{T}\mathbf{P}^{\mathsf{T}} + \mathbf{E}$$

背景：$\mathbf{X}$ 数据（样品 × 变量），$\mathbf{T}$ 得分，$\mathbf{P}$ 载荷，$\mathbf{E}$ 残差

/ background: $\mathbf{X}$ data (samples × variables), $\mathbf{T}$ scores, $\mathbf{P}$ loadings, $\mathbf{E}$ residual

图怎么读 / Reading the figure

- 照片：大型溞（体内可见一团黑色的卵）和从管道排出的废水。
  > Photos: *D. magna* (a dark cluster of eggs inside) and wastewater pouring from a pipe.
- 上：三维得分图，轴为 PC1 (29.2 %)、PC2 (17.4 %)、PC3 (9.2 %)。对照组（Control，红）聚成紧密一团，PC2 约 −1 至 −4；暴露组（Exposed，黄）较分散，PC2 约 0 至 5。两组主要沿 PC2 分开。
  > Top: 3D scores plot, axes PC1 (29.2 %), PC2 (17.4 %), PC3 (9.2 %). Controls (red) form a tight cluster at PC2 ≈ −1 to −4; exposed (yellow) scatter at PC2 ≈ 0 to 5. The groups separate mainly along PC2.
- 下：三维载荷图，轴为 Loading 1、2、3，每点一种代谢物。Loading 2 较高（约 0.2–0.3）的有谷氨酰胺、天冬氨酸、丙酮酸、酪氨酸；较低（约 −0.3）的有精氨酸、胆碱、尿苷；另有乳酸、缬氨酸、苯丙氨酸、丙氨酸、异亮氨酸、葡萄糖、脯氨酸、L-谷胱甘肽、赖氨酸、丝氨酸。
  > Bottom: 3D loadings plot, axes Loading 1, 2, 3, one point per metabolite. High on Loading 2 (≈ 0.2–0.3): glutamine, aspartic acid, pyruvic acid, tyrosine; low (≈ −0.3): arginine, choline, uridine; also lactic acid, valine, phenylalanine, alanine, isoleucine, glucose, proline, L-glutathione, lysine, serine.
- **背景** 按一般读法，两组沿 PC2 分开，Loading 2 两端的代谢物最能解释差异；幻灯片只说载荷图显示哪些代谢物最重要，没有说哪种升高或降低。
  > **Background** By the usual reading, since the groups split along PC2, metabolites at the ends of Loading 2 explain the difference best; the slide only says the loadings show which metabolites matter most, not which rise or fall.

考点速记 KEY POINTS

- 得分图看样品分组，载荷图看哪些代谢物驱动分组。
  Scores show grouping; loadings show which metabolites drive it.
- 陷阱：分组不一定出现在 PC1（29.2 %）上。
  Trap: the separation need not lie on PC1 (29.2 %).
- 废水暴露组 vs 对照组：沿 PC2（17.4 %）分开。
  Wastewater-exposed vs control: separated along PC2 (17.4 %).
- 背景：$\mathbf{X} = \mathbf{T}\mathbf{P}^{\mathsf{T}} + \mathbf{E}$。
  Background: $\mathbf{X} = \mathbf{T}\mathbf{P}^{\mathsf{T}} + \mathbf{E}$.

S74 · 活体实例：^{13}C 富集生物 + ^{12}C 示踪剂

S74 · In vivo example: ^{13}C-enriched organism + ^{12}C tracer

幻灯片要点 / What the slide says

还可以用一种不同的脉冲序列追踪特定化学键的形成。这里虾被 ^{13}C 富集，再用一个 ^{12}C 化合物作示踪剂（tracer）。这种方法只看到连在 ^{12}C 上的质子，因此消除了生物自身的背景，只显示烟碱（尼古丁）及其生物转化（biotransformation）产物。研究了烟碱转化为可替宁的过程，并得到一条动力学曲线。

> A different pulse sequence can also track the formation of specific bonds. Here the shrimp was enriched with ^{13}C and a ^{12}C compound was used as a tracer. The approach sees only protons bound to ^{12}C, so it eliminates the background from the organism and shows only the nicotine and the resulting biotransformation products. The biotransformation of nicotine into cotinine was studied and a kinetics curve was generated.

为什么只剩示踪剂 / Why only the tracer remains

背景 连在 ^{13}C 上的质子被 $^{1}J_{CH}$（通常 125–170 Hz）劈裂，序列可利用这个异核耦合把它们的信号相减抵消（^{12}C 过滤）；连在 ^{12}C 上的质子没有这个耦合，原样通过。生物被 ^{13}C 富集后，自身几乎所有 C–H 都被滤掉；示踪剂是 ^{12}C，它和它的转化产物（碳骨架来自示踪剂）都留下来。

> **Background** Protons on ^{13}C are split by $^{1}J_{CH}$ (typically 125–170 Hz), and the sequence uses this heteronuclear coupling to cancel their signals by subtraction (a ^{12}C filter); protons on ^{12}C have no such coupling and pass unchanged. With the organism ^{13}C-enriched, nearly all its own C–H signals are removed, while the ^{12}C tracer and its products (carbon skeleton from the tracer) remain.

生物自身：被滤掉（背景） / organism: filtered out (background)

$$^{1}H\text{–}^{13}C \quad (^{1}J_{CH} \approx 125\text{–}170\ \text{Hz})$$

示踪剂与产物：被观测（背景） / tracer and products: observed (background)

$$^{1}H\text{–}^{12}C \quad (I = 0)$$

^{12}C 自旋 $I = 0$：无耦合，本身也无信号 / ^{12}C has $I = 0$: no coupling, no signal of its own

图怎么读 / Reading the figure

- 左列：烟碱结构 → 虾（Biotransformation，生物转化）→ 可替宁（cotinine）结构。右上「Standard ^{1}H NMR of ^{13}C-H. azteca + ^{12}C-nicotine」：普通氢谱（4–2 ppm）挤满生物自身的峰。图中标签写 ^{13}C-H. azteca，幻灯片文字称 shrimp（虾）。

  > Left column: nicotine → shrimp (Biotransformation) → cotinine. Top right, "Standard ^{1}H NMR of ^{13}C-H. azteca + ^{12}C-nicotine": the ordinary proton spectrum (4–2 ppm) is crowded with the organism's own peaks. The label says ^{13}C-H. azteca; the slide text says shrimp.

- Ac：过滤后只剩烟碱，最高峰为 –N–CH_3（编号 3），另有 1、2 和 4,5 小峰，与旁边结构的编号对应。Ad：时间点 1 hour，出现可替宁的 –N–CH_3（3）及 5、4 峰。

  > Ac: after filtering only nicotine remains, tallest –N–CH_3 (label 3), with small peaks 1, 2 and 4,5 matching the numbered structure. Ad: at the 1 hour time point, cotinine's –N–CH_3 (3) and peaks 5, 4 appear.

- Ae：纵轴 Nicotine concentration (μg)，横轴 Time (hours) 0.1–1.0；烟碱由约 3.1 μg（0.1 h）降至约 0.3 μg（1.0 h），虚线拟合 R^2 0.93。

  > Ae: y-axis nicotine concentration (μg), x-axis time (hours) 0.1–1.0; nicotine falls from ≈ 3.1 μg (0.1 h) to ≈ 0.3 μg (1.0 h); dashed fit, R^2 0.93.

考点速记 KEY POINTS

- ^{13}C 富集生物 + ^{12}C 示踪剂 → 只看 ^{12}C 上的 ^{1}H。
  ^{13}C organism + ^{12}C tracer → only ^{1}H on ^{12}C is seen.
- 烟碱 → 可替宁，动力学曲线 $R^2 = 0.93$。
  Nicotine → cotinine; kinetics fit $R^2 = 0.93$.
- 消除生物背景，只剩烟碱及其转化产物。
  Organism background removed; only nicotine and products.
- 陷阱：观测的是 ^{12}C 上的质子，^{12}C 本身无信号。
  Trap: protons on ^{12}C are observed; ^{12}C itself is silent.

S75 · SSFP：用分辨率换灵敏度

S75 · SSFP: trading resolution for sensitivity

幻灯片要点 / What the slide says

- 稳态自由进动（steady-state free precession, SSFP）是一种牺牲分辨率换取灵敏度的方法。
  > Steady-state free precession (SSFP) is an approach that sacrifices resolution for sensitivity.
- 通常要等自旋弛豫回来才再打下一个脉冲（定量需要完全弛豫）。
  > Normally we wait for the spins to relax before pulsing them again (full relaxation is needed for quantitation).
- SSFP 先建立磁化的稳态，然后就能非常快地反复采样。
  > In SSFP we generate a steady state of magnetization that we can then sample very quickly.

机理 / Mechanism

背景 90° 脉冲后纵向磁化按 T_1 恢复。定量时要等约 $5T_1$，M_z 回到 M_0 的 99 % 以上，每次扫描才完全相同；T_1 若为数秒，每次扫描都要等很久。

> **Background** After a 90° pulse the longitudinal magnetization recovers with T_1. Quantitative work waits about $5T_1$ so that M_z is back above 99 % of M_0 and every scan is identical; with T_1 of several seconds, every scan waits a long time.

完全弛豫（背景） / full relaxation (background)

$$M_z(t) = M_0\left(1 - e^{-t/T_1}\right)$$

$t = 5T_1 \Rightarrow M_z \approx 0.993\,M_0$ / after $5T_1$, 99.3 % recovered

快速脉冲的稳态（背景，90°） / steady state, rapid 90° pulses (background)

$$M_{ss} = M_0\left(1 - e^{-T_R/T_1}\right)$$

$T_R \ll T_1$：$M_{ss} \ll M_0$，但每秒扫描多得多 / $T_R \ll T_1$: $M_{ss} \ll M_0$, but far more scans per second

背景 SSFP 以 $T_R \ll T_1$ 连续打脉冲。头几个脉冲后，每个 T_R 内恢复的量正好抵消每个脉冲消耗的量，磁化达到稳态 M_{ss}，此后每次扫描信号相同，可快速累加。真正的 SSFP 还保留并重聚横向磁化，稳态也与 T_2 和频率偏置有关；上式只是图中画法的简化。每次扫描信号变小，换来单位时间内多得多的扫描；代价是 T_R 短，采集时间（AQ）也只能短，分辨率下降（幻灯片 77）。

> **Background** SSFP pulses continuously with $T_R \ll T_1$. After the first few pulses the recovery during each T_R exactly balances what each pulse removes, the magnetization reaches the steady state M_{ss}, and from then on every scan gives the same signal and scans add up quickly. True SSFP also keeps and refocuses transverse magnetization, so its steady state also depends on T_2 and offset; the formula simplifies what the figure draws. Each scan carries less signal, but far more scans fit per unit time; the cost is that a short T_R forces a short acquisition time (AQ), lowering resolution (slide 77).

图怎么读 / Reading the figure

- 左：「1st RF」与「2nd RF」相隔很长的 TR（重复时间）。M_z 在第一个脉冲处由 M_0 掉到 0，再沿曲线一直恢复到虚线 M_0，才打第二个脉冲。
  > Left: "1st RF" and "2nd RF" are a long TR (repetition time) apart. M_z drops from M_0 to 0 at the first pulse and recovers all the way to the dashed M_0 before the second pulse.
- 右：「1st RF」到「4th RF」间距很短。每个脉冲后 M_z 掉到 0，只恢复到较低的虚线 M_{ss} 就被下一个脉冲打掉，形成锯齿状的稳态。
  > Right: "1st RF" to "4th RF" are closely spaced. After each pulse M_z drops to 0 and recovers only to the lower dashed line M_{ss} before the next pulse: a sawtooth steady state.

考点速记 KEY POINTS

- SSFP：牺牲分辨率，换取灵敏度。
  SSFP sacrifices resolution for sensitivity.
- SSFP：$T_R \ll T_1$，磁化稳定在 $M_{ss} < M_0$。
  SSFP: $T_R \ll T_1$; magnetization settles at $M_{ss} < M_0$.
- 常规：等完全弛豫（定量需要；背景：约 $5T_1$）。
  Normal: full relaxation for quantitation (background: $\approx 5T_1$).
- 稳态建立后快速采样、大量累加。
  Once steady: sample fast and average many scans.

S76 · SSFP：D1 降到 1 ms，扫描次数暴增

S76 · SSFP: D1 down to 1 ms, scans multiply

幻灯片要点 / What the slide says

- 关键优点：弛豫延迟（D1）可降到 1 ms，在同样时间内做极多次扫描。
  > The key advantage: D1 can drop to 1 ms, giving an extreme number of scans in the same time.
- 一个例子中，常规采集 16K 次扫描的时间相当于 380 000 次 SSFP 扫描。
  > In one example, 16K scans of normal acquisition corresponded to 380 000 SSFP scans.
- 这能给出好得多的 SNR。
  > This can give much better SNR.

能多出多少 SNR / How much SNR it buys

背景 每次扫描的信号相同而噪声随机，累加 N 次后 SNR 按 $\sqrt{N}$ 增长。同样时间内的扫描次数比约为 380 000 ÷ 16K ≈ 23–24，仅靠累加就能提高约 4.8–4.9 倍。每次 SSFP 扫描的信号比完全弛豫时小，实际增益还要扣掉这一点。

> **Background** The signal repeats each scan while noise is random, so after N scans SNR grows as $\sqrt{N}$. The scan ratio in the same time is 380 000 ÷ 16K ≈ 23–24, so averaging alone gives about 4.8–4.9 times more. Each SSFP scan carries less signal than a fully relaxed one, which reduces the real gain.

$$\mathrm{SNR} \propto \sqrt{N_{\mathrm{scans}}} \quad \Rightarrow \quad \sqrt{\frac{380\,000}{16\mathrm{K}}} \approx \sqrt{23\text{–}24} \approx 4.8\text{–}4.9$$

背景估算；16K 按 16 000 或 16 384 计结果相近

/ background estimate; 16K read as 16 000 or 16 384 gives about the same

图怎么读 / Reading the figure

- 左：蛋清（Egg white）样品放进低场（台式）NMR（Low-field NMR）。
  > Left: egg white loaded into a low-field (benchtop) NMR.
- 上谱「DEPT-45 ^{13}C NMR」：SNR = 2.68，几乎只有噪声，看不出峰。
  > Top, "DEPT-45 ^{13}C NMR": SNR = 2.68, essentially noise with no visible peaks.
- 下谱「^{13}C-SSFP NMR」：SNR = 65.29，横轴 ^{13}C Chemical Shift 180–0 ppm；约 178 ppm 一个峰，约 150–135 ppm 几个小峰，约 80–55 ppm 和约 45–20 ppm 两组宽峰（后者最高）。
  > Bottom, "^{13}C-SSFP NMR": SNR = 65.29, x-axis ^{13}C chemical shift 180–0 ppm; a peak at ≈ 178 ppm, small peaks at ≈ 150–135 ppm, and two broad groups at ≈ 80–55 and ≈ 45–20 ppm (the latter tallest).
- **背景** 两谱 SNR 之比 65.29 ÷ 2.68 ≈ 24，大于只按扫描次数估算的约 4.8 倍。但这是 DEPT-45 与直接 ^{13}C-SSFP 两种不同实验的比较，幻灯片也没说这张图就是 16K / 380 000 那个例子，不要把 24 倍全归于扫描次数。
  > **Background** The SNR ratio 65.29 ÷ 2.68 ≈ 24 exceeds the ≈ 4.8 estimated from scan count alone. But it compares two different experiments (DEPT-45 vs direct ^{13}C-SSFP), and the slide does not say this figure is the 16K / 380 000 example, so do not credit the whole 24× to the scan count.

考点速记 KEY POINTS

- SSFP：D1 = 1 ms，同时间扫描次数暴增。
  SSFP: D1 = 1 ms; vastly more scans in the same time.
- 背景：$\mathrm{SNR} \propto \sqrt{N_{\mathrm{scans}}}$ → 约 4.8 倍。
  Background: $\mathrm{SNR} \propto \sqrt{N_{\mathrm{scans}}}$ → ≈ 4.8×.
- 例：常规 16K 次 ↔ SSFP 380 000 次。
  Example: 16K normal scans ↔ 380 000 SSFP scans.
- 蛋清（低场）：DEPT-45 SNR 2.68 vs ^{13}C-SSFP 65.29。
  Egg white, low field: DEPT-45 2.68 vs ^{13}C-SSFP 65.29.

S77 · SSFP 的代价：截断与分辨率

S77 · The price of SSFP: truncation and resolution

幻灯片要点 / What the slide says

难点在于 SSFP 用短的采集时间效果最好，而短采集时间意味着截断（truncation），所以分辨率与 SNR 之间存在权衡。SSFP 对弛豫较快、本来就宽的信号效果更好。这里从常规 AQ 改为 30 ms 的 SSFP 会损失大量分辨率，而 120 ms 保留了大部分分辨率，同时 SNR 大幅提升。

> A challenge is that SSFP works best with short acquisition times (truncation), so there is a trade-off between resolution and SNR. It works better for faster-relaxing, already broad signals. Here, going from normal AQ to 30 ms SSFP loses lots of resolution, but 120 ms keeps most of the resolution and gives a great SNR boost.

为什么短 AQ 损失分辨率 / Why a short AQ costs resolution

背景 采集到 t_{aq} 就停，等于给 FID 乘了一个矩形窗；FT 后每条谱线都卷积上 sinc 形状，能分开的最小频率差约为 $1/t_{\mathrm{aq}}$。若 FID 在 t_{aq} 之前已衰减完（快弛豫、本来就宽的峰），截断几乎不丢信息，所以 SSFP 更适合宽信号。AQ 越短，T_{R} 越短，同样时间内扫描越多，SNR 越高。

> **Background** Stopping at t_{aq} multiplies the FID by a rectangular window; after the FT every line is convolved with a sinc shape, and the smallest resolvable frequency difference is about $1/t_{\mathrm{aq}}$. If the FID has decayed before t_{aq} (fast-relaxing, already broad peaks), truncation loses almost nothing, so SSFP suits broad signals. A shorter AQ means a shorter T_{R}, more scans in the same time and higher SNR.

$$\Delta\nu_{\min} \approx \frac{1}{t_{\mathrm{aq}}}: \qquad 30\ \mathrm{ms} \rightarrow 33\ \mathrm{Hz}, \qquad 120\ \mathrm{ms} \rightarrow 8.3\ \mathrm{Hz}$$

背景；截断还会在峰脚产生 sinc 摆动 / background; truncation also adds sinc wiggles at the peak foot

图怎么读 / Reading the figure

AQ	蓝列 Blue (a–e)	红列 Red (f–j)
30 ms	a) SNR 5628	f) SNR 6601
60 ms	b) SNR 3017	g) SNR 3384
90 ms	c) SNR 2007	h) SNR 2603
120 ms	d) SNR 1771	i) SNR 2157
常规（×4 显示） normal (shown ×4)	e) SNR 301	j) SNR 515
120 ms ÷ 常规（背景计算） 120 ms ÷ normal (background)	≈ 5.9	≈ 4.2

横轴 ¹H Chemical Shift 约 11–1 ppm；每条谱都有约 8.7 ppm 的一个尖峰和约 4–2 ppm 的一簇多重峰。AQ 越短 SNR 越高，但这簇峰越模糊，30 ms 时细节基本糊在一起；常规谱 e、j 纵向放大 ×4 才看得清。幻灯片没有说明蓝、红两列有何不同。

> x-axis ¹H chemical shift ≈ 11–1 ppm; each spectrum has a sharp peak at ≈ 8.7 ppm and a cluster of multiplets at ≈ 4–2 ppm. Shorter AQ gives higher SNR but a blurrier cluster, largely merged at 30 ms; the normal spectra e and j are enlarged ×4 to be visible. The slide does not say how the blue and red columns differ.

考点速记 KEY POINTS

- SSFP 需短 AQ → 截断；背景 $\Delta\nu_{\min} \approx 1/t_{\mathrm{aq}}$。
  SSFP needs short AQ → truncation; background $\Delta\nu_{\min} \approx 1/t_{\mathrm{aq}}$.
- 更适合快弛豫、本来就宽的信号。
  Best for fast-relaxing, already broad signals.
- 分辨率与 SNR 互相权衡。
  Resolution and SNR trade off.
- 陷阱：30 ms SNR 最高却最糊；120 ms 才是好折中。
  Trap: 30 ms has top SNR but blurs; 120 ms is the good compromise.

S78 · CRAFT：不做傅里叶变换的数据处理

S78 · CRAFT: processing without the Fourier transform

幻灯片要点 / What the slide says

- CRAFT = Complete Reduction to Amplitude Frequency Table（完全约化为振幅–频率表）。
  > CRAFT = Complete Reduction to Amplitude Frequency Table.
- 这是一种不用傅里叶变换（FT）来处理数据的方法。
  > It is an approach to process data without using the FT.
- 虽然 FT NMR 是标准做法，但它也带来一些难题；CRAFT 有助于克服其中一部分。
  > Though FT NMR is the standard, some challenges go along with it; CRAFT helps overcome some of them.

机理 / Mechanism

背景 FID 可以看成若干个衰减正弦信号之和。FT 把整条 FID 变成谱，再由人做相位校正、基线校正和积分。CRAFT 直接在时域拟合这个模型，得到每个成分的频率 ν_k、振幅 a_k、相位 ϕ_k 和衰减速率（对应线宽），结果是一张表，而不是一张需要人工处理的谱；原始 CRAFT 用贝叶斯分析完成这一拟合。

> **Background** The FID can be treated as a sum of decaying sinusoids. The FT turns the whole FID into a spectrum, which a person then phases, baseline-corrects and integrates. CRAFT fits that model directly in the time domain and returns each component's frequency ν_k, amplitude a_k, phase ϕ_k and decay rate (i.e. linewidth): a table, not a spectrum that needs manual processing; the original CRAFT does this fit by Bayesian analysis.

$$s(t) = \sum_{k=1}^{K} a_k \, e^{i\phi_k} \, e^{-t/T_{2,k}^{*}} \, e^{i2\pi\nu_k t} + \text{noise}$$

背景：表格每一行 = 一个成分 (ν_k, a_k)；振幅可直接用于定量

/ background: each table row = one component (ν_k, a_k); amplitudes feed quantitation directly

背景 FT 处理的难题包括：相位和基线要人工调，积分结果取决于调得好坏和积分范围；重叠峰难以分开积分；弱峰埋在噪声里。幻灯片 79、80 展示了其中两项：无法正常校相的倒峰，以及被噪声掩盖的弱峰。

> **Background** FT challenges include manual phasing and baseline correction, integrals that depend on how well these are done and on the limits chosen, overlapping peaks that are hard to integrate separately, and weak peaks buried in noise. Slides 79 and 80 show two of them: an inverted peak that cannot be phased normally, and weak peaks hidden by noise.

图怎么读 / Reading the figure

- 左上「FID (Free Induction Decay)」：横轴 time /s（0–1 s），一条衰减的振荡信号。右上「spectrum」：横轴 frequency /Hz（4000–0 Hz），约十条尖锐谱线。
  > Top left, "FID (Free Induction Decay)": x-axis time /s (0–1 s), a decaying oscillation. Top right, "spectrum": x-axis frequency /Hz (4000–0 Hz), about ten sharp lines.
- 绿箭头「CRAFT[1]」由 FID 直接指向中间的 Table，两列为 Frequency/ppm 和 Amplitude /a.u.：1.31 / 55.6、1.65 / 99.0、2.71 / 57.7、2.76 / 52.3、3.43 / 63.1、3.54 / 92.5、3.89 / 79.4 …… 8.92 / 86.0。
  > The green "CRAFT[1]" arrow goes from the FID straight to the Table, columns Frequency/ppm and Amplitude /a.u.: 1.31 / 55.6, 1.65 / 99.0, 2.71 / 57.7, 2.76 / 52.3, 3.43 / 63.1, 3.54 / 92.5, 3.89 / 79.4 … 8.92 / 86.0.
- 表格 → 绿箭头 →「quantitative interpretation」（定量解读）。紫、橙箭头把表格与谱和「Freq. domain · comfort zone for spectroscopists」（频域：波谱学家的舒适区）相连，大意是需要时表格也能画回谱图。
  > Table → green arrow → "quantitative interpretation". Purple and orange arrows link the table with the spectrum and "Freq. domain · comfort zone for spectroscopists": roughly, the table can be drawn back as a spectrum when wanted.

考点速记 KEY POINTS

- CRAFT = 完全约化为振幅–频率表，不用 FT。
  CRAFT = Complete Reduction to Amplitude Frequency Table; no FT.
- 输出是频率–振幅表，可直接定量。
  Output: a frequency–amplitude table, ready for quantitation.
- 背景：在时域拟合衰减正弦之和。
  Background: a time-domain fit of decaying sinusoids.
- FT 是标准，但有难题（校相、噪声等）。
  FT is standard but has problems (phasing, noise…).

S79 · CRAFT：子 FID 与倒峰的相位校正

S79 · CRAFT: sub-FIDs and phasing an inverted peak

幻灯片要点 / What the slide says

- FID 被拆成多个子 FID（sub-FID），每个可以单独处理。
  > The FID is broken into sub-FIDs, which can then be treated separately.
- 例如之前展示过的那张带一个倒峰（inverted peak）的谱：用常规方法不可能完成相位校正，而用 CRAFT 可以做到。
  > For example, the spectrum shown earlier with an inverted peak: it is impossible to phase normally, but in CRAFT you can do it.

为什么常规校相做不到 / Why normal phasing fails

背景 常规相位校正对整张谱只有两个参数：零阶 ϕ_0（所有峰相同）和一阶 ϕ_1（随频率线性变化）。若某个峰的相位不在这条直线上（例如差 180°，表现为倒峰），把它调正，其他峰就歪了，反之亦然。CRAFT 先按感兴趣区（ROI）把 FID 数字滤波成子 FID，每段只含该频段的信号、点数少，拟合更快更稳；每个成分有自己的相位 ϕ_k，所以每段可单独校相，或直接报告振幅。

> **Background** Normal phase correction has two parameters for the whole spectrum: zero order ϕ_0 (same for every peak) and first order ϕ_1 (linear in frequency). If one peak's phase is off that line (e.g. 180° off, appearing inverted), fixing it twists the others, and vice versa. CRAFT first filters the FID digitally into sub-FIDs by region of interest (ROI); each holds only that band's signals with fewer points, so fitting is faster and more stable. Every component has its own phase ϕ_k, so each region can be phased on its own, or its amplitude reported directly.

$$\phi(\nu) = \phi_0 + \phi_1 \, (\nu - \nu_{\text{ref}})$$

背景：常规校相只能沿这条直线；CRAFT 每个成分各有 ϕ_k

/ background: normal phasing can only follow this line; CRAFT gives each component its own ϕ_k

背景 若倒峰的负号有物理意义（例如来自编辑实验），把它调成正峰会丢掉这一信息；幻灯片没有说明这张谱为什么有倒峰。

> **Background** If the inversion carries physical meaning (e.g. from an editing experiment), phasing it positive discards that information; the slide does not say why this spectrum has an inverted peak.

图怎么读 / Reading the figure

- 上：左框是整条 FID，标「ROI」的箭头通向右框的 7 条子 FID，各有不同的振荡和衰减。
  > Top: the left box is the whole FID; an arrow labelled "ROI" leads to seven sub-FIDs on the right, each oscillating and decaying differently.
- 下：^{13}C 谱（¹³C Chemical Shift，约 170–0 ppm），峰标 –COO–（约 170 ppm，向下的倒峰）、–CH=CH–（约 125–130 ppm）、–CHO– 和 –CH_2O–（约 70–60 ppm）、–CH_2–（约 27–30 ppm，最高）、–CH_3（约 10–15 ppm）。
  > Bottom: a ^{13}C spectrum (≈ 170–0 ppm) labelled –COO– (≈ 170 ppm, pointing down: the inverted peak), –CH=CH– (≈ 125–130 ppm), –CHO– and –CH_2O– (≈ 70–60 ppm), –CH_2– (≈ 27–30 ppm, tallest), –CH_3 (≈ 10–15 ppm).
- 插图：¹H 谱（9–0 ppm），约 4.7 ppm 一个大峰、约 1.3 ppm 一个小峰；照片是管中一块橙红色样品（未注明是什么）。
  > Insets: a ¹H spectrum (9–0 ppm) with a large peak at ≈ 4.7 ppm and a small one at ≈ 1.3 ppm; a photo of an orange-red sample in a tube (not identified).

考点速记 KEY POINTS

- CRAFT 按 ROI 把 FID 拆成子 FID，分别处理。
  CRAFT splits the FID by ROI into sub-FIDs, treated separately.
- CRAFT 每个成分有自己的相位 → 能校相。
  CRAFT gives each component its own phase → phasing works.
- 常规校相只有 ϕ_0, ϕ_1：带倒峰的谱无法整体校正。
  Normal phasing has only ϕ_0, ϕ_1: an inverted peak can't be fixed.
- 例：¹³C 谱中 –COO–（约 170 ppm）是倒峰。
  Example: –COO– (≈ 170 ppm) is inverted in the ¹³C spectrum.

S80 · CRAFT：「SNR」提升与更多的峰

S80 · CRAFT: "SNR" improvement and more peaks

幻灯片要点 / What the slide says

另一种可能是「SNR」提升。CRAFT 在某种程度上能去掉噪声，使原本被掩盖的信号显现出来。这里 CRAFT 看到的峰约为标准一维谱的 5 倍。CRAFT 也可以与 SSFP 结合。

> Another possibility is "SNR" improvement. CRAFT can kind of remove noise, which allows signals to be seen that would otherwise be hidden. Here CRAFT has ~5 times more peaks than are visible in the standard 1D. It can also be combined with SSFP.

为什么「SNR」加引号 / Why "SNR" is in quotes

背景 CRAFT 输出的是拟合出的成分表，画回谱图时只含模型中的峰，噪声大多不在模型里，谱看起来几乎没有噪声。这种「SNR」不是测量本身的 SNR，不能与普通谱直接比较，图中 (C) 也没给 SNR 数值。同理，在极低 SNR 下模型也可能给出并不存在的成分，需要对照验证（幻灯片没有讨论这一点）。

> **Background** CRAFT outputs a table of fitted components; drawn back as a spectrum it holds only the modelled peaks, and most of the noise is not in the model, so the spectrum looks nearly noise-free. This "SNR" is not the measurement's own SNR and cannot be compared directly with an ordinary spectrum; panel (C) gives no SNR value. Likewise, at very low SNR a model can return components that are not real, so results need checking (the slide does not discuss this).

(A) 1D，12 h / (A) 1D, 12 h	(B) SSFP，12 h / (B) SSFP, 12 h	(C) CRAFT，12 h / (C) CRAFT, 12 h
$\mathrm{SNR} = 3.93$	$\mathrm{SNR} = 15.61$	13 peaks
3 个峰 / 3 peaks	7 个峰 / 7 peaks	未给 SNR / no SNR given

图怎么读 / Reading the figure

- $^{19}\mathrm{F}$ 谱，横轴 ^{19}F Chemical Shift −40 至 −180 ppm：(A) 红、(B) 绿、(C) 蓝；(C) 另标「4.7x num. peaks VS 0.6x SSFP」。约 −65 ppm 的高峰标 I.S.（内标），约 −120 ppm 的高峰标 F−（氟离子）。

  > $^{19}\mathrm{F}$ spectra, x-axis −40 to −180 ppm: (A) red, (B) green, (C) blue; (C) is also labelled "4.7x num. peaks VS 0.6x SSFP". The tall peak at ≈ −65 ppm is I.S. (internal standard), the one at ≈ −120 ppm F− (fluoride).

- 右侧放大框（−110 至 −130 ppm）：A 只有 F− 一个峰；B 多出几个小峰；C 的小峰最多、最清楚。**背景** SNR 之比 15.61 ÷ 3.93 ≈ 4.0；峰数 3 → 7 → 13。

  > Insets (−110 to −130 ppm): A shows only F−; B adds a few small peaks; C shows the most and clearest. **Background** SNR ratio 15.61 ÷ 3.93 ≈ 4.0; peak count 3 → 7 → 13.

编者注 / Editor's note

图中标签为「4.7x num. peaks」，而 13 ÷ 3 ≈ 4.3；幻灯片文字说「~5 times」。两种说法都照录，答题可用幻灯片原话「约 5 倍」。「0.6x SSFP」的含义和 (C) 是否基于 SSFP 数据，幻灯片都没有说明。

> The figure label says "4.7x num. peaks", while 13 ÷ 3 ≈ 4.3; the slide text says "~5 times". Both are quoted; in an answer, the slide's "~5 times" is safe. The slide explains neither "0.6x SSFP" nor whether (C) was computed from SSFP data.

考点速记 KEY POINTS

- CRAFT 能「去噪」，显出被噪声掩盖的弱峰。
  CRAFT "removes" noise, revealing hidden weak peaks.
- 文字「~5 倍」，图标「4.7x」，13/3 ≈ 4.3。
  Text "~5×", label "4.7x", 13/3 ≈ 4.3.
- 12 h 峰数：1D 3 → SSFP 7 → CRAFT 13。
  Peaks in 12 h: 1D 3 → SSFP 7 → CRAFT 13.
- 可与 SSFP 结合；CRAFT「SNR」≠ 测量 SNR。
  Combines with SSFP; CRAFT "SNR" ≠ measured SNR.

公式与符号总表（一）· §1–§4

Formula and notation summary (1) · §1–§4

左列为幻灯片编号；标 **背景** 的是课件没有印出、但图背后的标准关系。

Slide numbers on the left; items marked **Background** are not printed on the slides but are the standard relation behind a figure.

S	项目 Item	公式或数值 Formula or value
	§1 溶剂峰抑制 · Solvent suppression	
8	WATERGATE 序列 / sequence	$(\pi/2)_x$;　$G_1\left[p_1\ \tau\ p_2\ \tau\ p_3\ \tau\ p_4\ \tau\ p_5\ \tau\ p_6\right]G_2$
8	角度与相位 / angles, phases	$p_1=p_6=20.79°$, $p_2=p_5=62.28°$, $p_3=p_4=131.58°$;　$p_1,p_2,p_3=x$; $p_4,p_5,p_6=-x$
8	**背景** 3-9-19 / 3-9-19 train	$3\alpha, 9\alpha, 19\alpha$, $26\alpha=180°$ $(\alpha\approx 6.92°)$；水净 $0°$, $\Delta\nu=1/(2\tau)$ 处净 $180°$ / $26\alpha=180°$ ($\alpha\approx 6.92°$); water net $0°$; net $180°$ at $\Delta\nu=1/(2\tau)$
11–12	近溶剂损失；延迟 / near-solvent loss; delays	约 0.4 ppm 抹除、至多 1.1 ppm 衰减；模块间几百 μs，低场更长 / ≈ 0.4 ppm obliterated, ≤ 1.1 ppm attenuated; a few hundred μs, longer at low field
	§2–§3 多维 NMR、NUS 与超快 · nD NMR, NUS and ultrafast	
17–18	三维；H–C–Si / 3D; H–C–Si	t_1,t_2,t_3; F_1,F_2,F_3;　${}^1J_{CH}=140$ Hz, $J_{CSi}=59$ Hz; $\delta_{Si}=9.95,\ 7.99,\ 2.25$ ppm
20	NUS 省时 / NUS saving	NUS 25 %：64 增量 3.75 min ≈ 常规 256 增量 15 min / NUS 25 %: 64 increments, 3.75 min ≈ conventional 256, 15 min
21–23	超快 NMR / ultrafast	$\Delta, 2\Delta, \ldots, n\Delta$：$n$ 次 vs 1 次扫描；COSY 21 min vs 0.2 s；薄荷醇 4.5 min vs 100 ms / n scans vs 1 scan; COSY 21 min vs 0.2 s; menthol 4.5 min vs 100 ms
	§4 MAS、HR-MAS、固体与 CMP · MAS, HR-MAS, solids, CMP	
25	魔角 / magic angle	$\theta_m=\arccos\frac{1}{\sqrt{3}}\approx 54.74°$;　$P_2(\cos\theta)=\frac{1}{2}\left(3\cos^2\theta-1\right)=0$ （**背景**；幻灯片误称「一阶」） / (background; the slide says "first order")
25–27	转速；**背景** 边带 / spin rate; sidebands	$\nu_{rot}\approx$ 1–15 kHz;　$\nu_{SSB}=\nu_{iso}\pm k\,\nu_{rot}$, $k=1,2,\ldots$;　500 Hz vs 2500 Hz
24	转子 / rotors	$0.7, 1.3, 3.2, 4, 7$ mm $\rightarrow 111, 67, 24, 15, 7$ kHz; $0.59, 3, 46.7, 99, 359.6\ \mu$L
28–29	转速极限 / spin limits	4 mm 固体约 15 kHz；细胞约 3500 Hz 破裂；0.5 mm 达 150 kHz（幻灯片「~6 million RPM」，应为 9.0×10^6 rpm） / 4 mm solids ≈ 15 kHz; cells break ≈ 3500 Hz; 150 kHz in 0.5 mm (slide "~6 million RPM", really 9.0×10^6 rpm)
33	谱编辑 / spectral editing	180 ms 内移动 $>1\ \mu$m / $<1\ \mu$m；T_2 / I-T_2 ^{13}C CP-MAS / moves $>$ / $<1\ \mu$m in 180 ms; T_2 / I-T_2 ^{13}C CP-MAS

考点速记 KEY POINTS

- 魔角 $54.74°$：$P_2(\cos\theta)=0$（二阶）。
  Magic angle $54.74°$: $P_2(\cos\theta)=0$ (second order).
- WATERGATE：$26\alpha=180°$，水净翻转 $0°$。
  WATERGATE: $26\alpha=180°$; water's net rotation $0°$.
- 边带在 $\nu_{iso}\pm k\nu_{rot}$，转得越快离得越远。
  Sidebands at $\nu_{iso}\pm k\nu_{rot}$; faster spin, further out.
- NUS 25 %（64 增量）≈ 常规 4 倍时间的结果。
  NUS 25 % (64 increments) ≈ a conventional run 4× longer.

公式与符号总表（二）· §5–§8

Formula and notation summary (2) · §5–§8

S	项目 Item	公式或数值 Formula or value
	§5 探头与线圈尺寸 · Probe and coil size	
37–38	管与体积 / tubes, volumes	5 mm 600 µL、3 mm 150 µL、1.7 mm 28 µL（S38 文字 30 µL，约为 5 mm 的 1/20） / 5 mm 600 µL, 3 mm 150 µL, 1.7 mm 28 µL (S38 text: 30 µL, ≈ 20× less than 5 mm)
39, 48	质量灵敏度；样品量 / mass sensitivity; amount	$S_\mathrm{m} = a\,d^{-b}$（拟合）；大探头 $V \propto d^2$ / $S_\mathrm{m} = a\,d^{-b}$ (fit); larger probe $V \propto d^2$
43, 46	**背景** 章动；植入线圈 / nutation; implant	$\theta = \gamma B_1 t_\mathrm{p}$，$S \propto \sin\theta$；450 nL，图 $\times 76$、文字约 100 倍 / $\theta = \gamma B_1 t_\mathrm{p}$, $S \propto \sin\theta$; 450 nL, figure $\times 76$, text "~100×"
	§6 混合物与选择性实验 · Mixtures and selective experiments	
55	仅氨基酸 NMR / amino-acid-only	$(\mathrm{H})\mathrm{C}_\beta\mathrm{C}_\alpha\mathrm{CO}(\mathrm{C}_\alpha)\mathrm{H}_\alpha$；　$^{\beta}\mathrm{HC}\!-\!^{\alpha}\mathrm{HC}\!-\!\mathrm{C}\!=\!\mathrm{O}$
59, 66	GEMSTONE；聚焦 / GEMSTONE; focusing	$5.255,\ 5.285,\ 5.305$ ppm；　SNR $234 \to 1647,\ 102 \to 918,\ 114 \to 1055,\ 21 \to 265$
	§7 活体 NMR · In vivo NMR	
71	存活实验转速 / survival spin rates	50 Hz 对比 2.5 kHz；**背景** $a = (2\pi\nu_\mathrm{rot})^2 r$，加速度之比 2500 / 50 Hz vs 2.5 kHz; **Background** $a = (2\pi\nu_\mathrm{rot})^2 r$, acceleration ratio 2500
73–74	PCA；同位素过滤 / PCA; isotope filter	PC1 29.2 %、PC2 17.4 %、PC3 9.2 %；$^{13}\mathrm{C}$ 生物 + $^{12}\mathrm{C}$ 示踪，$R^2 = 0.93$ / PC1 29.2 %, PC2 17.4 %, PC3 9.2 %; $^{13}\mathrm{C}$ organism + $^{12}\mathrm{C}$ tracer, $R^2 = 0.93$
	§8 SSFP 与 CRAFT · SSFP and CRAFT	
75	**背景** 恢复与稳态 / recovery, steady state	$M_z(t) = M_0\left(1 - e^{-t/T_1}\right)$；　$M_\mathrm{ss} = M_0\left(1 - e^{-T_\mathrm{R}/T_1}\right)\ (90^\circ)$
76	SSFP 速度 / SSFP speed	$D_1 = 1$ ms；16K ↔ 380 000 次；SNR 2.68 → 65.29；**背景** SNR $\propto \sqrt{N_\mathrm{scans}}$ / $D_1 = 1$ ms; 16K ↔ 380 000 scans; SNR 2.68 → 65.29; **Background** SNR $\propto \sqrt{N_\mathrm{scans}}$
77	AQ；**背景** 截断 / AQ; truncation	$30, 60, 90, 120$ ms；　$\Delta\nu_\mathrm{min} \approx 1/t_\mathrm{aq}$
78–79	**背景** CRAFT 模型；校相 / model; phasing	$s(t) = \sum_k a_k e^{i\phi_k} e^{-t/T_{2,k}^*} e^{i2\pi\nu_k t}$；　$\phi(\nu) = \phi_0 + \phi_1(\nu - \nu_\mathrm{ref})$
80	CRAFT 结果 / CRAFT result	SNR 3.93（1D）、15.61（SSFP）；峰 3、7、13；图「4.7x」、13/3 ≈ 4.3、文字「~5 倍」 / SNR 3.93 (1D), 15.61 (SSFP); peaks 3, 7, 13; label "4.7x", 13/3 ≈ 4.3, text "~5 times"

考点速记 KEY POINTS

- 大探头样品量 $V \propto d^2$；小线圈质量灵敏度高。
  Larger probe $V \propto d^2$; small coils win on mass sensitivity.
- SSFP：$D_1 = 1$ ms；背景 $\Delta\nu_\mathrm{min} \approx 1/t_\mathrm{aq}$。
  SSFP: $D_1 = 1$ ms; background $\Delta\nu_\mathrm{min} \approx 1/t_\mathrm{aq}$.
- 活体：50 Hz 存活，2.5 kHz 致死；背景 $a \propto \nu_\mathrm{rot}^2$。
  In vivo: 50 Hz survive, 2.5 kHz kills; background $a \propto \nu_\mathrm{rot}^2$.
- CRAFT 峰数 3 → 13（文字「~5 倍」）。
  CRAFT peaks 3 → 13 (text "~5 times").

全书回顾（一）· §1–§4

Chapter recap (1) · §1–§4

§1 溶剂峰抑制（S2–13） / §1 Solvent suppression (S2–13)

- 大溶剂峰决定接收机增益，样品峰损失 SNR。先选不出峰的溶剂（氘代；^{19}F 可能不需要），否则用实验抑制。

  A huge solvent peak sets the receiver gain and costs sample SNR. First pick a silent solvent (deuterated; ^{19}F may need none), otherwise suppress experimentally.

- 由弱到强：presat → PURGE（中等、简单、内含 presat）→ WATERGATE、W5、JRS。WET 用梯度，可同时抑制多个溶剂峰且不扰可交换峰。

  Weak to strong: presat → PURGE (medium, easy, contains presat) → WATERGATE, W5, JRS. WET uses gradients, suppresses several solvent peaks at once and spares exchangeables.

- 三项代价：可交换质子被衰减（presat 类）；最激进时约 0.4 ppm 抹除、至多 1.1 ppm 衰减；长序列有弛豫损失，低场更重。

  Three costs: exchangeables attenuated (presat-type); at most aggressive ≈ 0.4 ppm obliterated, up to 1.1 ppm attenuated; relaxation loss in long sequences, worse at low field.

§2 多维 NMR（S14–18） / §2 Higher-dimensional NMR (S14–18)

- 3D 峰容量更高，主要用于蛋白质和混合物；小分子结构解析通常不需要，3D 以上多为展示。

  3D has higher peak capacity and serves mainly proteins and mixtures; small-molecule elucidation rarely needs it, and above 3D is mostly showing off.

- 常用投影代替整个立方体；可把实验组合（HSQC-TOCSY、HMQC-COSY）或加入别的核（H–C–Si）。代价是时间与成本；活体 3D 因弛豫信号弱，离体好得多。

  Projections stand in for the full cube; experiments can be combined (HSQC-TOCSY, HMQC-COSY) or other nuclei added (H–C–Si). The cost is time and money; in vivo 3D is weak from relaxation, ex vivo much better.

§3 NUS 与超快 NMR（S19–23） / §3 NUS and ultrafast NMR (S19–23)

- NUS 少采 t_1 增量，NUS 25 %（3.75 min）≈ 常规 15 min；代价是 SNR（加权采样可缓解）和伪峰。

  NUS samples fewer t_1 increments: NUS 25 % (3.75 min) ≈ conventional 15 min; costs SNR (weighted sampling helps) and possible artifacts.

- 超快 NMR 在管内不同位置同时采集所有增量，一次扫描（COSY 0.2 s vs 21 min）；但 SNR 差、需高浓度、对样品不均匀敏感，峰可能虚假或缺失。

  Ultrafast NMR acquires all increments at once in different parts of the tube, in one scan (COSY 0.2 s vs 21 min); but SNR is poor, high concentration is needed, it is sensitive to inhomogeneity, and peaks can be spurious or missing.

§4 MAS、HR-MAS、固体与 CMP（S24–36） / §4 MAS, HR-MAS, solids, CMP (S24–36)

- 魔角 54.74°（$P_2 = 0$）旋转可消除多种致宽作用；0 Hz 为粉末谱，边带间距等于转速，大峰（水）的边带最明显。

  Spinning at 54.74° ($P_2 = 0$) cancels many broadening interactions; 0 Hz gives a powder spectrum, sidebands sit one spin rate apart and are worst for huge peaks (water).

- 转速越快越好但越难：4 mm 固体约 15 kHz，细胞约 3500 Hz 破裂；小转子更快，但贵且难用。

  Faster is better but harder: 4 mm to ≈ 15 kHz for solids, cells break at ≈ 3500 Hz; smaller rotors spin faster but are expensive and awkward.

- HR-MAS：凝胶与半固体、慢转、带梯度；固体探头：高功率、快转；CMP：一个探头测所有相，配谱编辑（扩散编辑、T_2 / I-T_2 ^{13}C CP-MAS 过滤）。

  HR-MAS: gels and semi-solids, slow spinning, gradient; solids probe: high power, fast spinning; CMP: all phases in one probe, with spectral editing (diffusion editing, T_2 / I-T_2 ^{13}C CP-MAS filters).

考点速记 KEY POINTS

- 含 presat 的方法会衰减 OH、NH。
  Methods containing presat attenuate OH and NH.
- NUS 少采增量；超快单次扫描但 SNR 差。
  NUS skips increments; ultrafast is one scan but low SNR.
- 3D 换峰容量，代价是时间。
  3D buys peak capacity at the price of time.
- 只要固体 → 固体探头；跨相变化 → CMP。
  Solids only → solids probe; changes between phases → CMP.

全书回顾（二）· §5–§8

Chapter recap (2) · §5–§8

§5 探头与线圈尺寸（S37–49） / §5 Probe and coil size (S37–49)

- 小管（1.7 mm 约 30 µL）少稀释、质量灵敏度高；微线圈适合质量受限样品，阵列可让暴露组与对照同测。
  > Small tubes (1.7 mm, ≈ 30 µL) dilute less and give better mass sensitivity; microcoils suit mass-limited samples; arrays measure exposed and control together.
- 微线圈：B_1 不均匀、样品永不反转（COSY 无信号）、放样难；普通样品（牛奶）不如 5 mm。
  > Microcoils: poor B_1 homogeneity, the sample never inverts (no COSY signal), hard placement; ordinary samples (milk) do better in 5 mm.
- 大线圈：浓度灵敏度好，样品量 $\propto d^2$（雨水等稀样）；但功率大、不均匀、发热、质量灵敏度差，8 mm 已基本淘汰。
  > Large coils: good concentration sensitivity, amount $\propto d^2$ (dilute samples, e.g. rainwater); but more power, inhomogeneity, heating, poor mass sensitivity; 8 mm largely phased out.

§6 混合物与选择性实验（S50–66） / §6 Mixtures and selective experiments (S50–66)

- 选择性 TOCSY：激发一个峰，看其耦合链（峰不能重叠）。单重态：过滤近等价自旋对，无需成形脉冲，有泄漏。
  > Selective TOCSY: excite one peak, see its coupled chain (no overlap allowed). Singlet state: filters near-equivalent spin pairs, no shaped pulses, some breakthrough.
- 仅氨基酸：只选 βHC–αHC–C=O，切趾去掉蛋白质，乳酸也通过。GEMSTONE：梯度空间编码挑出重叠多重峰，快弛豫样品损失大。
  > Amino-acid-only: passes βHC–αHC–C=O, apodization removes proteins, lactic acid passes too. GEMSTONE: gradient encoding picks overlapping multiplets; big loss for fast-relaxing samples.
- DREAMTIME：位移、耦合、J 都对才通过，每个化合物两个峰；可多重选择、加 TOCSY、接 HSQC，聚焦提升 SNR。
  > DREAMTIME: right shifts, coupling and J to pass, two peaks per compound; multiple selection, TOCSY block, selective HSQC, focusing for SNR.

§7 活体 NMR（S67–74） / §7 In vivo NMR (S67–74)

- 无损 → 可测活体；会动、维持生命、弛豫快 → 峰形差；自由游动 → 扩散实验不灵。
  > Nondestructive → living organisms; movement, life support, fast relaxation → poor lineshape; swimming spoils diffusion experiments.
- 流动系统：锁场、塞子、注入与抽吸。活体 CMP：转子帽开孔、超慢速 MAS（50 Hz 存活、2.5 kHz 致死），代价是边带。
  > Flow system: lock, plugs, injection and suction. In vivo CMP: holed rotor cap, ultra-slow MAS (50 Hz survives, 2.5 kHz kills), sidebands as the cost.
- 应用：过程监测（鲍鱼暴露与恢复）、PCA（废水 vs 对照）、^{13}C 生物 + ^{12}C 示踪（烟碱 → 可替宁）。
  > Uses: process monitoring (abalone exposure, recovery), PCA (wastewater vs control), ^{13}C organism + ^{12}C tracer (nicotine → cotinine).

§8 SSFP 与 CRAFT（S75–80） / §8 SSFP and CRAFT (S75–80)

- SSFP：不等完全弛豫，稳态下快速采样（D1 = 1 ms；16K ↔ 380 000 次），以分辨率换灵敏度；短 AQ 截断，120 ms 是好折中，宜用于快弛豫的宽信号。
  > SSFP: no full relaxation, fast sampling of a steady state (D1 = 1 ms; 16K ↔ 380 000 scans), resolution for sensitivity; short AQ truncates, 120 ms is a good compromise, best for fast-relaxing broad signals.
- CRAFT：不用 FT，把 FID 约化为振幅–频率表；子 FID 分别处理，能给带倒峰的谱校相；「去噪」显出弱峰（13 vs 3 个），可与 SSFP 结合。
  > CRAFT: no FT, the FID becomes an amplitude–frequency table; sub-FIDs treated separately, so an inverted-peak spectrum can be phased; "removes noise" to reveal weak peaks (13 vs 3); combines with SSFP.

考点速记 KEY POINTS

- 质量灵敏度看小线圈，浓度灵敏度看大线圈。
  Mass sensitivity: small coils; concentration: large coils.
- 活体：无损、过程监测；转得慢才能活。
  In vivo: nondestructive, process monitoring; spin slowly.
- 选择越精确，丢的信号越多（聚焦可补回）。
  More selective, more signal lost (focusing helps).
- SSFP 换灵敏度；CRAFT 换掉 FT。
  SSFP buys sensitivity; CRAFT replaces the FT.

方法选择速查表（一）· §1–§3

Which technique for which problem (1) · §1–§3

只用课件说过或图上看得到的内容，括号为幻灯片编号；「—」表示课件没有说。

Only what the deck states or shows, with slide numbers in brackets; "—" means the deck says nothing.

问题 Problem	方法 Technique	为什么 Why	代价 / 限制 Cost / limit
溶剂峰一般，要最简单 ordinary solvent peak, simplest	presat	长而弱的选择性脉冲饱和溶剂；多数样品往往够用（S4–5） long, weak selective pulse saturates the solvent; often enough (S4–5)	衰减可交换质子（S10）；对湖水不够（S9） attenuates exchangeables (S10); insufficient for lake water (S9)
非专家操作，要省事 non-expert, little tuning	PURGE	用梯度，序列几乎不用调；好于 presat（S7） gradients, very little to adjust; better than presat (S7)	中等强度；内含 presat → 衰减可交换质子（S7） medium; contains presat → attenuates exchangeables (S7)
多个溶剂峰（酒精饮料），要保留 OH、NH several solvent peaks (alcoholic drinks), keep OH, NH	WET	梯度；多峰同时抑制；不扰可交换峰；激发前完成，弛豫不是问题（S5–6） gradients; several peaks at once; exchangeables untouched; done before excitation, so relaxation is no concern (S5–6)	宽带 WET 抹掉整段区域，区内样品峰一起丢（S6 图） broadband WET clears a whole region, sample peaks included (S6 fig.)
溶剂峰很糟；极稀水样（湖水） very bad solvent peak; very dilute water (lake water)	WATERGATE, W5 (SPR-W5-WATERGATE), JRS	脉冲串翻转除水以外的一切，梯度把水散相；抑制极好（S8–9、S13） train flips all but water, a gradient dephases water; excellent suppression (S8–9, S13)	最激进时约 0.4 ppm 抹除、至多 1.1 ppm 衰减（S11）；长序列有弛豫损失，低场更重（S12） most aggressive: ≈ 0.4 ppm obliterated, up to 1.1 ppm attenuated (S11); relaxation in long sequences, worse at low field (S12)
谱峰太挤（蛋白质、混合物） too crowded (proteins, mixtures)	3D NMR	峰容量更高；用投影代替立方体（S14–15） higher peak capacity; projections replace the cube (S14–15)	时间长、成本高，小分子少用（S15、S17）；活体信号弱（S16） long and costly, rare for small molecules (S15, S17); weak in vivo (S16)
多维实验太慢 nD experiment too slow	NUS	少采 t_1 增量；NUS 25 % ≈ 常规 4 倍时间的结果（S19–20） fewer t_1 increments; NUS 25 % ≈ a conventional run 4× longer (S19–20)	SNR 下降（加权采样可缓解）；可能有伪峰（S20） SNR loss (weighted sampling helps); possible artifacts (S20)
一次扫描拿到 2D 2D in a single scan	超快 NMR（UF） ultrafast NMR (UF)	在管内不同位置同时采集所有增量（S21）；COSY 0.2 s vs 21 min（S22） all increments at once in different parts of the tube (S21); COSY 0.2 s vs 21 min (S22)	SNR 差、需高浓度、对样品不均匀敏感、峰可能虚假或缺失（S23） poor SNR, high concentration, sensitive to inhomogeneity, spurious or missing peaks (S23)

考点速记 KEY POINTS

- 先问三件事：溶剂峰多糟？要不要 OH、NH？要不要溶剂旁的峰？（S13）
  Ask: how bad is the solvent? Need OH, NH? Need peaks near it? (S13)
- 要 OH、NH → 避开 presat、PURGE。
  Need OH, NH → avoid presat and PURGE.
- NUS 省时但损 SNR；UF 最快但最不可靠。
  NUS saves time at some SNR; UF is fastest but least reliable.
- 3D 只在峰太挤时才值得花时间。
  3D is worth its time only when peaks are too crowded.

方法选择速查表（二）· §4–§5

Which technique for which problem (2) · §4–§5

问题 Problem	方法 Technique	为什么 Why	代价 / 限制 Cost / limit
凝胶、半固体、生物组织 gels, semi-solids, biological tissue	HR-MAS	慢转、带梯度；与静态相比差别巨大（S30）；适合组织样品（S34） slow spinning, gradient; a drastic gain over static (S30); good for tissue (S34)	测不了真固体（S30）；细胞约 3500 Hz 破裂（S28） no true solids (S30); cells break at ≈ 3500 Hz (S28)
只关心固体 only solids matter	固体探头（固体 NMR） solids probe (solid-state NMR)	功率更高、转得更快（S34）；快转提高分辨率（S29、S31） higher power, faster spinning (S34); fast spinning improves resolution (S29, S31)	峰宽、分辨率很少达到溶液水平、弛豫快、需高功率脉冲（S31）；小转子贵且难用（S29） broad lines, resolution rarely solution-like, fast relaxation, high-power pulses (S31); small rotors costly and awkward (S29)
样品含所有相，或关心相间变化 all phases, or changes between phases	CMP NMR + 谱编辑 CMP NMR + spectral editing	一个探头测所有相，功率承受高、有梯度（S32）；扩散编辑与 T_2 / I-T_2 ^{13}C CP-MAS 过滤分出各相（S33） one probe for all phases, high power handling, gradients (S32); diffusion editing and T_2 / I-T_2 ^{13}C CP-MAS filters separate phases (S33)	只要固体时不如固体探头（S34） for solids only, a solids probe is better (S34)
样品量少或很贵（蛋白质） little or precious sample (proteins)	小管（1.7 mm、1 mm），超低温探头 small tubes (1.7 mm, 1 mm), cryoprobe	少稀释；质量灵敏度更高（S37–39）；1.7 mm 约 30 µL，约为 5 mm 的 1/20（S38） less dilution; better mass sensitivity (S37–39); 1.7 mm ≈ 30 µL, ≈ 20× less than 5 mm (S38)	—
质量极受限（单个卵、大型溞卵、植入脑内） extremely mass-limited (single ova, *D. magna* eggs, brain implant)	微线圈；微线圈阵列 microcoil; microcoil array	质量灵敏度最好（S40）；植入：图 ×76，文字约 100 倍（S46）；阵列同时测暴露组与对照，减少差异（S42） best mass sensitivity (S40); implant: figure ×76, text ≈ 100× (S46); arrays measure exposed and control together, less variability (S42)	B_1 不均匀、样品不反转、COSY 无信号（S43–44）；放样难，普通样品无益（S45） poor B_1 homogeneity, never inverts, no COSY signal (S43–44); hard placement, no gain for ordinary samples (S45)
样品多但很稀（雨水） plenty of dilute sample (rainwater)	大线圈（8、10、20 mm） larger coils (8, 10, 20 mm)	样品量 $\propto d^2$（S48）；浓度灵敏度好（S47） amount $\propto d^2$ (S48); good concentration sensitivity (S47)	功率大、场不均匀、样品发热、质量灵敏度差，加倍样品不等于加倍信号；8 mm 已基本淘汰（S49） more power, poorer homogeneity, heating, poor mass sensitivity, double sample ≠ double signal; 8 mm largely phased out (S49)

考点速记 KEY POINTS

- 凝胶/组织 → HR-MAS；只要固体 → 固体探头；跨相 → CMP。
  Gels/tissue → HR-MAS; solids only → solids probe; phases → CMP.
- 组织样品转速不超过约 3500 Hz。
  Keep tissue samples below ≈ 3500 Hz.
- 样品少 → 小线圈（质量灵敏度）。
  Little sample → small coil (mass sensitivity).
- 样品多但稀 → 大线圈（浓度灵敏度）。
  Plenty but dilute → large coil (concentration sensitivity).

方法选择速查表（三）· §6–§7

Which technique for which problem (3) · §6–§7

问题 Problem	方法 Technique	为什么 Why	代价 / 限制 Cost / limit
混合物中一个化合物的自旋体系 one compound's spin system in a mixture	选择性 TOCSY selective TOCSY	激发一个共振峰，只看其耦合链（S51–52） excite one resonance, see only its coupled chain (S51–52)	目标峰不能重叠（S59）；强度与纯品不同（S52） target must not overlap (S59); intensities differ from the pure compound (S52)
含近等价自旋对的化合物 compounds with near-equivalent spin pairs	单重态 NMR singlet-state NMR	单重态作过滤，可加 TOCSY 模块；无需成形脉冲，旧谱仪也能做（S53–54） singlet filter, optional TOCSY block; no shaped pulses, works on old hardware (S53–54)	不完美，仍有泄漏（S54） not perfect: breakthrough remains (S54)
生物样品中的氨基酸 amino acids in a biological sample	仅氨基酸 NMR amino-acid-only NMR	频带选择性脉冲只选 βHC–αHC–C=O，切趾去掉蛋白质，埋住的峰显出（S55–57） band-selective pulses pass only βHC–αHC–C=O, apodization removes proteins, buried peaks appear (S55–57)	乳酸结构相似，也会通过（S57） lactic acid has a similar motif and passes too (S57)
目标多重峰与别的峰重叠 target multiplet overlaps others	GEMSTONE	像 UF 一样用梯度对管内不同部分编码，重叠也能挑出（S59–60） gradients encode parts of the tube, as in UF; picks it out despite overlap (S59–60)	灵敏度损失，快弛豫样品尤其大（S60） sensitivity penalty, large for fast-relaxing samples (S60)
精确挑出指定化合物（活体、血液、尿液） exactly the chosen compounds (in vivo, blood, urine)	DREAMTIME（多重选择、TOCSY、HSQC、聚焦） DREAMTIME (multiple selection, TOCSY, HSQC, focusing)	位移、耦合、J 都对才通过，每个化合物两个峰（S62）；同相磁化可接入几乎任何实验（S65）；聚焦提升 SNR（S66） passes only with the right shifts, coupling and J, two peaks per compound (S62); in-phase output feeds nearly any experiment (S65); focusing raises SNR (S66)	选择过程丢信号（S66）；TOCSY 模块丢很多（S64） selection loses signal (S66); the TOCSY block loses a lot (S64)
活体生物随时间的响应 a living organism's response over time	活体 NMR（流动系统；活体 CMP）+ PCA in vivo NMR (flow system; in vivo CMP) + PCA	无损，同一个体走完暴露与恢复（S67、S72）；PCA 区分暴露组与对照，载荷指出关键代谢物（S73） nondestructive, the same individual through exposure and recovery (S67, S72); PCA separates exposed from control, loadings name key metabolites (S73)	峰形差、弛豫快（S68）；扩散实验不灵（S69）；CMP 需开孔、慢转，边带成问题（S71） poor lineshape, fast relaxation (S68); diffusion experiments fail (S69); CMP needs a hole and slow spinning, sidebands (S71)
在生物背景中追踪给入的化合物 tracking a dosed compound against the organism	^{13}C 富集生物 + ^{12}C 示踪剂 ^{13}C-enriched organism + ^{12}C tracer	只看连在 ^{12}C 上的质子，生物背景消失，可得动力学曲线（S74） sees only protons on ^{12}C, removing the organism's background; gives a kinetics curve (S74)	生物须先 ^{13}C 富集，并用不同的脉冲序列（S74） the organism must be ^{13}C-enriched, with a different pulse sequence (S74)

考点速记 KEY POINTS

- 峰不重叠 → 选择性 TOCSY；重叠 → GEMSTONE。
  No overlap → selective TOCSY; overlap → GEMSTONE.
- 活体：无损，可监测暴露与恢复。
  In vivo: nondestructive; tracks exposure and recovery.
- 选得越精确，丢的信号越多；DREAMTIME 用聚焦补回。
  Sharper selection loses more signal; DREAMTIME refocuses it.
- ^{13}C 生物 + ^{12}C 示踪：只见示踪剂。
  ^{13}C organism + ^{12}C tracer: only the tracer shows.

方法选择速查表（四）· §8 与全书通用的取舍

Which technique for which problem (4) · §8 and recurring trade-offs

问题 Problem	方法 Technique	为什么 Why	代价 / 限制 Cost / limit
信号弱、时间有限 weak signal, limited time	SSFP	不等完全弛豫，建立稳态后快速采样；D1 = 1 ms，同时间 16K ↔ 380 000 次扫描（S75–76） no wait for full relaxation, fast sampling of a steady state; D1 = 1 ms, 16K ↔ 380 000 scans in the same time (S75–76)	牺牲分辨率：短 AQ 截断，120 ms 是好折中，宜用于快弛豫的宽信号（S77）；定量通常需要完全弛豫（S75） costs resolution: short AQ truncates, 120 ms is a good compromise, best for fast-relaxing broad signals (S77); quantitation normally needs full relaxation (S75)
谱中有倒峰，无法正常校相 an inverted peak, the spectrum cannot be phased	CRAFT（子 FID） CRAFT (sub-FIDs)	FID 拆成子 FID 分别处理，常规做不到的校相可以完成（S79） the FID is split into sub-FIDs treated separately, so the phasing that fails normally can be done (S79)	—
弱峰被噪声掩盖 weak peaks hidden by noise	CRAFT	某种程度上「去噪」，显出原本看不到的峰：13 个 vs 1D 的 3 个（S80） "kind of removes noise", showing peaks otherwise hidden: 13 vs 3 in the 1D (S80)	「SNR」加了引号；峰数标注「4.7x」与 13/3 ≈ 4.3 不一致，文字为「~5 倍」（S80） "SNR" is in quotes; the label "4.7x" disagrees with 13/3 ≈ 4.3, the text says "~5 times" (S80)
要直接用于定量的结果 results ready for quantitation	CRAFT	不用 FT，直接得到频率–振幅表，用于定量解读（S78） no FT; a frequency–amplitude table straight from the FID for quantitative interpretation (S78)	—
又弱又想看到更多峰 weak and need more peaks	SSFP + CRAFT	两者可结合（S80）；12 h 内 1D 3 个峰、SSFP 7 个、CRAFT 13 个（S80） the two can be combined (S80); in 12 h: 1D 3 peaks, SSFP 7, CRAFT 13 (S80)	继承 SSFP 的分辨率代价（S77） inherits SSFP's resolution cost (S77)

全书反复出现的取舍 / Trade-offs that recur through the deck

- 抑制或选择越强，丢的信号越多：溶剂旁的峰（S11）、GEMSTONE（S60）、DREAMTIME 的 TOCSY 与选择（S64、S66）。
  Stronger suppression or selection loses more signal: peaks near the solvent (S11), GEMSTONE (S60), DREAMTIME's TOCSY and selection (S64, S66).
- 越快越要付 SNR、伪峰或分辨率：NUS（S20）、超快 NMR（S23）、SSFP（S77）。
  Speed is paid for in SNR, artifacts or resolution: NUS (S20), ultrafast NMR (S23), SSFP (S77).
- 转速越快，分辨率越好、边带越远，但细胞和活体受不了（S28、S29、S71）。
  Faster spinning gives better resolution and more distant sidebands, but cells and organisms cannot take it (S28, S29, S71).
- 线圈越小，质量灵敏度越好；越大，浓度灵敏度越好（S39、S47）。序列越长，弛豫损失越多（S12、S60）。
  Smaller coils give better mass sensitivity, larger ones better concentration sensitivity (S39, S47). Longer sequences lose more to relaxation (S12, S60).

考点速记 KEY POINTS

- 要灵敏度、不在乎分辨率 → SSFP。
  Need sensitivity, can spare resolution → SSFP.
- 更强的选择或抑制 = 更少的信号。
  Stronger selection or suppression = less signal.
- 倒峰校相、弱峰、定量表 → CRAFT。
  Inverted-peak phasing, weak peaks, quantitative table → CRAFT.
- 更快的采集 = SNR、伪峰或分辨率的代价。
  Faster acquisition = a cost in SNR, artifacts or resolution.